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TONGHANG E-DRIVE

Selection Manual for Servo Series Products

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■ CRAFTSMANSHIP AND QUALITY
■ INTELLIGENT MANUFACTURING FOR THE FUTURE
High performance industrial automation system service provider

TONGHANG



Brand concept

Let customers have confidence to choose domestic brands
Let customers use the cost of inverter, enjoy the performance of AC spindle servo
Let customers enjoy the use of high-performance permanent magnet synchronous servo
Let customers sprint industry 4.0 with our help

Enterprise tenet

Ingenuity, quality, intelligence, focus on Perfection



COMPANY INFORMATION

Zhejiang (Hangzhou) Hanghang Electric Drive Technology Co., Ltd., a high-performance industrial automation system service provider, is a national high-tech enterprise that integrates research and development, production, sales, and service. Since its establishment in 2017, sales performance has soared year after year, approaching 200 million yuan by the end of 2023. The company has established an elite R&D team, which includes dozens of high-level talents such as doctors and masters, and owns dozens of invention patents and utility model patents.

The company specializes in industrial automation products such as servo drives and servo motors. Our products cover low, medium, and high voltage series servo drives and servo motors, with a wide power range from 100W to 22kW. We have successfully developed various bus type servo drives such as EtherCAT, CANOpen, MECHATROLINK II, and MECHATROLINK III. At the same time, a tapping machine servo system and an industrial fast door servo control system were also launched. In addition, we can customize and develop specialized servo series products according to customer industry needs.

In terms of production, the company has a modern clean production workshop of nearly 5000m². While introducing advanced production equipment, we actively develop specialized production equipment and have built multiple automated production lines, achieving an annual production capacity of over 500000 shafts. In terms of quality management, the company always adheres to the principle of "quality first", and strictly conducts more than 30 rigorous processes such as raw material inspection, PCBA testing, motor withstand voltage testing, noise testing, high temperature aging, whole machine load testing, and finished product warehousing testing to ensure that each product meets industry quality standards before leaving the factory. In terms of delivery time, the company relies on its 2500m² finished goods warehouse to ensure sufficient and stable supply of goods, thereby providing customers with solid and reliable delivery guarantees, ensuring timely and worry free cooperation every time.

To quickly respond to market demand, the company has established offices and after-sales service centers in Beijing, Tianjin, Shandong, Hebei, Henan, Jiangsu, Anhui, Shanghai, Zhejiang, Fujian, Guangdong, Jiangxi, Hunan, Hubei, Shaanxi and other places. In addition, the company actively expands overseas markets, and its products are exported to Germany, Russia, Türkiye, Iran, India, Vietnam, Pakistan, Dubai, South Korea and other countries and regions, so as to provide high-quality products and services for global customers.





Corporate mission

Responsibility

Become a responsible enterprise for the society and the best partner for customers Become a platform for employees to ensure their quality of life



Corporate vision

Refine on

Long term development, sustainable operation, and dedication to the community.

Management philosophy

Sincere service

Build a high quality project with customer satisfaction, Create a beautiful and harmonious social environment.

Management philosophy

Sincere service

Good faith management, casting fine products Contribute to the society and benefit the employees.

Employment criteria

Filial piety and loyalty
Emotional intelligence

Taking filial piety as the root and loyalty as the foundation Take emotion as the muscle and wisdom as the bone

Corporate values

Honesty, responsibility, democracy, sharing, cooperation and innovation

Base on the world with honesty and take responsibility for oneself Building a team with democracy and driving interests with sharing Overcome difficulties and dangers through cooperation and open up the future through innovation



Corporate Culture

The company has strong technical force, servo engineers and professional drive technicians, has formed a strong professional and technical talent team, superior production conditions, advanced technology. The company has complete production and testing equipment, strict quality management system and advanced testing means to ensure product quality.



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SERVO DRIVE SELECTION AND CONFIGURATION TABLE



Servo drive model	model	Inputs	analog quantity	With feedback	RS485	CANOpen	M2bus	M3bus	EtherCAT
T3a/T3L	Double board pulse type	√	√	√	√	○	○	○	○
T3D	Single board 17/23 bit absolute value type	√	√	√	√	○	○	○	○
T3DF/C30G	Single board pulse incremental type	√	√	√	√	○	○	○	○
T5a	485 absolute value type	√	√	√	√	○	○	○	○
T5ML	M2 bus type	○	○	○	○	○	√	○	○
T6M	M3 bus type	○	○	○	○	○	○	√	○
T6E/T6DE	EtherCAT type	○	○	○	○	○	○	○	√
T3DC	CANOpen type	○	○	○	○	√	○	○	○
S3a	Spindle type	√	√	√	√	○	○	○	○
T3M	Maixin pulse type	√	√	√	○	○	○	○	○
T3G	Wide pulse type	√	○	○	○	○	○	○	○
T3C	Double board 17/23 bit absolute value type	√	√	√	√	○	○	○	○

√ Indicating standard configuration ○ Indicates no configuration

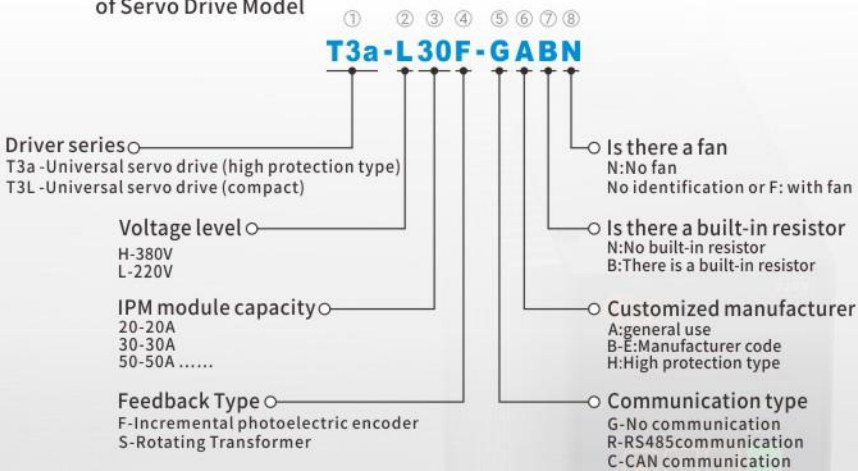
T3a/T3L Series Universal Servo Drivers

The T3a/T3L series digital AC servo drive is a domestically leading universal AC servo drive product independently developed and produced by Zhejiang Hanghang Electric Drive Technology Co., Ltd. This series of products adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability, widely used for servo shaft drive in food processing, packaging machinery, textile machinery and other related automation industry machinery.

Product Features

- ◆ The driver instruction type is pulse type, and the motor position feedback is pulse incremental encoder
- ◆ Adopting high-performance ARM main control chip, with excellent control performance
- ◆ T3a series fully sealed design, good protection, strong anti-interference ability
- ◆ T3L series compact design saves installation space
- ◆ Select a new industrial grade IPM module with strong overload driving capability
- ◆ Integrating position control, speed control, and torque control
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Specification Description of Servo Drive Model



UNIVERSAL SERVO DRIVE STANDARD SPECIFICATIONS AND PERFORMANCE PARAMETERS

model		T3a/L-L20	T3a/L-L30	T3a-L40	T3a-L50	T3a-L75	T3a-H25	T3a-H50	T3a-H75
Input power supply	main power supply	Single phase/Three phase220VAC -15%~+10% 50/60Hz					Threephase380VAC-15%~+10%50/60Hz		
environment	temperature	Work: 0℃~40℃				Storage: -40℃~50℃			
	humidity	Job: 40%~80% (no condensation)				Storage: less than 93% (no condensation)			
	Atmospheric pressure	86kPa ~ 106 kPa							
Protection level		IP20							
control model		Position, speed, torque, position/speed, speed/torque, position/torque							
Digital input		6 programmable input terminals (optically isolated)							
Digital output		4 programmable output terminals (photoelectric isolation)							
Communication method		Rs485communication							
encoder Signal output	signal type	A, B, Z differential output, Z signal collector open circuit output							
position	Input frequency	Differential input: ≤ 500kHz(kpps); Single ended input: ≤ 200kHz(kpps)							
	Command mode	Pulse+direction; Forward/reverse pulse; Orthogonal pulse							
	Electronic gear ratio	1~32767/1~32767							
speed	Simulate command input	± 10VDC, input impedance 10k Ω							
	Command acceleration and deceleration	Parameter settings P-060/P-061							
	Instruction source	Analog quantity, internal speed command							
torque	Simulate command input	-10V~+10V, input impedance 10k Ω							
	Torque limit	Parameter settings P-065/P-066							
	Instruction source	Analog quantity, internal torque command							
Monitoring function		Speed, current position, position deviation, motor torque, motor current, command pulse frequency, etc							
Protection function		Overspeed, overvoltage, overcurrent, overload, abnormal braking, abnormal encoder, position deviation, instruction overlocking, etc							
characteristic	Speed frequency response	≥800Hz							
	Speed fluctuation rate	< ±0.03%(load0~100%); < ±0.02%(power supply-15% ~ +10%)							

T3a SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

Driver size one (model T3a-L20)

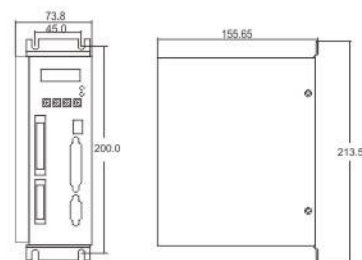


Figure 1 Size diagram of T3a-L20 servo drive (unit: mm)

Driver size two (models T3a-L30)

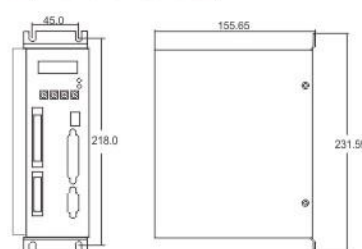


Figure 2 Dimension diagram of T3a-L30 servo drive (unit: mm)

Driver size three (model T3a-L40/T3a-H25)

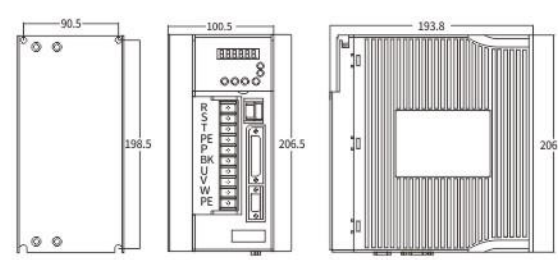


Figure 3 Size diagram of T3a-L40/T3a-H25 servo drive (unit: mm)

Driver size four (models T3a-L50/L75, T3a-H50/H75)

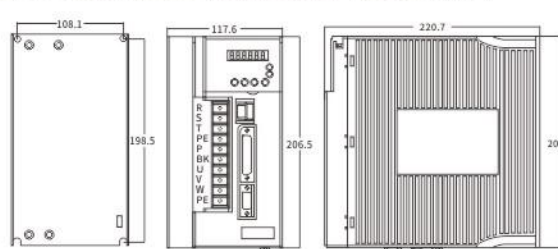


Figure 4 Size diagram of T3a-L50/L75 and T3a-H50/H75 servo drives (unit: mm)

T3L SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

Driver size one (model T3L-L15/L20)

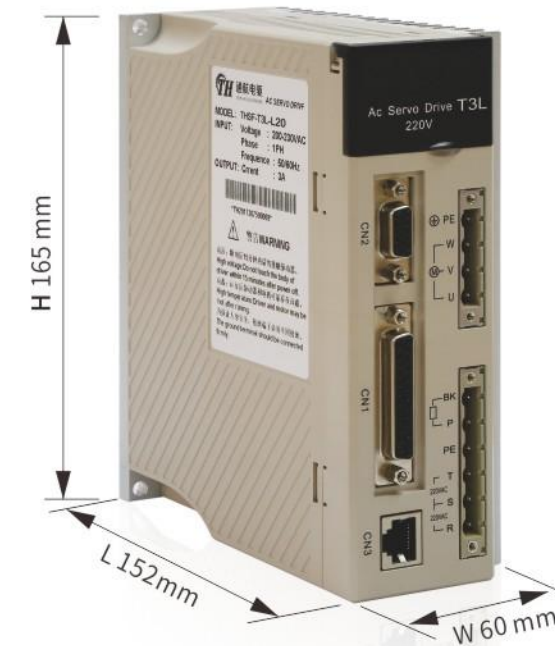


Figure 1 Installation dimension diagram of T3L-L15/L20 (unit: mm)

Driver size two (model T3L-L30)

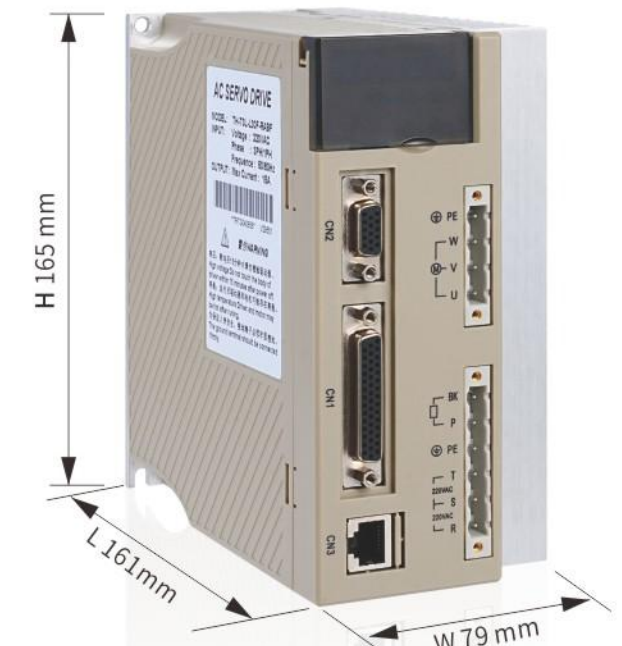
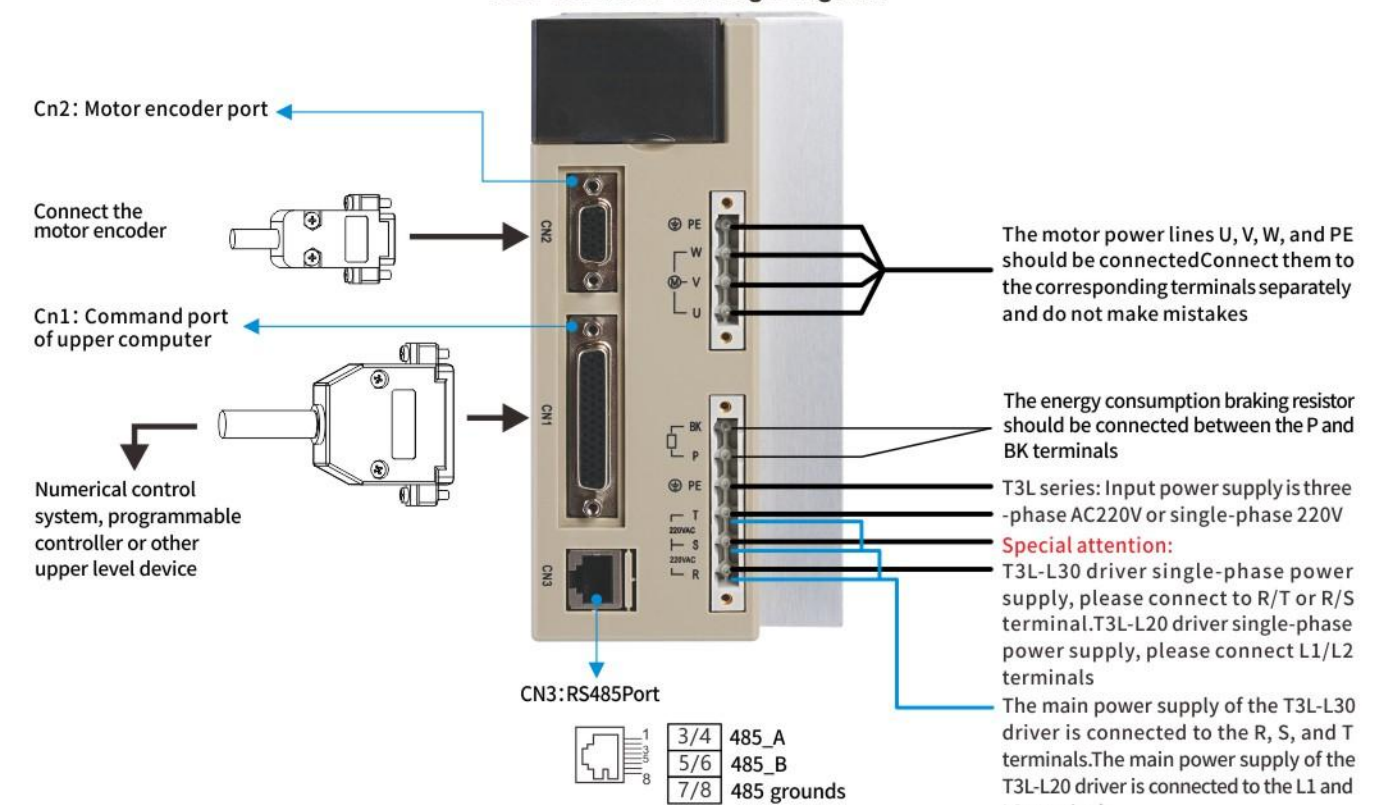


Figure 2 Installation dimension diagram of T3L-L30 (unit: mm)

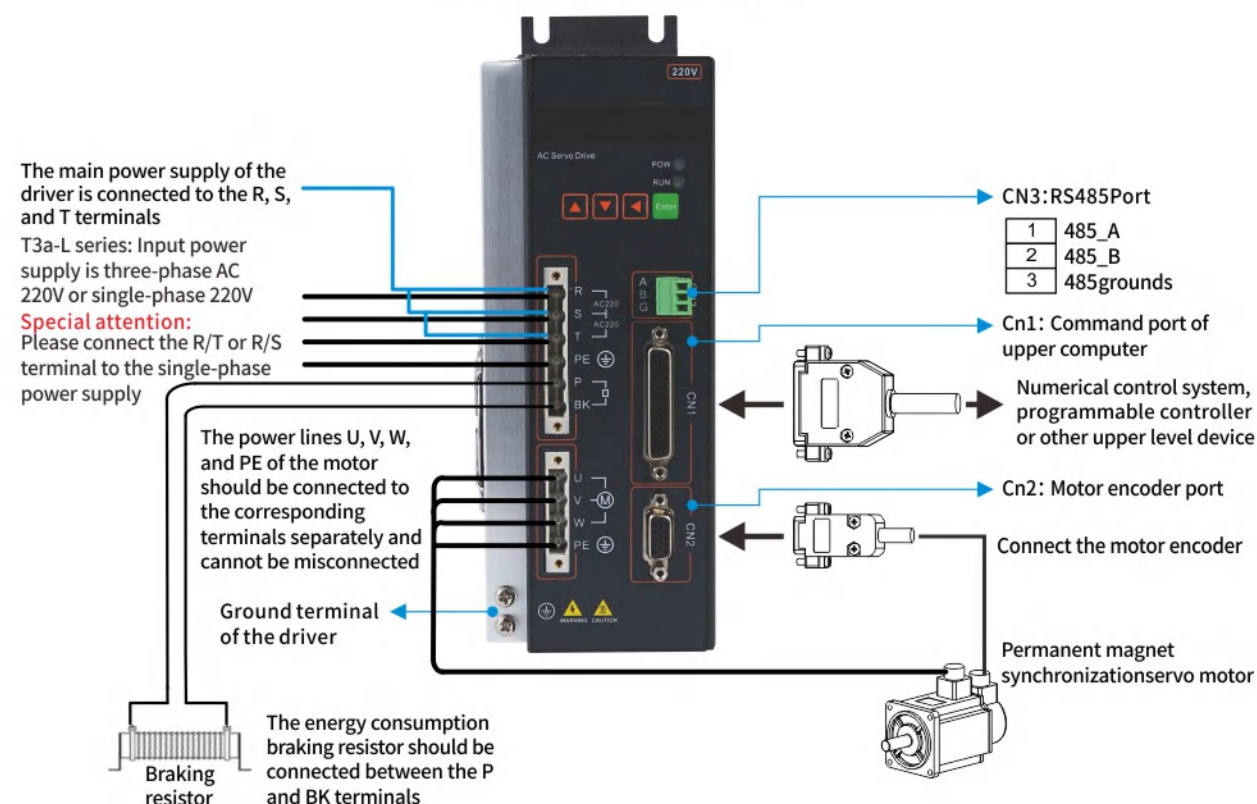
T3L SERVO DRIVE WIRING DIAGRAM

T3L-L20/L30 Wiring Diagram

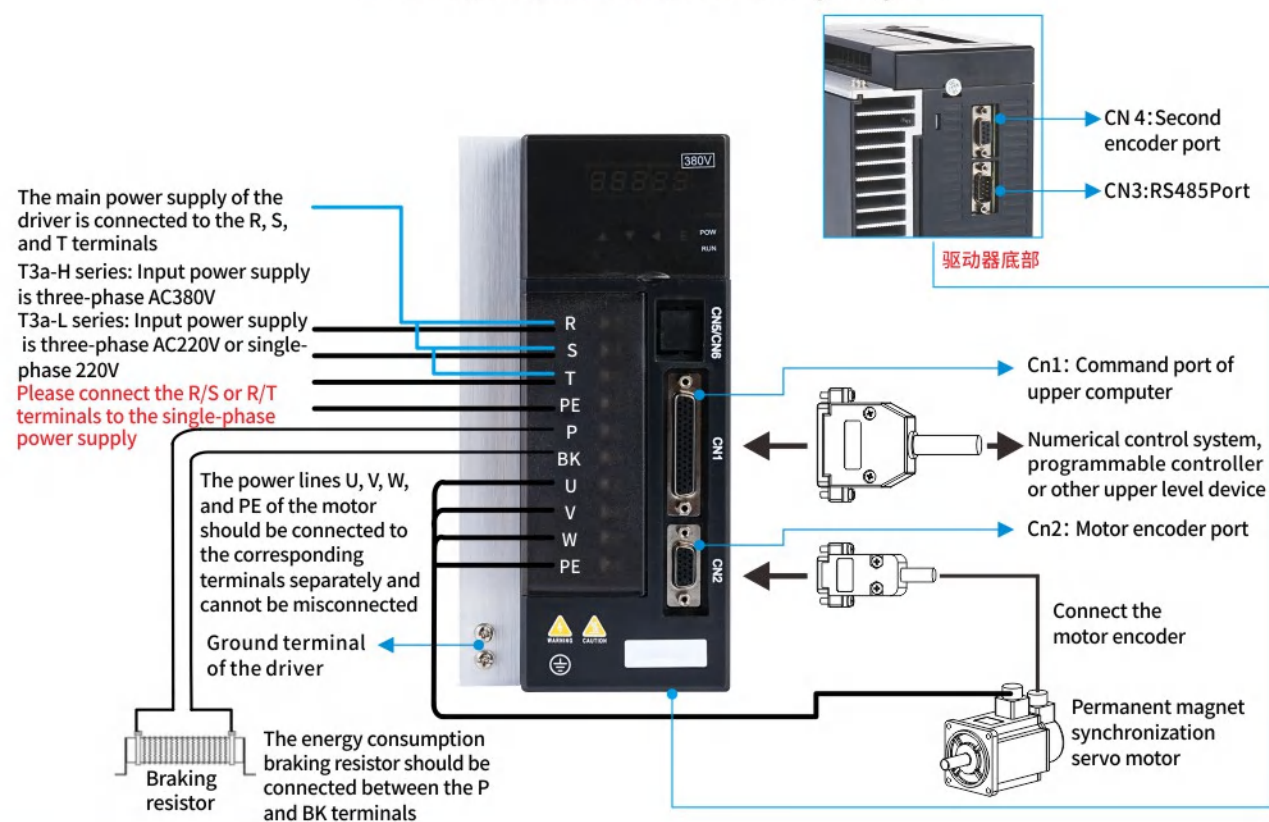


T3a SERVO DRIVE WIRING DIAGRAM

T3a-L20/L30 wiring diagram

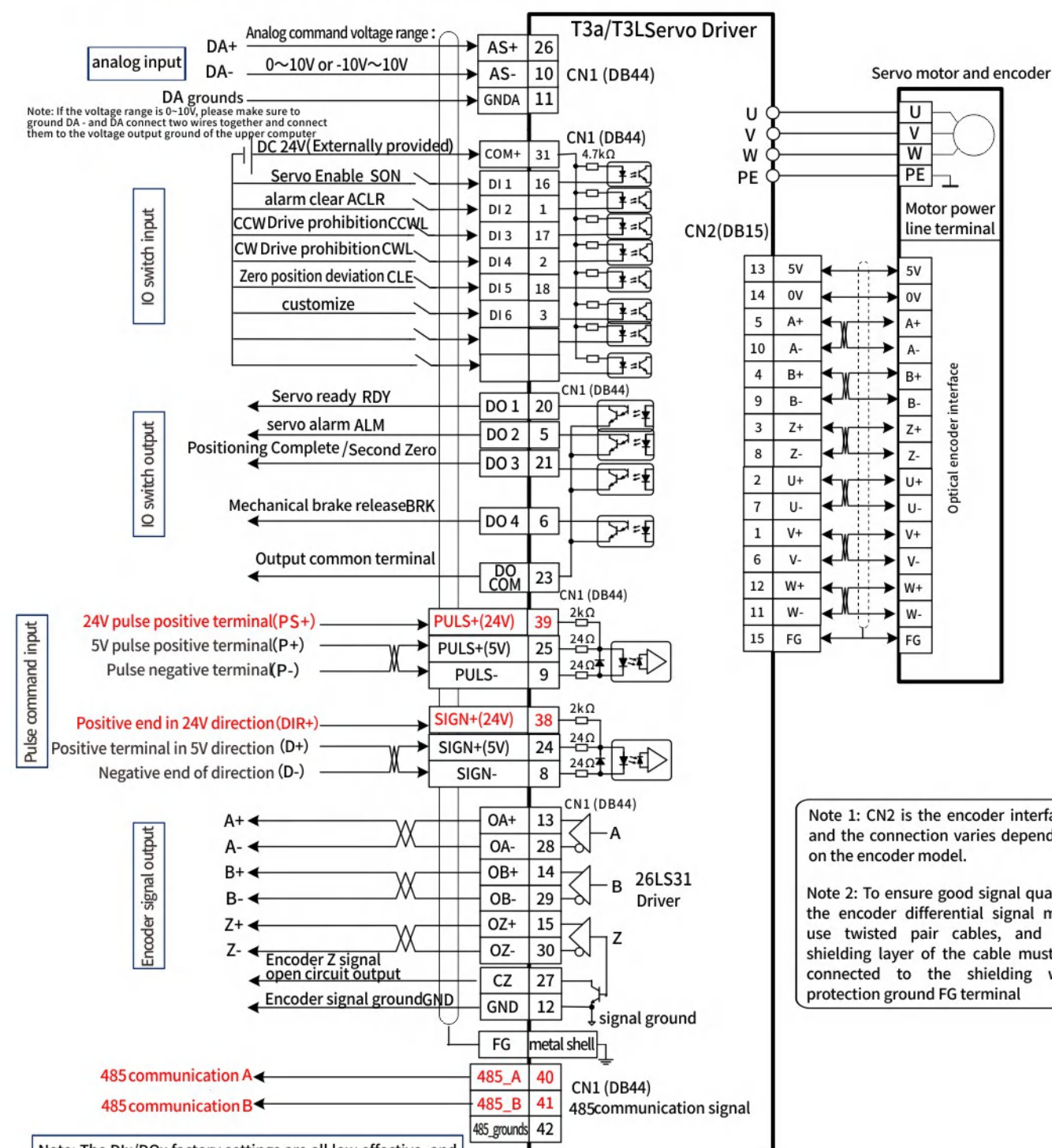


T3a-L50/L75/H25/H50/H75 wiring diagram



CONTROL WIRING DIAGRAM

This side signal is connected to the upper computer controller



INTRODUCTION TO SERVO DRIVE PORTS

Main circuit terminal

name	Terminal symbol	detailed description
Main circuit power supply	R、S、T or L1、L2 (T3L-L20)	Connect external AC power supply: three-phase AC380V (85%~110%) 50/60Hz. Single phase/three-phase AC220V (85%~110%) 50/60Hz
Braking resistor terminal	P,BK	When using an external braking resistor in parallel with an internal braking resistor, please pay attention to selecting the resistance and power to avoid burning out the driver.
Motor connection terminal	U	Output to motor U-phase power supply
	V	Output to motor V-phase power supply
	W	Output to motor W-phase power supply
Grounding terminal		Grounding terminal of motor casing
	or PE	Driver grounding terminal

CONTROL CIRCUIT TERMINAL

type	name	function	detailed description
External 24V control power supply	COM+	DC24V control power output terminal	The positive pole of the power supply at the input terminal is used to drive the optocoupler at the input terminal, DC12~24V, Current ≥ 100mA.
	DO COM	Control power supply 0V terminal	Power ground of input terminal
Control signal input (programmable)	DI1(SON)	Enable input	DI1~DI6 digital input functions are all programmable. The examples shown in this table are factory default configurations, which can be flexibly configured by driver software parameters and can be adapted to various types of upper computers, such as CNC systems PLC、Motion controllers, etc.
	DI2(ACLR)	Alarm clearing	
	DI3(CCWL)	CCW drive prohibited	
	DI4(CWL)	CW drive prohibited	
	DI5(CLE)	Zero position deviation	
Control signal output (programmable)	DI6	custom	All DO1~DO4 digital output functions are programmable, and the examples shown in this table are factory default configurations. They can be flexibly configured by driver software parameters and can be adapted to various types of upper computers, such as CNC systems PLC、Motion controllers, etc.
	DO1(RDY)	Servo ready	
	DO2(ALM)	Alarm output	
	DO3(COIN)	Positioning completed	
Analog command input	DO4(BRK)	Brake output	Positive connection with the analog command output of the upper computer
	AS+	Simulate command input positive	
	AS-	Simulate instruction input negative	
Pulse quantity command input	GNDA	Analog signal ground	Negative connection to the analog command output of the upper computer
	PULS+	Command pulse PLUS input	Connect to the analog signal ground of the upper computer
	PULS-	Command pulse MINUS input	External command pulse input terminal. Note: The pulse input mode is set by parameter P035 and supports three types of pulse input modes
Encoder output signal (sent to the upper computer)	SIGN+	Command pulse SIGN input	◆P035=0: Command pulse+direction mode (default state);
	SIGN-	Command pulse SIGN input	◆P035=1: CCW/CW instruction pulse mode;
	OA+ / OA -	Encoder A-phase signal output	◆P035=2:Orthogonal pulse mode.
	OB+ / OB -	Encoder B-phase signal output	◆Encoder ABZ signal differential drive output (26LS31 output, equivalent to Rs422);
	OZ+ / OZ -	Encoder Z-phase signal output	◆Non isolated output (non insulated).
Encoder input signal (from spindle motor)	CZ	Encoder Z-phase collector electrode open circuit output	◆The Z-phase signal of the encoder is output by an open collector. When the Z-phase signal of the encoder appears, it outputs ON (output conduction), otherwise it outputs OFF (output cutoff);
	GND	Encoder signal ground	◆Non isolated output (non insulated);
	A+/A-	Encoder A-phase signal input	Servo motor encoder signal input terminal, different types of encoder signals have different definitions. The table shows the photoelectric encoder, and subsequent products support photoelectric encoders, rotary encoders, sine cosine encoders, and absolute value encoders
	B+/B-	Encoder B-phase signal input	
	Z+/Z-	Encoder Z-phase signal input	
	U+/U-	Encoder U phase signal input	
	V+/V-	Encoder V-phase signal input	
Grounding terminal	W+/W-	Encoder W-phase signal input	Connect to the shielded wire of the signal cable
	5V/GNDD	Encoder 5V power/ground	
Grounding terminal		PE / FG	Shielded wire protection ground

DRIVER ACCESSORIES (BRAKING RESISTOR)

T3a-H series (380V servo) brake resistor configuration table (standard 380V servo does not include built-in resistors, external brake resistors must be connected, except for customized models)

drive model	(Light load type) Recommended configuration (infrequent start stop)	(Heavy duty type) Recommended configuration (frequent start stop)
T3a-H25F	≥ 50 ohms, power greater than 1000W	≥ 50 ohms, power greater than 1500W
T3a-H50F	≥ 50 ohms, power greater than 1000W	≥ 50 ohms, power greater than 1500W
T3a-H75F	≥ 30 ohms, power greater than 1500W	≥ 30 ohms, power greater than 2000W

T3a-L/T3L series (220V servo) brake resistor configuration table (standard 220V servo includes built-in resistor, external brake resistor can be selected according to load conditions)

drive model	(Light load type) Recommended configuration (using built-in resistor)	(Heavy duty type) Recommended configuration (with external resistor connected in parallel)
T3a/T3L-L20F	47 euros (built-in)	≥ 47 euros, 500W
T3a/T3L-L30F	47 euros (built-in)	≥ 47 euros, 500W
T3a-L40F	47 euros (built-in)	≥ 47 euros, 500W
T3a-L50F	47 euros (built-in)	≥ 47 euros, 500W
T3a-L75F	47 euros (built-in)	≥ 20 euros, 1000W

TECHNICAL PARAMETERS OF SUPPORTING SERVO MOTOR

Table x T3a-H25/T3a-H50/T3a-H75 Driver Compatible Motor Models

Motor code (P-002)	Adaptive driver (AC 380V)	Servo motor model (380V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
0	T3a-H25	110ST-M02030	0.6	2.0	2	3
1		110ST-M04020	0.8	2.0	4	3
2		110ST-M04030	1.2	3.0	4	3
3		110ST-M05030	1.5	4.5	5	3
4		110ST-M06020	1.2	3.0	6	3
5		110ST-M06030	1.8	4.5	6	3
6		130ST-M04025	1.0	2.6	4	3
7		130ST-M05025	1.3	3.0	5	3
8		130ST-M06025	1.5	3.7	6	3
9		130ST-M07725	2.0	4.7	7.7	3
10		130ST-M10010	1.0	2.5	10	3
11		130ST-M10015	1.5	3.5	10	3
12		130ST-M10020	2.0	5.1	10	3
13		130ST-M10025	2.6	5.9	10	3
14		130ST-M15015	2.3	5.0	15	3
15		130ST-M15025	3.8	7.4	15	2.8
16		130ST-M17020	3.5	7.4	17	2.8
17		150ST-M15020	3.0	6.8	15	3
18	T3a-H25/T3a-H50	150ST-M15025	3.8	9.5	15	2.2/2.9
19	T3a-H25/T3a-H50	150ST-M18020	3.6	8.5	18	2.5/3
20	T3a-H25/T3a-H50	150ST-M23020	4.7	12	23	1.8/2.3
21	T3a-H25/T3a-H50	150ST-M27015	4.2	11	27	1.9/2.5
22	T3a-H50/T3a-H75	150ST-M27020	5.5	14.5	27	2/2.7
23	T3a-H75	150ST-M27025	6.8	17	27	2.3
24	T3a-H25	180ST-M17215	2.7	6.5	17.2	3
25	T3a-H25	180ST-M19015	3.0	7.5	19	2.8
26	T3a-H25/T3a-H50	180ST-M21520	4.5	9.5	21.5	2.2/2.9
27	T3a-H25	180ST-M27010	2.9	7.5	27	2.8
28	T3a-H25/T3a-H50	180ST-M27015	4.3	10	27	2.1/2.8
29	T3a-H25/T3a-H50	180ST-M35010	3.7	10	35	2.1/2.8
30	T3a-H25/T3a-H50	180ST-M35015	5.5	12	35	1.8/2.3
31	T3a-H75	180ST-M48015	7.5	20	48	1.95

TECHNICAL PARAMETERS OF SUPPORTING SERVO MOTOR

Table x T3x-L20/T3x-L30 Driver Compatible Motor Models

Motor code (P-002)	Adaptive driver (AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
20	T3x-L20	40ST-M00130	0.05	0.61	0.16	3
20		40ST-M00330	0.1	0.9	0.32	3
21		60ST-M00630	0.2	1.2	0.6	3
22		60ST-M01330	0.4	2.8	1.3	3
23		60ST-M01930	0.6	3.5	1.9	3
27		80ST-M01330	0.4	2.0	1.3	3
28		80ST-M02430	0.75	3.0	2.4	2.5
29		80ST-M03520	0.73	3.0	3.5	2.5
30		80ST-M04025	1.0	4.4	4	2.5
31		90ST-M02430	0.75	3	2.4	3
32		90ST-M03520	0.73	3	3.5	3
33		90ST-M04025	1.0	4	4	2.5
34	T3x-L20/L30	110ST-M02030	0.6	2.5	2	2.5/3
35	T3x-L20/L30	110ST-M04030	1.2	5	4	2/3
39	T3x-L20/L30	110ST-M04020	0.8	3.5	4	3/3
36	T3x-L30	110ST-M05030	1.5	6	5	2.5
42	T3x-L30	110ST-M05050	2.5	8.5	5	1.8
37	T3x-L30	110ST-M06020	1.2	4.5	6	3
38	T3x-L30	110ST-M06030	1.8	6	6	2.5
53	T3x-L30	110ST-M06040	2.4	8	6	1.9
44	T3x-L20/L30	130ST-M04025	1	4	4	2.5/3
45	T3x-L20/L30	130ST-M05025	1.3	5	5	2/3
46	T3x-L30	130ST-M06025	1.5	6	6	2.5
41	T3x-L30	130ST-M06030	1.8	7	6	2.2
47	T3x-L30	130ST-M07720	1.6	6	7.7	2.5
47	T3x-L30	130ST-M07725	2.0	7.5	7.7	2
48	T3x-L30	130ST-M07730	2.4	9	7.7	1.6
40	T3x-L20/L30	130ST-M10010	1.0	4.5	10	2.2/3
49	T3x-L30	130ST-M10015	1.5	6	10	2.5
43	T3x-L30	130ST-M10020	2	8.5	10	1.7
50	T3x-L30	130ST-M10025	2.6	10	10	1.5
51	T3x-L30	130ST-M15025	3.8	13.5	15	1.3
52	T3x-L30	130ST-M15015	2.3	9.5	15	1.6

Note: T3x represents T3a T3L

Table x T3a-L40/T3a-L50/T3a-L75 Driver Compatible Motor Models

Motor code (P-002)	Adaptive driver (AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
0	T3a-L40	130ST-M07720	1.6	6	7.7	3
0	T3a-L40	130ST-M07725	2.0	7.5	7.7	3
1	T3a-L40/L50	130ST-M07730	2.4	9	7.7	2.3/3
2	T3a-L40	130ST-M10015	1.5	6	10	3
3	T3a-L40/L50	130ST-M10025	2.6	10	10	2.1/2.8
4	T3a-L40/L50	130ST-M12020	2.4	10	12	2.1/2.8
5	T3a-L40/L50	130ST-M15015/130ST-M17015	2.3	9.5	15	2.2/3
6	T3a-L50/L75	130ST-M15025/130ST-M23015	3.9	13.5	15	1.6/2.3
7	T3a-L50/L75	150ST-M15020	3.0	14	15	2.0/2.8
8	T3a-L50/L75	150ST-M15025	3.8	17	15	1.7/2.4
9	T3a-L50/L75	150ST-M18020	3.6	17	18	1.7/2.4
10	T3a-L50/L75	150ST-M23020	4.7	21	23	1.4/1.9
11	T3a-L50/L75	150ST-M27020	5.4	24	27	1.4/1.9
12	T3a-L40/L50	180ST-M17215	2.7	10.5	17.2	2/2.7
13	T3a-L50/L75	180ST-M19015	3	12	19	2.3/3
14	T3a-L50/L75	180ST-M21520	4.5	16	21.5	1.8/2.5
15	T3a-L50/L75	180ST-M27010	2.9	12	27	2.3/3
16	T3a-L50/L75	180ST-M27015	4.3	16	27	1.8/2.4
17	T3a-L50/L75	180ST-M35010	3.7	16	35	1.8/2.4
18	T3a-L75	180ST-M35015	5.5	24	35	1.6
19	T3a-L75	180ST-M48015	7.5	32	48	1.2

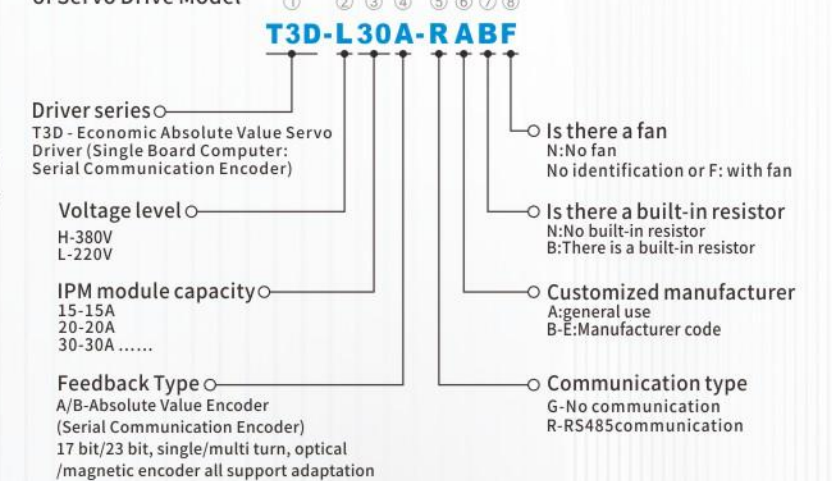
T3D Series Single Board Absolute Value Servo Driver

The T3D series universal servo drive adopts advanced fully digital control and AC motor vector control theory, and is compatible with serial communication absolute value encoders. The system has excellent performance and high reliability, and is widely used in various general automation equipment.

Product Features

- ◆ The driver instruction type is pulse type, and the motor position feedback is pulse incremental encoder
- ◆ Adopting high-performance ARM main control chip, with excellent control performance
- ◆ Suitable for 17 bit/23 bit, single/multi turn serial communication encoders, with high positioning accuracy and good servo shaft rigidity
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Compact single board design, saving installation space
- ◆ Integrating position control, speed control, and torque control
- ◆ Choose the latest industrial grade IPM module with strong overload driving capability
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Can drive various types of permanent magnet synchronous servo motors

Specification Description of Servo Drive Model

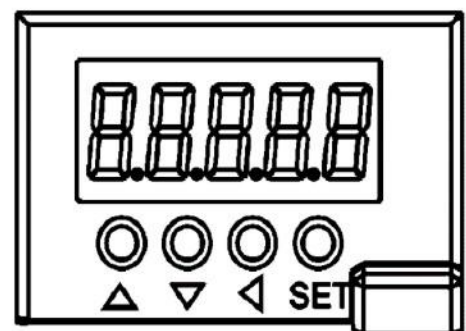


STANDARD SPECIFICATIONS AND PERFORMANCE PARAMETERS FOR SERVO DRIVES

model		L15	L20	L30	H10	H15	H25
Input power supply	main power supply	single phase220VAC -15%~+10% 50/60Hz/three-phase220VAC -15%~+10% 50/60Hz(L30Model support)			three-phase380VAC -15%~+10% 50/60Hz		
environment	temperature	Work: 0℃~40℃ Storage: -40℃~50℃					
	humidity	Job: 40%~80% (no condensation) Storage: below 93% (no condensation)					
	Atmospheric pressure	86kPa~106 kPa					
Protection level		IP20					
control model		Position, speed, torque, position/speed, speed/torque, position/torque					
Digital input		4 programmable input terminals (optically isolated)					
Digital output		4 programmable output terminals (photoelectric isolation)					
encoder Signal output	signal type	A, B, Z differential output Z signal collector open circuit output RS485 communication reading feedback encoder position Encoder feedback pulse signal output/position command synchronous pulse signal output					
position	Input frequency	T3D: Differential input: ≤1.5MHz(kpps); Single ended input: ≤200kHz(kpps)					
	Command mode	T3D: Pulse+symbol; Forward/reverse pulse; Orthogonal pulse					
	Electronic gear ratio	1~32767/1~32767					
speed	Simulate command input	0~10VDC, input impedance 10kΩ					
	Command acceleration and deceleration	Parameter settings P-060/P-061					
	Instruction source	Analog quantity, internal speed command					
torque	Simulate command input	0~+10V or -10V~+10V, input impedance 10kΩ					
	Torque limit	Parameter settings P-065/P-066					
	Instruction source	Analog quantity, internal torque command					
Monitoring function		Speed, current position, position deviation, motor torque, motor current, command pulse frequency, etc					
Protection function		Overspeed, overvoltage, overcurrent, overload, abnormal braking, abnormal encoder, position deviation, instruction overlocking, etc					
characteristic	Speed frequency response	≥1.6kHz					
	Speed fluctuation rate	<±0.03% (load 0~100%)<±0.02% (power supply -15%~+10%)					

INTRODUCTION TO THE OPERATION PANEL

The T3D series panel consists of 5 LED digital tube displays and 4 buttons $\blacktriangle$ 、 $\blacktriangledown$ 、 $\blacktriangleleft$ 、 SET , used to display various system statuses and set parameters. The operations are all hierarchical operations, which are expanded layer by layer from the main menu. The operation panel is shown in the following figure:



symbol	name	function
POW	Power light (none)	The decimal light on the second digital tube from the left indicates servo undervoltage
RUN	Running light (none)	The decimal light on the first digital tube from the left indicates that the servo is enabled
$\blacktriangle$	add keys	Add serial numbers or numerical values; Long pressing has a repeating effect.
$\blacktriangledown$	Reduce key	Reduce the serial number or numerical value; Long pressing has a repeating effect.
$\blacktriangleleft$	Exit button	Menu exit; Operation cancelled.
SET	Confirm button	Menu access; Parameter modification confirmation or operation confirmation.

T3D SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

Driver size one (model T3D-L15/L20)



Figure 1 Installation dimension diagram of T3D-L15/L20 (unit: mm)

Driver size three (model T3D-H10/H15/H25)

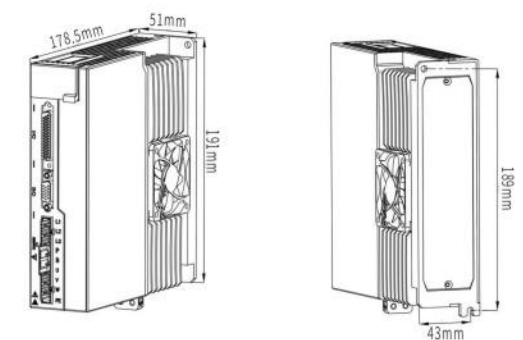


Figure 3 Installation dimension diagram of T3D-H10/H15/H25 (unit: mm)

Driver size two (model T3D-L30)

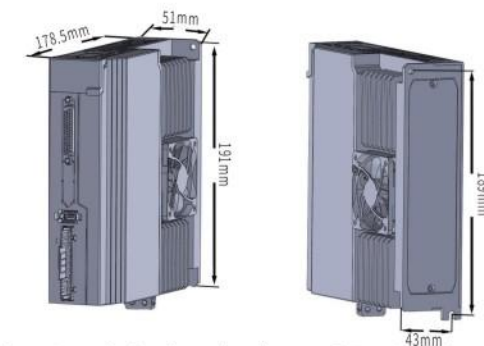
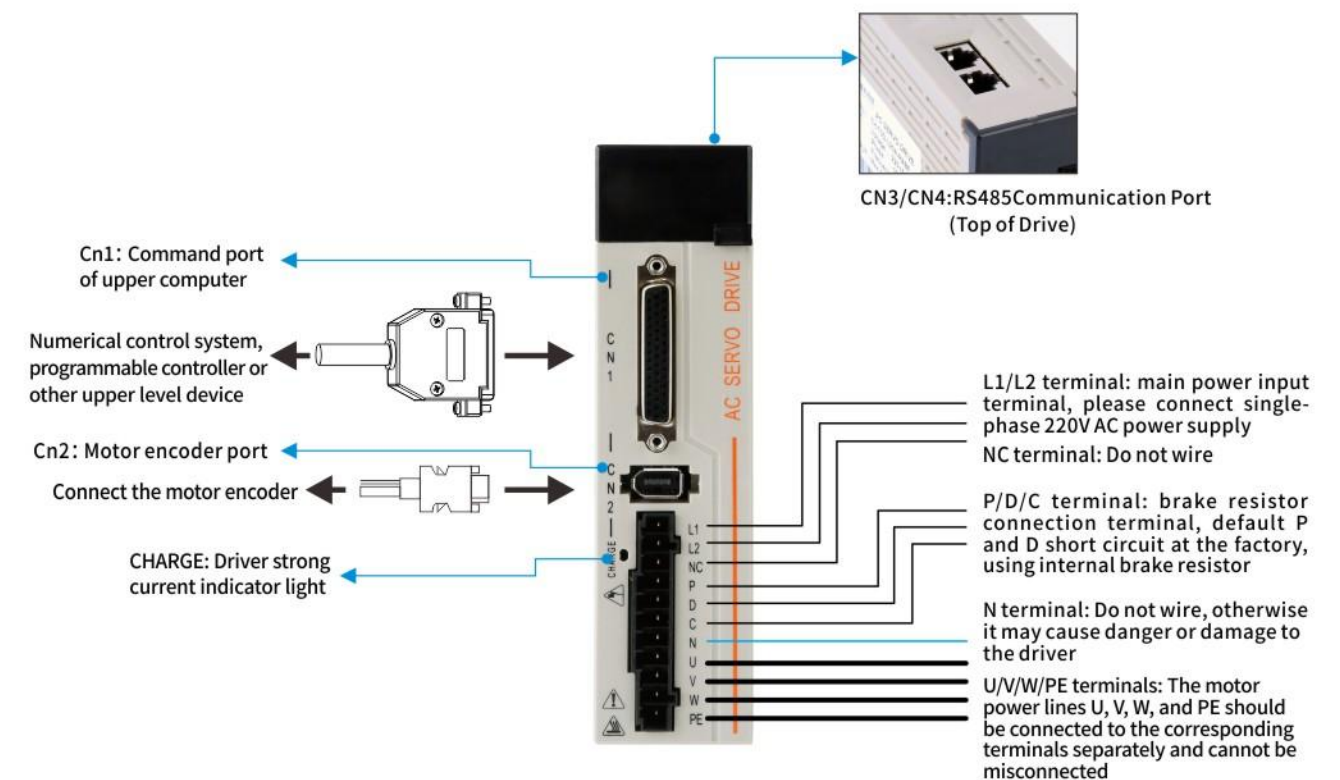


Figure 2 Installation dimension diagram of T3D-L30 (unit: mm)

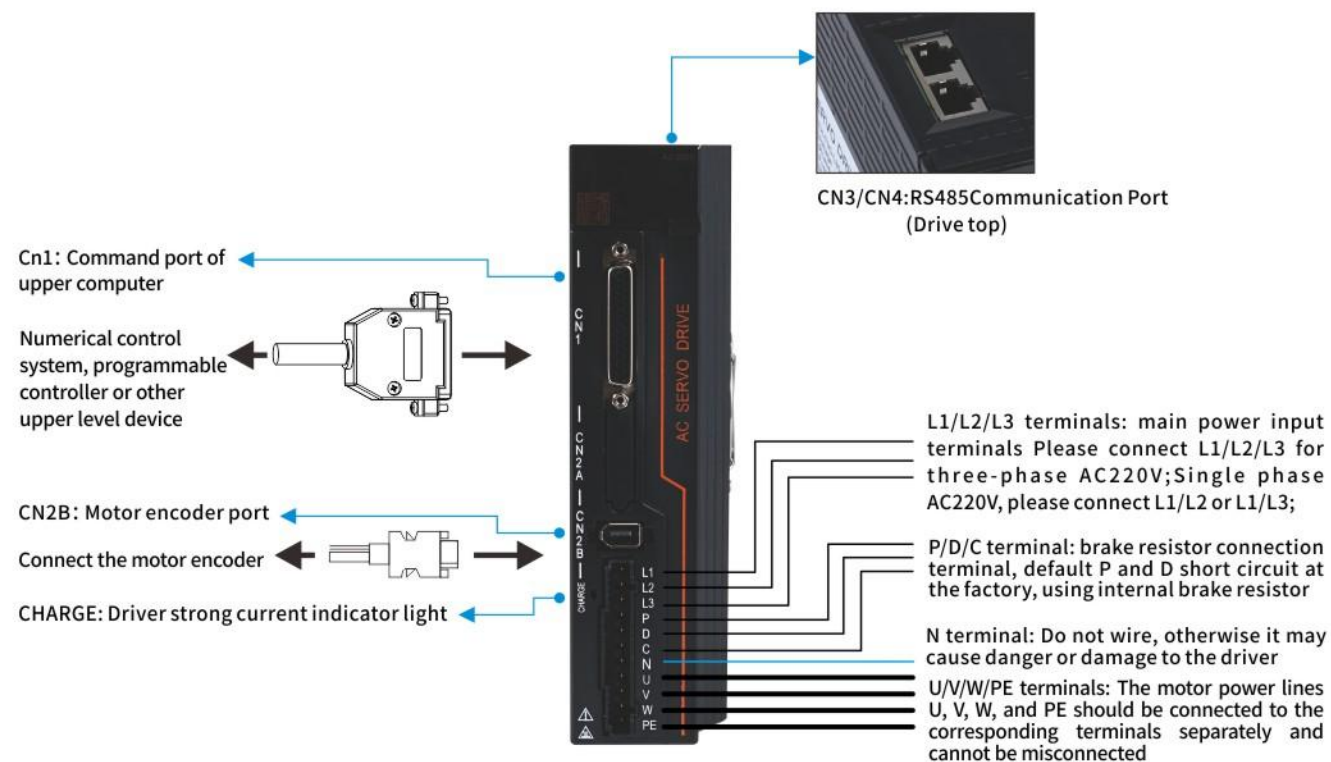
T3D SERVO DRIVE WIRING DIAGRAM

T3D-L15A/L20A/B Wiring diagram

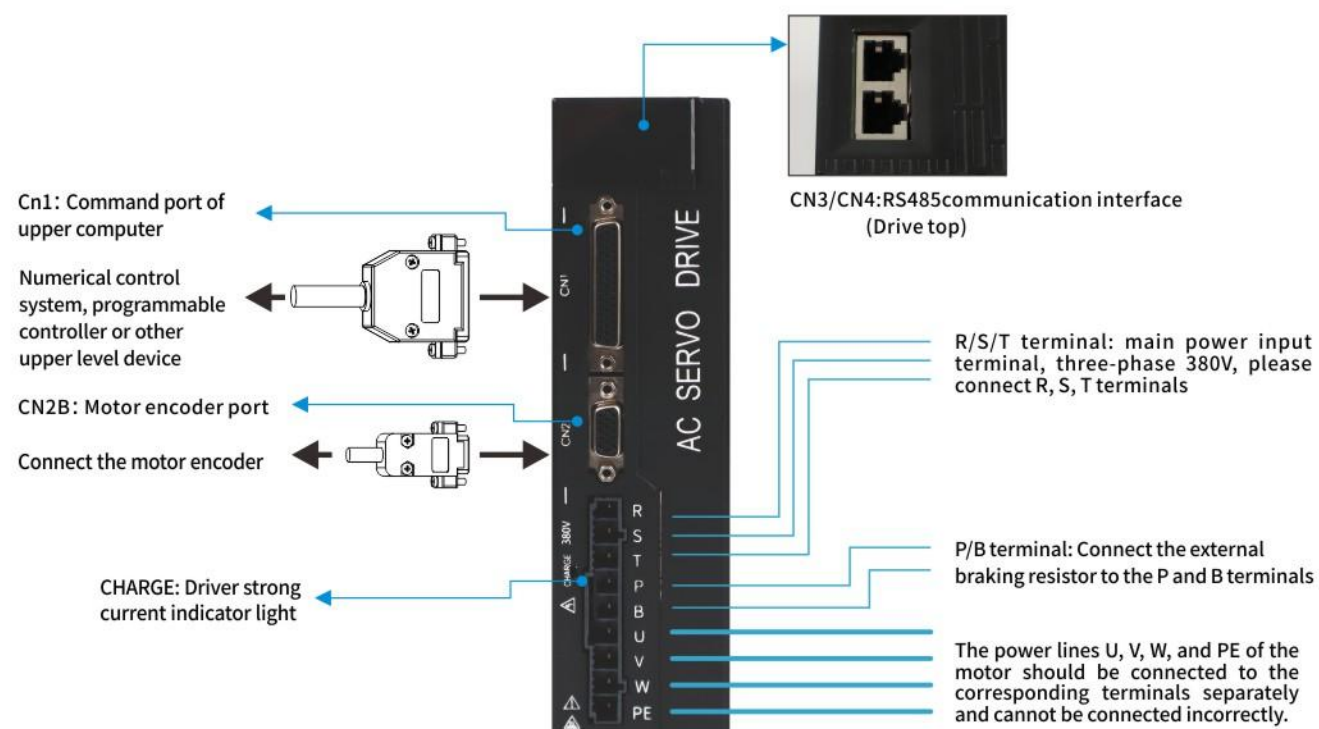


T3D SERVO DRIVE WIRING DIAGRAM

T3D-L30A Wiring diagram

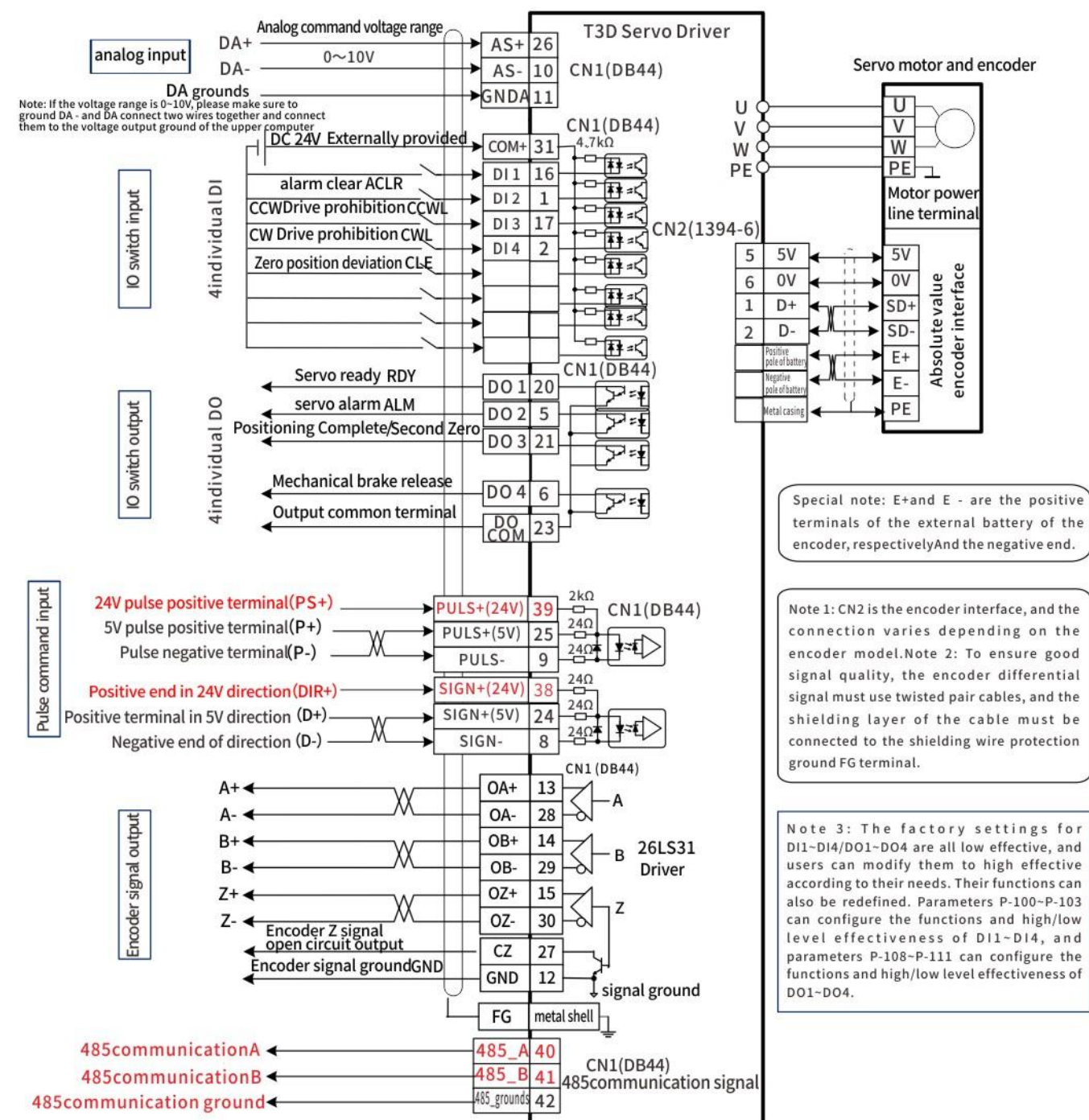


T3D-H10/H15/25 Wiring diagram



CONTROL WIRING DIAGRAM

This side signal is connected to the upper computer controller



Special attention:

If the pulse command signal is a 24V power supply, PULS+ must be connected to pin 39 of CN1, and SIGN+ must be connected to pin 38 of CN1, otherwise it may damage the pulse command port;
If the pulse command signal is a 5V power supply, PULS+ must be connected to pin 25 of CN1, and SIGN+ must be connected to pin 24 of CN1, otherwise there may be abnormal pulse reception.

LIST OF COMPATIBILITY BETWEEN 5-POLE MOTORS AND T3D DRIVERS

Standard motor adapter table (5-pole motor, 17 bit absolute value magnetic encoder)

(Motor series code P-099: **5-nR5**)

Table T3D Driver Compatible Motor Models

Motor code (P-002)	Adapted Driver (AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
0	L15	40AST-A1C00330	0.1	1.1	0.32	3
1	L15	60AST-A1C00630	0.2	1.1	0.64	3
2	L15	60AST-A1C01330	0.4	2.8	1.27	2.8
	L15	60AST-A1C01930	0.6	3.6	1.9	2
3	L15	80AST-A1C01330	0.4	2.8	1.27	2.5
4	L20	80AST-A1C02430	0.75	4.5	2.39	2.5
	L20	80AST-A1C03230	1.0	4.8	3.2	2.5
5	L20	80AST-A1C03220	0.67	5.6	3.2	2
	L20	80AST-A1C04025	1.0	4.5	3.8	2.5
6	L20/L30	110AST-A1C04220	0.88	4.5	4.2	2.2/3
7	L20/L30	110AST-A1C05420	1.1	5.5	5.4	2/2.7
8	L30	110AST-A1C06420	1.3	6.5	6.4	2.5
9	L30	110AST-A1C07520	1.6	8.0	7.5	2
10	L30	110AST-A1C04230	1.3	6.5	4.2	2.5
11	L30	110AST-A1C05430	1.7	8.2	5.4	2
12	L30	110AST-A1C06425	1.7	9.5	6.4	1.7
13	L30	130AST-A1C05415	0.85	6.5	5.4	2.5
14	L30	130AST-A1C06415	1.0	8.0	6.4	2
15	L30	130AST-A1C07515	1.2	9.0	7.5	1.7
16	L30	130AST-A1C08415	1.3	9.5	8.4	1.7
17	L30	130AST-A1C09615	1.5	10.0	9.6	1.5

T3D-H SERIES DRIVER MOTOR ADAPTATION LIST (5-pole, 23 bit absolute encoder)

(Motor series code P-099: **5-nR5**)

Motor code (P-002)	Adapted Driver (AC 380V)	Servo motor model (380V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
40	T3D-H10/ T3D-H15	40AST-AM00330(380V)	0.1	0.6	0.32	3
41		60AST-AM00630(380V)	0.2	0.7	0.64	3
42		60AST-AM01330(380V)	0.4	1.6	1.27	3
43		80AST-AM01330(380V)	0.4	1.8	1.27	2.7/3
44		80AST-AM02430(380V)	0.75	2.8	2.39	1.7/2.5
45		80AST-AM04025(380V)	1	2.6	3.8	1.9/2.6
46	T3D-H25	100AST-AM03230(380V)	1	4.5	3.2	2.6
47		100AST-AM04930(380V)	1.5	5.1	4.9	2.4
48		100AST-AM06430(380V)	2.0	6.6	6.4	1.8
49		110AST-AM04220(380V)	0.88	4.6	4.2	2.6
50		110AST-AM06420(380V)	1.3	7	6.4	1.7
51		130AST-AM05415(380V)	0.85	4	5.4	3
52		130AST-AM08415(380V)	1.3	6	8.4	2
53		130AST-AM09615(380V)	1.5	7.5	9.6	1.6
54		130AST-AM11515(380V)	1.8	8.5	11.5	1.4
55		130AST-AM12715(380V)	2	7.4	12.7	1.6
56		130AST-AM14615(380V)	2.3	10	14.6	1.2
57		130AST-AM17815(380V)	2.8	11	17.8	1
58		130AST-AM09630(380V)	3	9	9.6	1.3
59		130AST-AM12730(380V)	4	12	12.7	1
60		180AST-AM18615(380V)	2.9	11.8	18.6	1
61		180AST-AM28415(380V)	4.4	15.7	28.4	
62		180AST-AM35015(380V)	5.5	20.6	35	
63		180AST-AM48015(380V)	7.5	25.7	48	
64		180AST-AM83015(380V)	13	26.5	83	
65		180AST-AM18620(380V)	3.9	13.5	18.6	
66		180AST-AM28420(380V)	5.9	17.5	28.4	
67		180AST-AM35020(380V)	7.3	22.5	35	
68		180AST-AM48005(380V)	2.5	10.5	48	

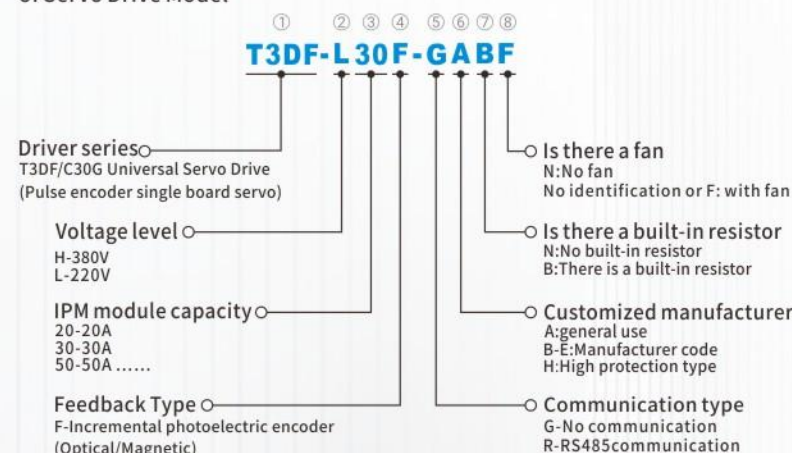
T3DF/C30G Series Single Board Universal Servo Driver

The T3DF/C30G series digital AC servo drive is a general-purpose AC servo drive product independently developed and produced by Zhejiang Hanghang Electric Drive Technology Co., Ltd., which has a leading level in China. This series of products adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability, widely used for servo shaft drive in food processing, packaging machinery, textile machinery and other related automation industry machinery.

Product Features

- ◆ The driver instruction type is pulse type, and the motor position feedback is pulse incremental encoder
- ◆ Adopting high-performance ARM main control chip, with excellent control performance
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Single board design, saving installation space
- ◆ Select a new industrial grade IPM module with strong overload driving capability
- ◆ Integrating position control, speed control, and torque control
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Specification Description of Servo Drive Model



T3DF SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

Driver size one (model T3DF-L30 installation dimension diagram)

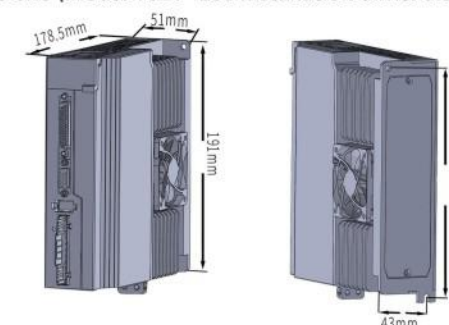


Figure 1 Installation dimension diagram of T3DF-L30 (unit: mm)

Driver size two (model T3DF-H10/H15/H25)

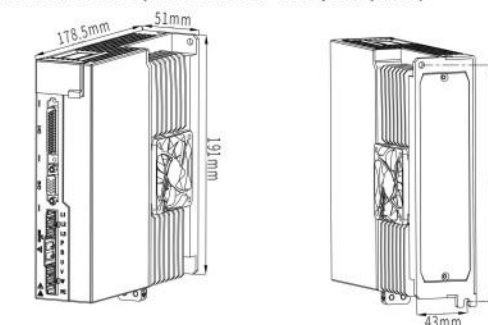


Figure 2 Installation dimension diagram of T3DF-H10/H15/H25 (unit: mm)

C30G EXTERNAL DIMENSIONS OF SERVO DRIVER (MM)

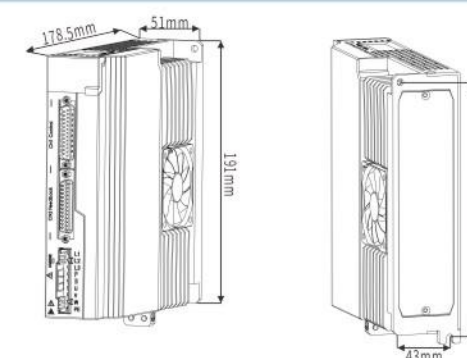
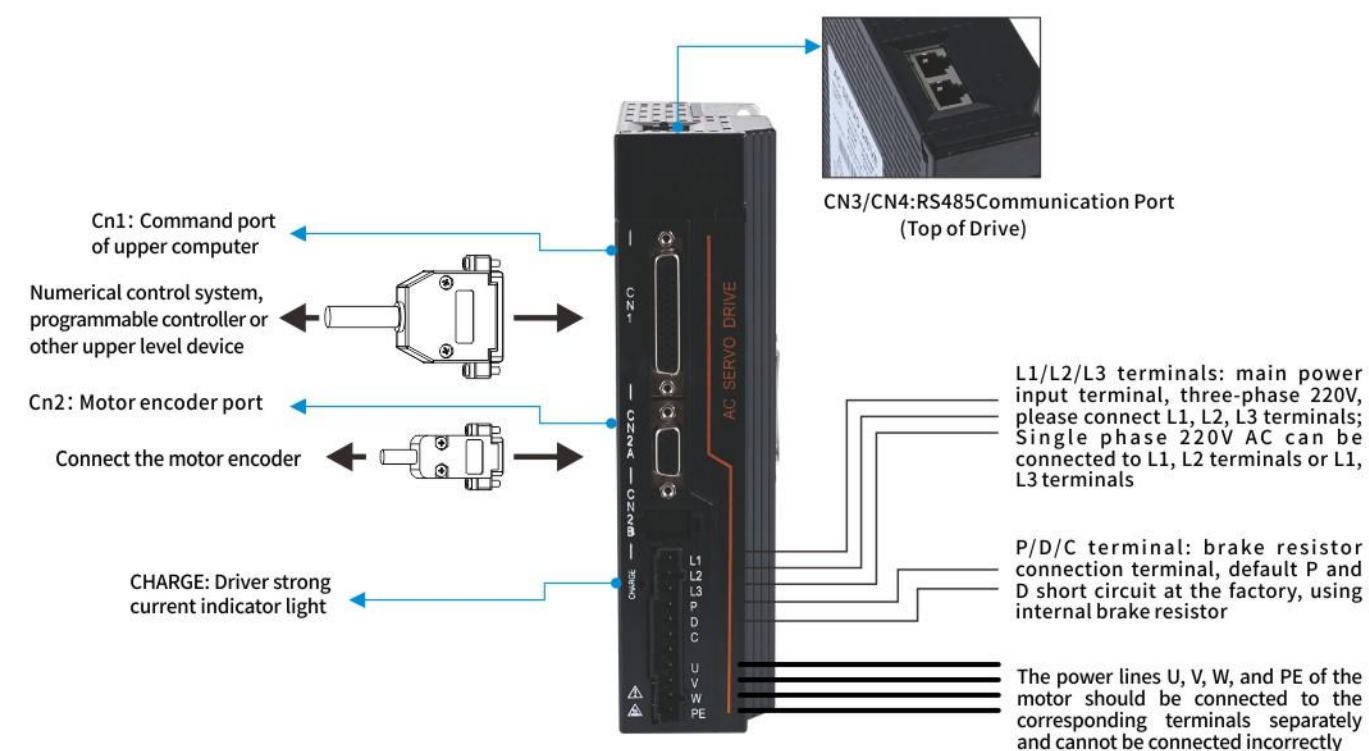


Figure 3 Installation dimension diagram of C30G-L20/L30 (unit: mm)

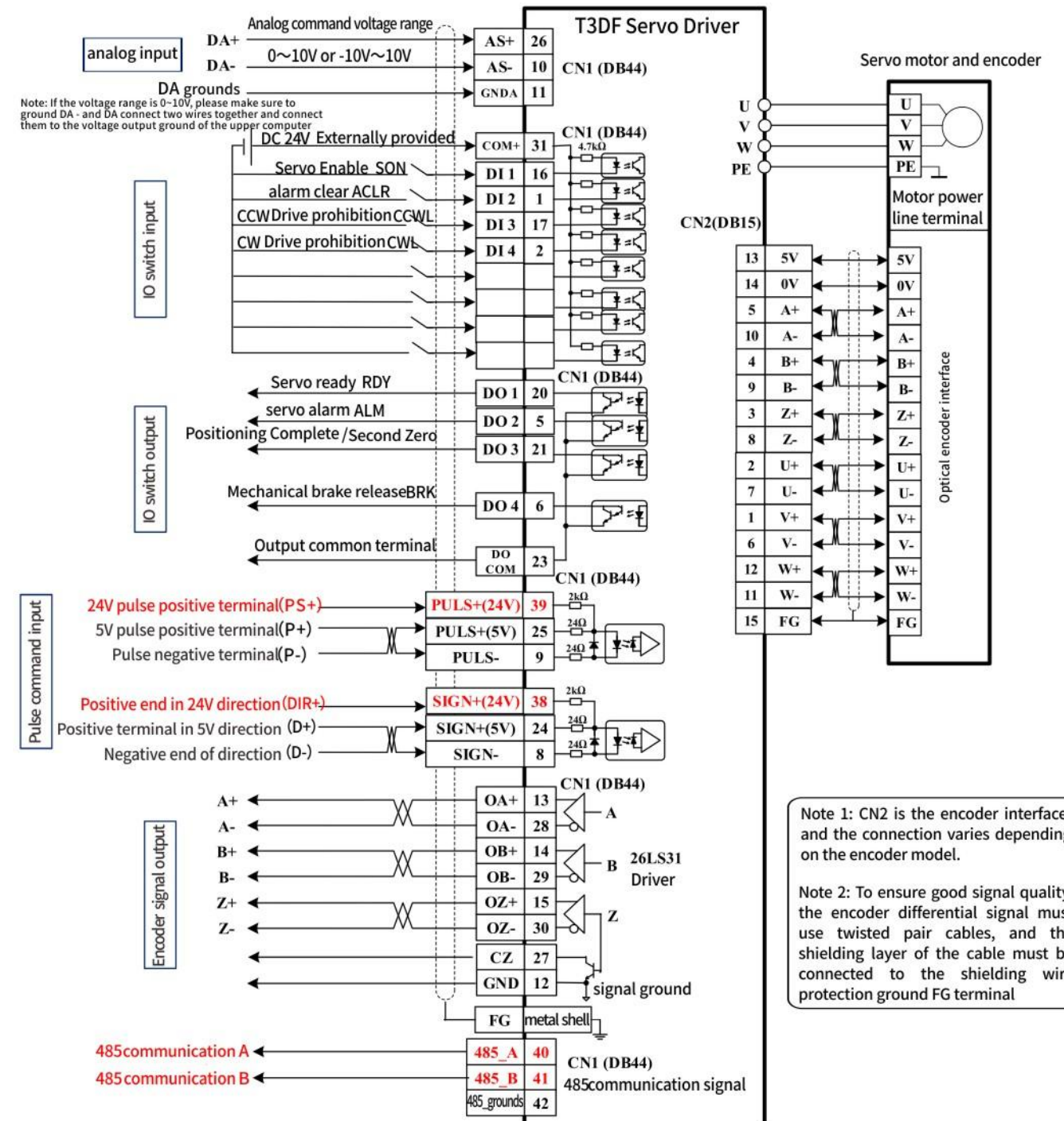
T3DF SERVO DRIVE WIRING DIAGRAM

T3DF-L30Wiring Diagram



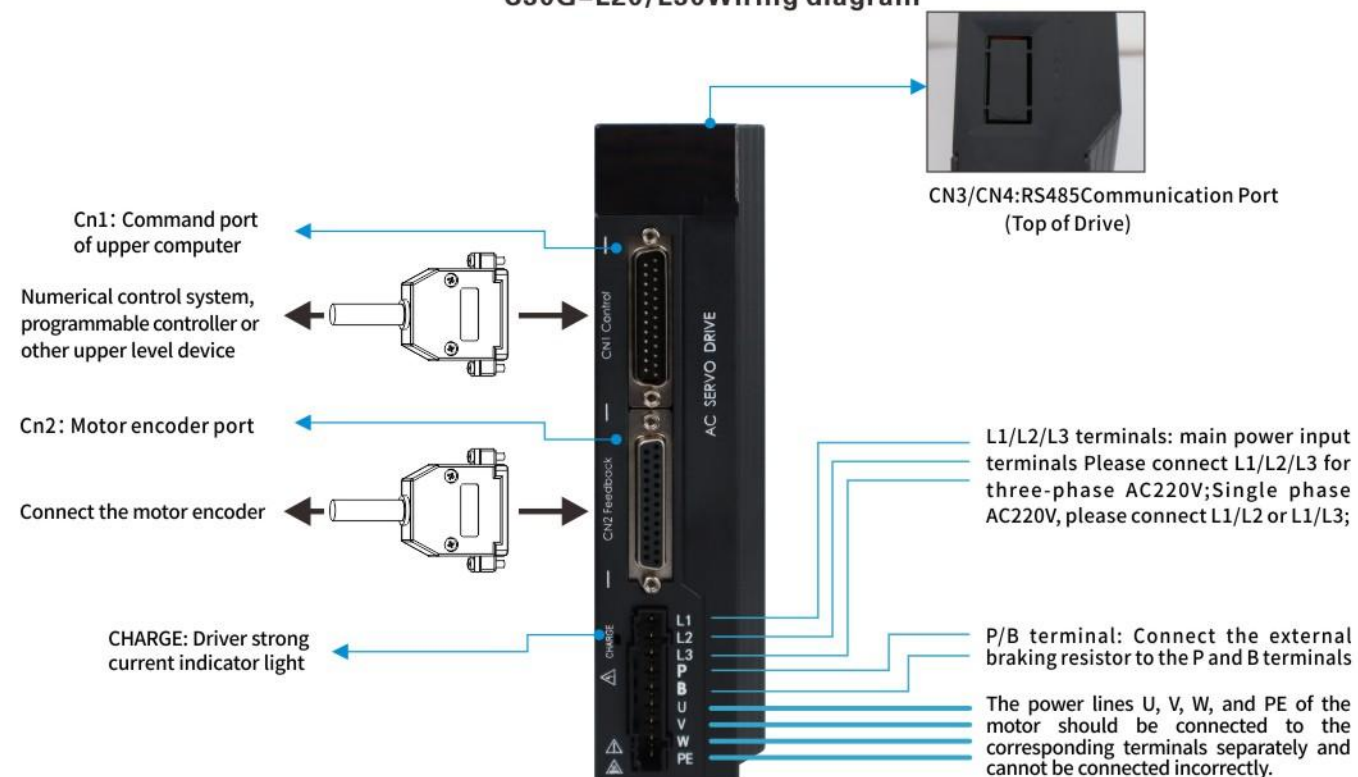
CONTROL WIRING DIAGRAM

This side signal is connected to the upper computer controller



Special attention: If the pulse command signal is a 24V power supply, PULS+ must be connected to pin 39 of CN1, and SIGN+ must be connected to pin 38 of CN1, otherwise it may damage the pulse command port; If the pulse command signal is a 5V power supply, PULS+ must be connected to pin 25 of CN1, and SIGN+ must be connected to pin 24 of CN1, otherwise there may be abnormal pulse reception;

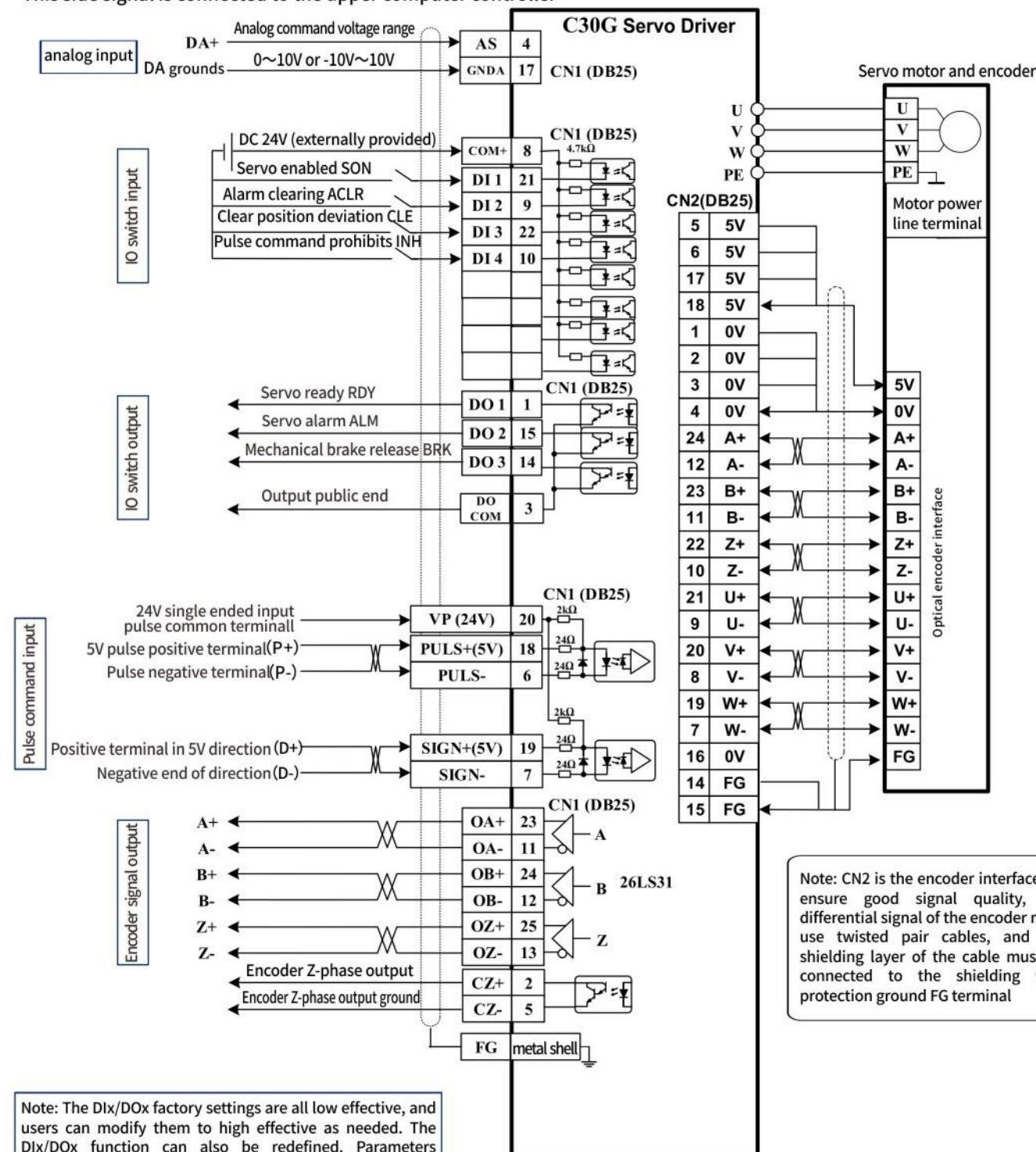
C30G-L20/L30Wiring diagram



Note 2: To ensure good signal quality, the encoder differential signal must use twisted pair cables, and the shielding layer of the cable must be connected to the shielding wire protection ground FG terminal

CONTROL WIRING DIAGRAM

This side signal is connected to the upper computer controller



Note: The DIx/DOx factory settings are all low effective, and users can modify them to high effective as needed. The DIx/DOx function can also be redefined. Parameters P100~P103 configure the DIx function and high/low level effectiveness, while parameters P108~P110 configure the DOx function and high/low level effectiveness.

Special attention: If the pulse command is a 24V single ended signal, the 24V power supply must be connected to pin 20 of CN1. Incorrect connection may damage the pulse command port; If the pulse command signal is a 5V power supply, PULS+ must be connected to pin 18 of CN1, and SIGN+ must be connected to pin 19 of CN1, otherwise there may be abnormal pulse reception.

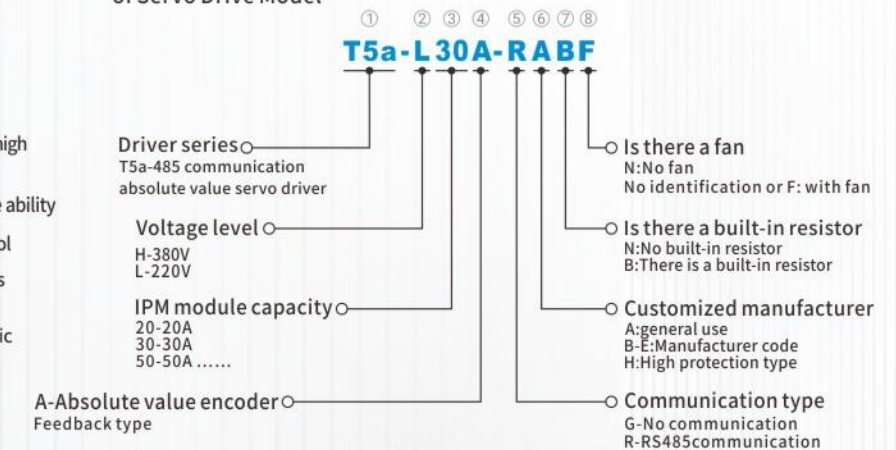
T5a Series Absolute Value Servo Drivers

The T5a series absolute value AC servo drive is independently developed and produced by Zhejiang Hanghang Electric Drive Technology Co., Ltd. It is a high-precision servo drive product with leading domestic level. This series of products adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability. It is widely used for servo shaft drive in CNC machine tools, textile machinery, and other general automation industry machinery.

Product Features

- ◆ The driver instruction type is pulse type, and the motor position feedback is pulse incremental encoder
- ◆ Adopting high-performance ARM main control chip, with excellent control performance
- ◆ Suitable for 17 bit or 23 bit multi turn absolute encoders, with high positioning accuracy and good servo shaft rigidity
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Integrating position control, speed control, and torque control
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Specification Description of Servo Drive Model



ABSOLUTE VALUE SERVO DRIVE STANDARD SPECIFICATIONS AND PERFORMANCE PARAMETERS

model		L20	L30	L40	L50	L75	H25	H50	H75
Input power supply	main power supply	Single phase/Three phase220VAC -15%~+10% 50/60Hz					Three phase380VAC -15%~+10% 50/60Hz		
	temperature	Work: 0 °C~40 °C Storage:- 40°C~50°C							
environment	humidity	Job: 40%~80% (no condensation) Storage: below 93% (no condensation)							
	Atmospheric pressure	86kPa~106 kPa							
Protection level		IP20							
control model		Position, speed, torque, position/speed, speed/torque, position/torque							
Digital input		8 programmable input terminals (optically isolated)							
Digital output		4 programmable output terminals (photoelectric isolation)							
encoder Signal output	signal type	A, B, Z differential output/Z signal collector open circuit output /RS485 communication reading feedback encoder position							
position	Input frequency	Differential input: ≤ 500kHz(kpps); Single ended input: ≤ 200kHz(kpps)							
	Command mode	Pulse+symbol; Forward/reverse pulse; Orthogonal pulse							
	Electronic gear ratio	1~32767/1~32767							
speed	Simulate command input	± 10VDC, input impedance 10k Ω							
	Command acceleration and deceleration	Parameter settings P-060/P-061							
	Instruction source	Analog quantity, internal speed command							
torque	Simulate command input	-10V~+10V, input impedance 10k Ω							
	Torque limit	Parameter settings P-065/P-066							
	Instruction source	Analog quantity, internal torque command							
Monitoring function		Speed, current position, position deviation, motor torque, motor current, command pulse frequency, etc							
Protection function		Overspeed, overvoltage, overcurrent, overload, abnormal braking, abnormal encoder, position deviation, instruction overlocking, etc							
characteristic	Speed frequency response	≥ 1.6kHz							
	Speed fluctuation rate	± 0.03% (load 0~100%) < ± 0.02% (power supply -15%~+10%)							

T5a SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

Driver size one (model T5a-L20)

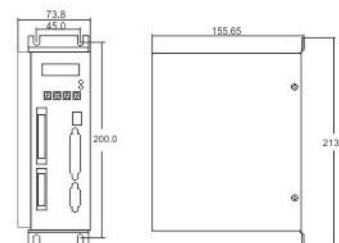


Figure 1 Dimension diagram of T5a-L20 servo drive (unit: mm)

Driver size three (model T5a-L40/T5a-H25)

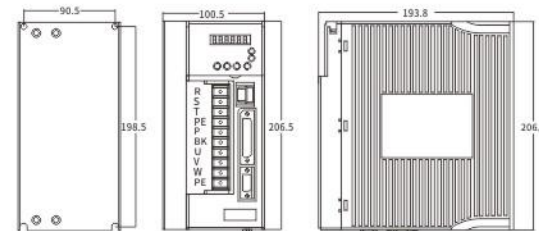


Figure 3 Dimension diagram of T5a-L40/T5a-H25 servo drive (unit: mm)

Driver size two (model T5a-L30)

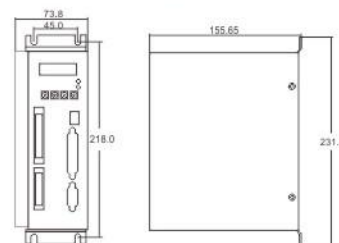


Figure 2 Dimension diagram of T5a-L30 servo drive (unit: mm)

Driver size four (models T5a-L50/L75, T5a-H50/H75)

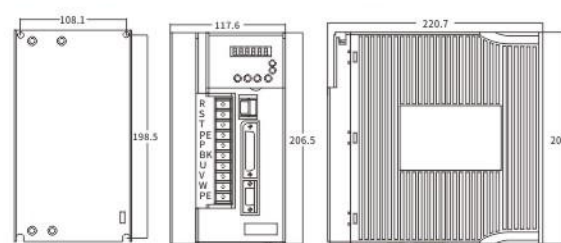
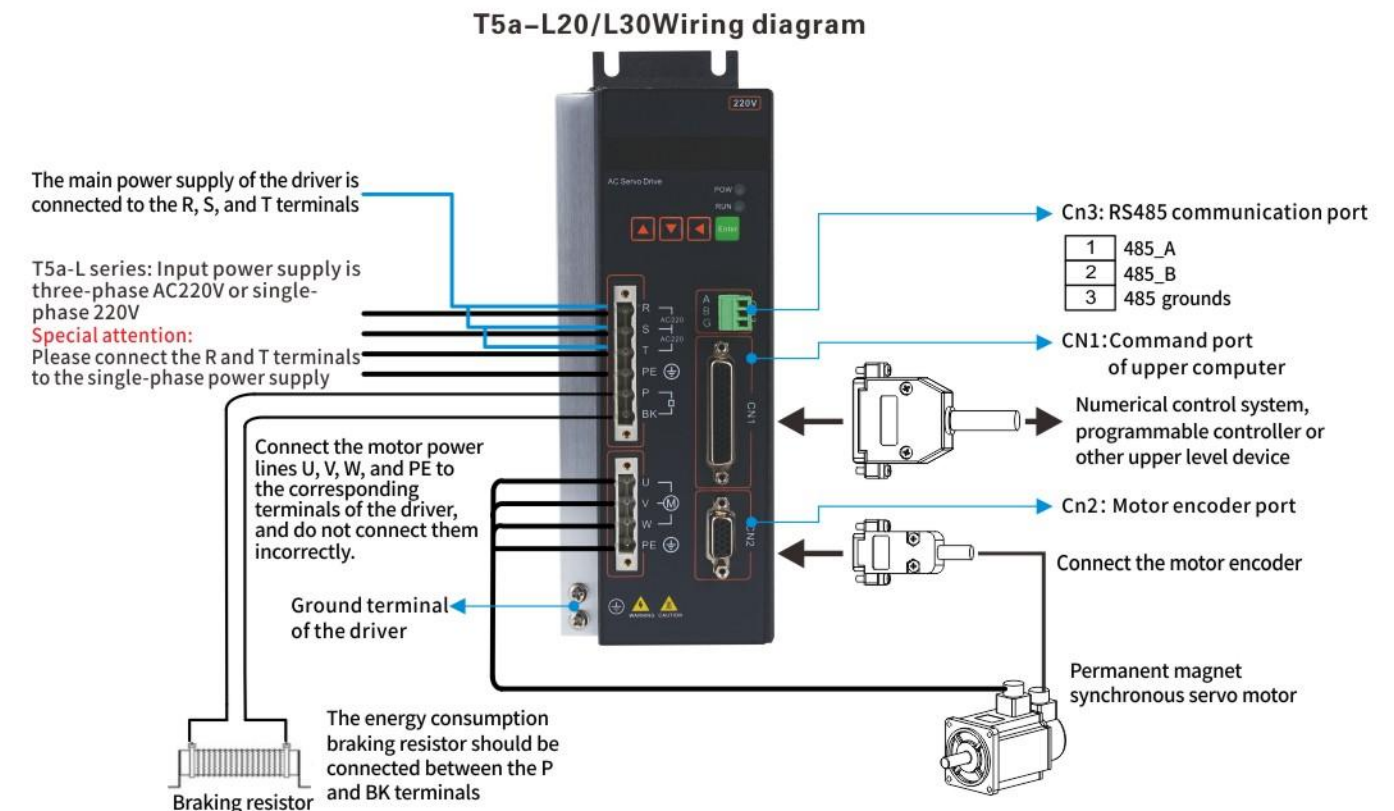
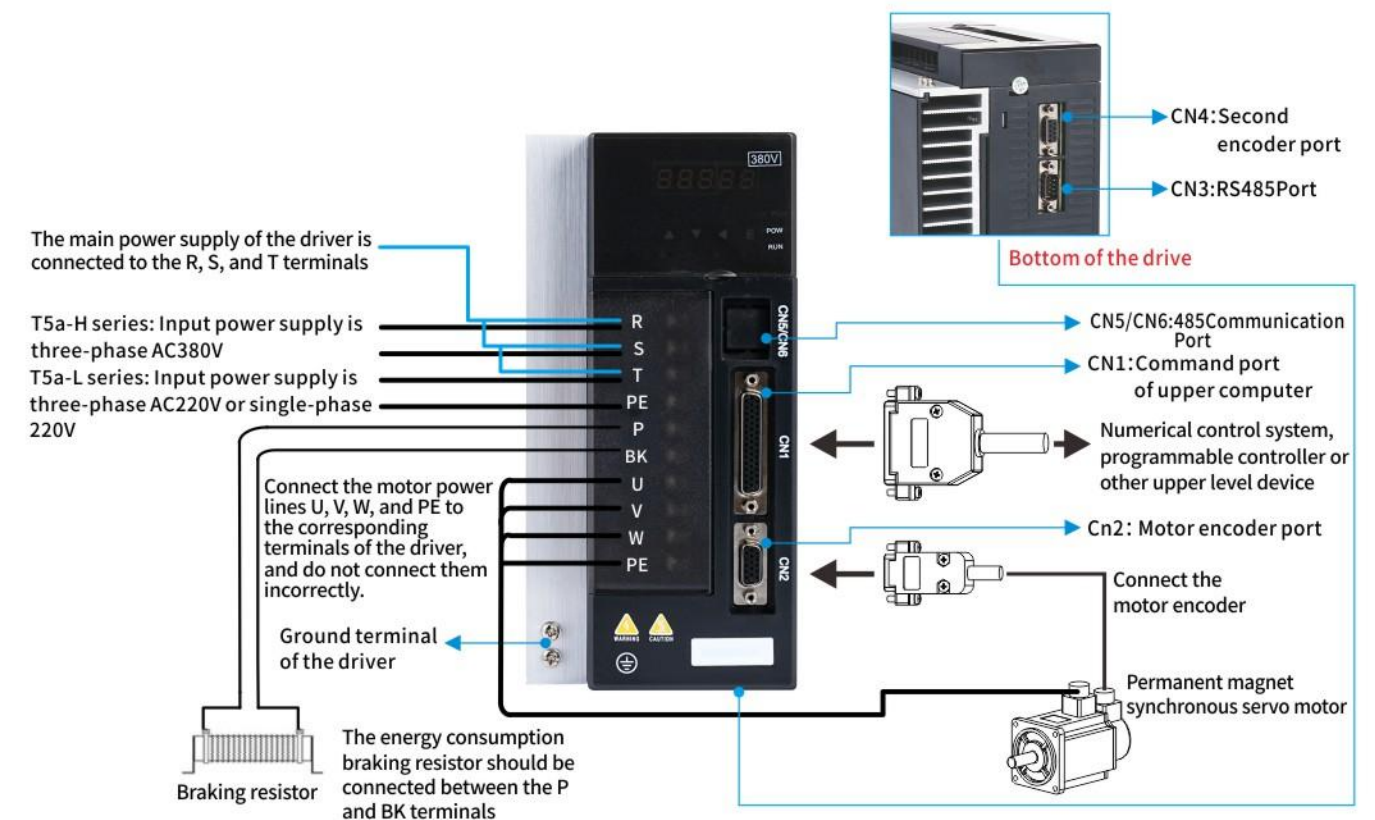


Figure 4 Dimension diagram of T5a-L50/L75 and T5a-H50/H75 servo drives (unit: mm)

SERVO DRIVE WIRING DIAGRAM

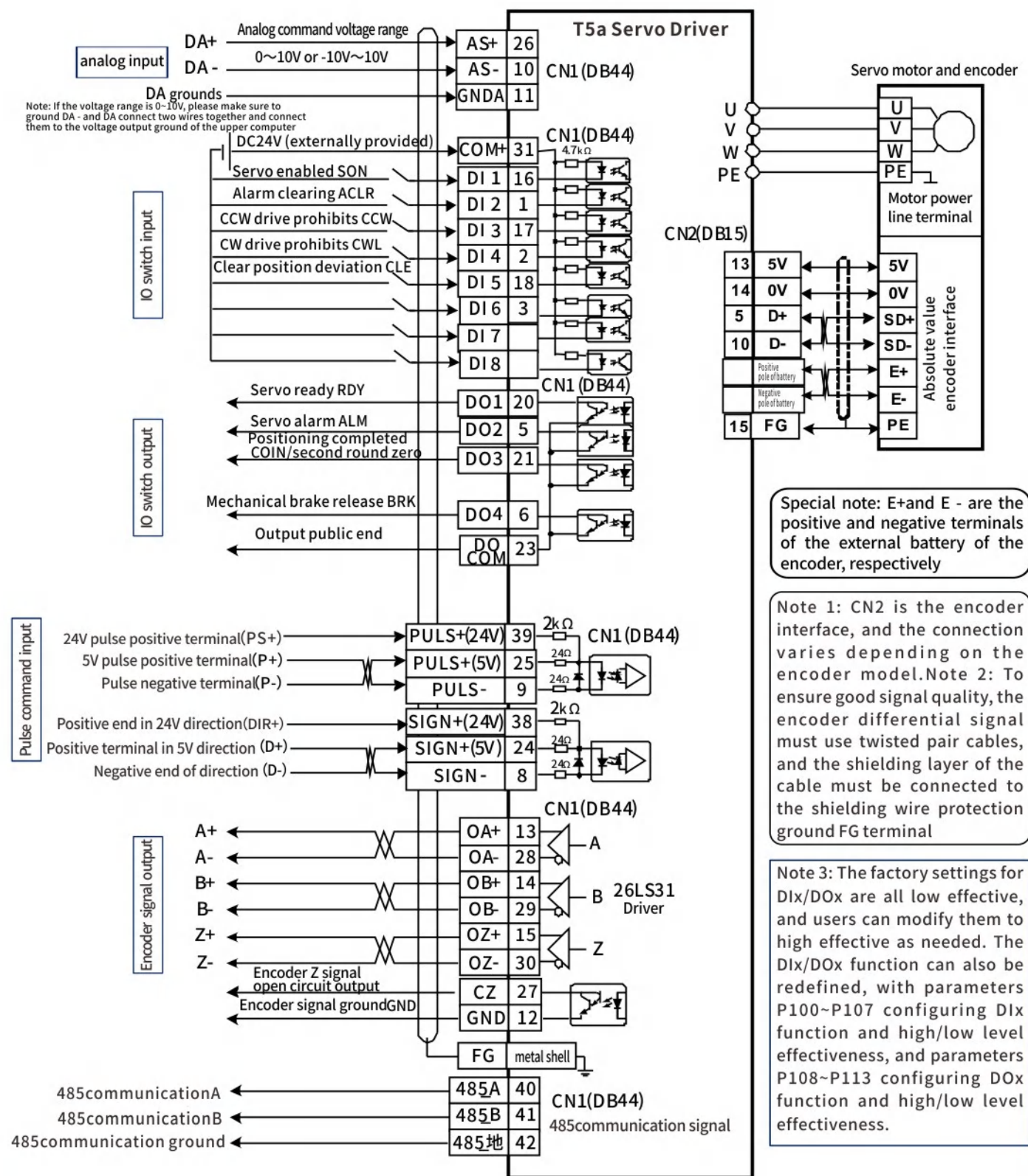


T5a-L40/L50/L75/H25/H50/H75Wiring diagram



CONTROL WIRING DIAGRAM

This side signal is connected to the upper computer controller



SUPPORTING ABSOLUTE VALUE SERVO MOTOR TECHNICAL PARAMETERS

Table x N series 220V motor and T5a driver compatibility list (4-pole motor)

(Motor series code P-099: **6-nA4**)

Motor model code(P-002)	Adapt to driver model specifications(AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
1	L20	40ST-AM00130	0.05	1.2	0.159	3
2	L20	40ST-AM00330	0.1	1.8	0.3	3
3	L20	60ST-AM00630	0.2	1.2	0.637	3
4	L20	60ST-AM01330	0.4	2.8	1.27	3
5	L20	60ST-AM01930	0.6	3.5	1.91	3
6	L20	80ST-AM01330	0.4	2	1.27	3
7	L20	80ST-AM02430	0.75	3	2.39	3
8	L20	80ST-AM03520	0.73	3	3.5	3
9	L20	80ST-AM04025	1.0	4.4	4	2.5
10	L20	90ST-AM02430	0.75	3	2.4	3
11	L20	90ST-AM03520	0.73	3	3.5	3
12	L20	90ST-AM04025	1.0	4	4	2.5
13	L20	110ST-AM02030	0.6	2.5	2	3
14	L20	110ST-AM04020	0.8	3.5	4	3
15	L20/L30	110ST-AM04030	1.2	5.0	4	2/3
16	L20/L30	110ST-AM05030	1.5	6.0	5	1.7/2.5
17	L20/L30	110ST-AM06020	1.2	4.5	6	2.2/3
18	L20/L30	110ST-AM06030	1.8	6.0	6	1.7/2.5
19	L20	130ST-AM04025	1.0	4.0	4	2.5
20	L20/L30	130ST-AM05025	1.3	5.0	5	2/3
21	L20/L30	130ST-AM06025	1.5	6.0	6	1.7/2.5
22	L30	130ST-AM07725	2.0	7.5	7.7	2
23	L30	130ST-AM07730	2.4	9.0	7.7	1.8
24	L20/L30	130ST-AM10010	1.0	4.5	10	2.2/3
25	L20/L30	130ST-AM10015	1.5	6.0	10	1.7/2.5
26	L30/L40	130ST-AM10025	2.6	10	10	1.5/2
27	L30/L40	130ST-AM15015	2.3	9.5	15	1.6/2.1
28	L40/L50	130ST-AM15025	3.8	13.5	15	1.5/2.5
29	L50	150ST-AM15020	3.0	14	15	2.2
30	L50/L75	150ST-AM15025	3.8	17	15	1.7/2.5
31	L50/L75	150ST-AM18020	3.6	17	18	1.7/2.5
32	L50/L75	150ST-AM23020	4.7	21	23	1.5/2
33	L75	150ST-AM27020	5.5	27	27	1.5
34	L30/L40	180ST-AM17215	2.7	10.5	17.2	1.5/2
35	L40/L50	180ST-AM19015	3.0	12	19	1.7/2.5
36	L50/L75	180ST-AM21520	4.5	16	21.5	2/2.5
37	L40/L50	180ST-AM27010	2.9	12	27	1.7/2.5
38	L50/L75	180ST-AM27015	4.3	16	27	2/2.5
39	L50/L75	180ST-AM35010	3.7	16	35	2/2.5
40	L75	180ST-AM35015	5.5	24	35	1.7
41	L100	180ST-AM48015	7.5	32	48	2
42	L100	220ST-AM67010	7.0	31.3	67	2
43	L150	220ST-AM95515	11	67	95.5	2

SUPPORTING ABSOLUTE VALUE SERVO MOTOR TECHNICAL PARAMETERS

Table X: Adaptation List of A-series Motors and T5a Drivers (5-pole Motors)

(Motor series code P-099: **5-nR5**)

Motor model code(P-002)	Adapt to driver model specifications(AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
1	L20	60AST-AM00630LB	0.2	1.1	0.64	3
2	L20	60AST-AM01330LB	0.4	2.1	1.27	3
3	L20	80AST-AM01330LB	0.4	2.3	1.27	3
4	L20	80AST-AM02430LB	0.75	4	2.39	2.5
5	L20//L30	80AST-AM03220LB	0.67	5.6	3.2	2/3
6	L20/L30	110AST-AM04220LB	0.88	4.5	4.2	2.2/3
7	L20/L30	110AST-AM05420LB	1.1	5.5	5.4	2/2.7
8	L30	110AST-AM06420LB	1.3	6.5	6.4	2.5
9	L30/L40	110AST-AM07520LB	1.6	8.0	7.5	2/2.5
10	L30	110AST-AM04230LB	1.3	6.5	4.2	2.5
11	L30/L40	110AST-AM05430LB	1.7	8.2	5.4	2/2.5
12	L30/L40	110AST-AM06425LB	1.7	9.5	6.4	1.7/2
13	L30	130AST-AM05415LB	0.85	6.5	5.4	2.5
14	L30/L40	130AST-AM06415LB	1.0	8.0	6.4	2/2.5
15	L30/L40	130AST-AM07515LB	1.2	9.0	7.5	1.7/2
16	L30/L40	130AST-AM08415LB	1.3	9.5	8.4	1.7/2
17	L30/L40	130AST-AM09615LB	1.5	10.0	9.6	1.5/2
18	L50	130AST-AM11515LB	1.8	14.0	11.5	2
19	L50	130AST-AM14615LB	2.3	16.0	14.6	2
20	L30/L40	130AST-AM05430LB	1.7	9.5	5.4	1.7/2
21	L40/L50	130AST-AM06430LB	2.0	11.5	6.4	1.7/2.5
22	L40/L50	130AST-AM07530LB	2.4	12.0	7.5	1.5/2.5
23	L50	130AST-AM08430LB	2.6	13.5	8.4	2.2
24	L50	130AST-AM09625LB	2.5	16	9.6	1.9
25	L50/L75	130AST-AM11520LB	2.4	17.8	11.5	1.7/2.2
26	L75	130AST-AM14620LB	3.1	23.0	14.6	1.8
27	L30/L40	180AST-AM17215LB	2.7	10.8	17.2	1.5/2
29	L50/L75	180AST-AM27015LB	4.3	17.5	27	1.7/2.2
30	L75	180AST-AM48015LB	7.5	30	48	1.4

SUPPORTING ABSOLUTE VALUE SERVO MOTOR TECHNICAL PARAMETERS

Table x T5a-H25/T5a-H50/T5a-H75 driver compatible motor models (4-pole, 23 bit absolute encoder)

(Motor series code P-099: **5-nR4**)

Motor model code(P-002)	Adapt to driver model specifications(AC 380V)	Servo motor model (380V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
51	T5a-H25	110ST-AM02030 (380V)	0.6	2.0	2	3
52		110ST-AM04020 (380V)	0.8	2.0	4	3
53		110ST-AM04030 (380V)	1.2	3.0	4	3
54		110ST-AM05030 (380V)	1.5	4.5	5	3
55		110ST-AM06020 (380V)	1.2	3.0	6	3
56		110ST-AM06030 (380V)	1.8	4.5	6	3
57		130ST-AM04025 (380V)	1.0	2.6	4	3
58		130ST-AM05025 (380V)	1.3	3.0	5	2.5
59		130ST-AM06025 (380V)	1.5	3.7	6	3
60		130ST-AM07720 (380V)				
61		130ST-AM07725 (380V)	2.0	4.7	7.7	2.5
62		130ST-AM07730 (380V)				
63		130ST-AM10010 (380V)	1.0	2.5	10	3
64		130ST-AM10015 (380V)	1.5	3.5	10	3
65		130ST-AM10020 (380V)	2.0	5.1	10	3
66		130ST-AM10025 (380V)	2.6	5.9	10	3
67		130ST-AM15015 (380V)	2.3	5.0	15	3
68		130ST-AM15025 (380V)	3.8	7.4	15	2.5
69		130ST-AM17020 (380V)	3.5	7.4	17	2.5
70	T5a-H25/T5a-H50	130ST-AM20015 (380V)				
71		130ST-AM20025 (380V)				
72		130ST-AM25015 (380V)				
73		130ST-AM25025 (380V)				
74	T5a-H50	150ST-AM15020 (380V)	3.0	6.8	15	3
75		150ST-AM15025 (380V)	3.8	9.5	15	3
76		150ST-AM18020 (380V)	3.6	8.5	18	3
77	T5a-H75	150ST-AM23020 (380V)	4.7	12	23	3
78		150ST-AM27015 (380V)	4.2	11	27	3
79		150ST-AM27020 (380V)	5.5	14.5	27	2.5
80		150ST-AM27025 (380V)	6.8	17	27	2.1
81		180ST-AM17215 (380V)	2.7	6.5	17.2	3
82		180ST-AM19015 (380V)	3.0	7.5	19	3
83		180ST-AM21520 (380V)	4.5	9.5	21.5	3
84		180ST-AM27010 (380V)	2.9	7.5	27	3
85		180ST-AM27015 (380V)	4.3	10	27	3
86		180ST-AM35010 (380V)	3.7	10	35	3
87		180ST-AM35015 (380V)	5.5	12	35	3
88		180ST-AM48015 (380V)	7.5	20	48	1.9

SUPPORTING ABSOLUTE VALUE SERVO MOTOR TECHNICAL PARAMETERS

Table x T5a-H25/T5a-H50/T5a-H75 driver compatible motor models (5-pole, 23 bit absolute encoder)

(Motor series code P-099: **5-nR5**)

Motor model code(P-002)	Adapt to driver model specifications(AC380V)	Servo motor model (380V)	Rated power (kW)	Rated current (A)	Rated torque (Nm)	Overload multiple
40	T5a-H25	40AST-AM00330(380V)	0.1	0.6	0.32	3
41		60AST-AM00630(380V)	0.2	0.7	0.64	3
42		60AST-AM01330(380V)	0.4	1.6	1.27	3
43		80AST-AM01330(380V)	0.4	1.8	1.27	3
44		80AST-AM02430(380V)	0.75	2.8	2.39	3
45		80AST-AM04025(380V)	1	2.6	3.8	3
46		100AST-AM03230(380V)	1	4.5	3.2	3
47		100AST-AM04930(380V)	1.5	5.1	4.9	3
48		100AST-AM06430(380V)	2.0	6.6	6.4	3
49		110AST-AM04220(380V)	0.88	4.6	4.2	3
50		110AST-AM06420(380V)	1.3	7	6.4	3
51		130AST-AM05415(380V)	0.85	4	5.4	3
52		130AST-AM08415(380V)	1.3	6	8.4	3
53		130AST-AM09615(380V)	1.5	7.5	9.6	2.8
54	T5a-H50	130AST-AM11515(380V)	1.8	8.5	11.5	3
55		130AST-AM12715(380V)	2	7.4	12.7	3
56		130AST-AM14615(380V)	2.3	10	14.6	2.9
57		130AST-AM17815(380V)	2.8	11	17.8	2.6
58	T5a-H75	130AST-AM09630(380V)	3	9	9.6	3
59		130AST-AM12730(380V)	4	12	12.7	3
60		180AST-AM18615(380V)	2.9	11.8	18.6	3
61		180AST-AM28415(380V)	4.4	15.7	28.4	2.4
62		180AST-AM35015(380V)	5.5	20.6	35	1.8
63		180AST-AM48015(380V)	7.5	25.7	48	1.5
64		180AST-AM83015(380V)	13	26.5	83	1.4
65		180AST-AM18620(380V)	3.9	13.5	18.6	2.8
66		180AST-AM28420(380V)	5.9	17.5	28.4	2.2
67		180AST-AM35020(380V)	7.3	22.5	35	1.7
68		180AST-AM48005(380V)	2.5	10.5	48	3

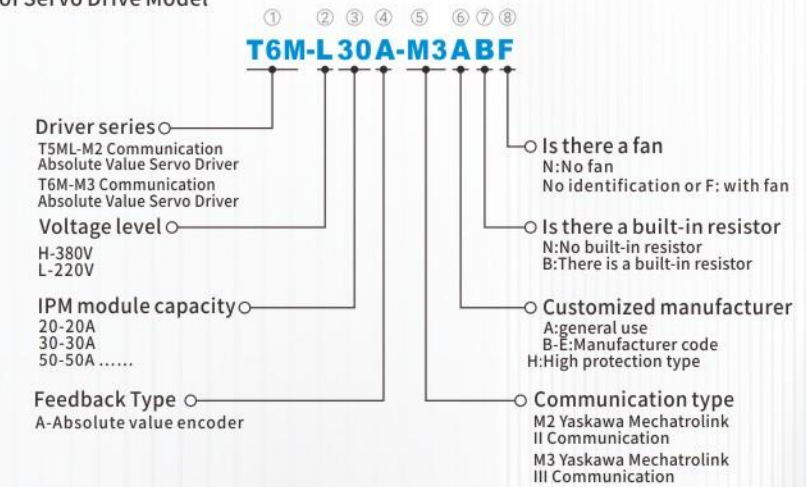
T5ML (M2) /T6M (M3)Bus absolute value servo driver

The T5ML (M2)/T6M (M3) series bus absolute value AC servo drive adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability, widely used in metal processing CNC machine tools and other general automation machinery.

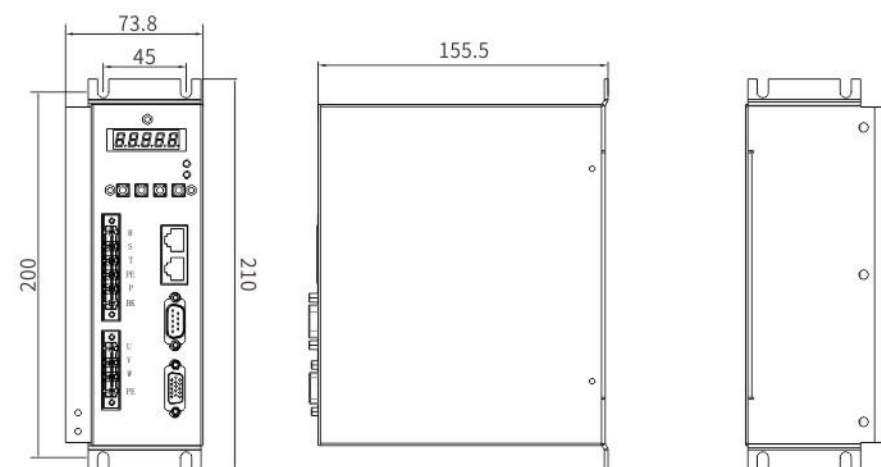
Product Features

- ◆ Adopting high-performance ARM control chips, with excellent control performance
- ◆ Control instructions are transmitted through the MECHATROLINK-II/III bus, making it easy to achieve high-speed and high-precision control
- ◆ Suitable for 17 bit/23 bit multi turn absolute encoders, with high positioning accuracy and good servo shaft rigidity
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Choose the latest industrial grade IPM module with strong overload driving capability
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Specification Description of Servo Drive Model



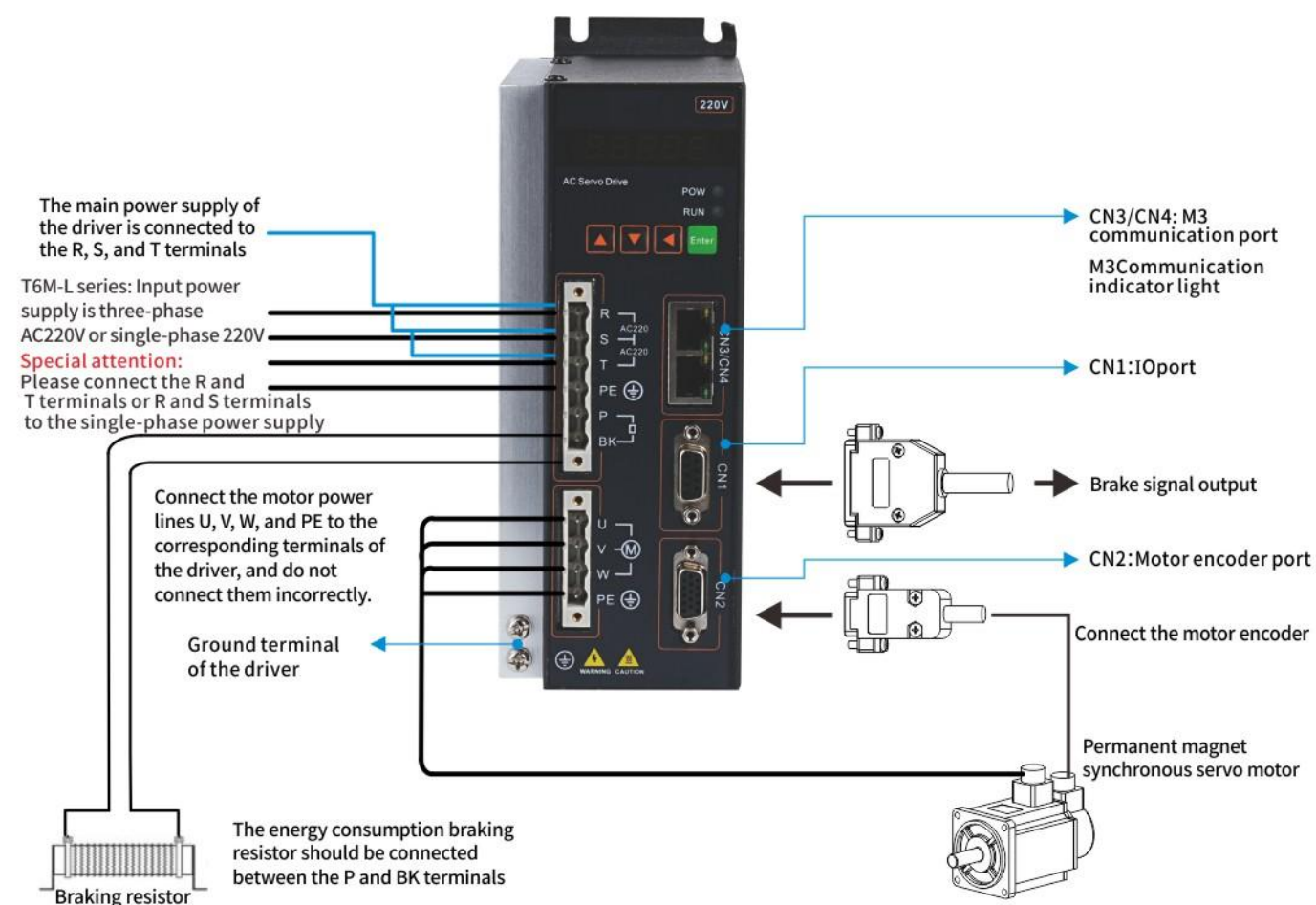
T6M SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)



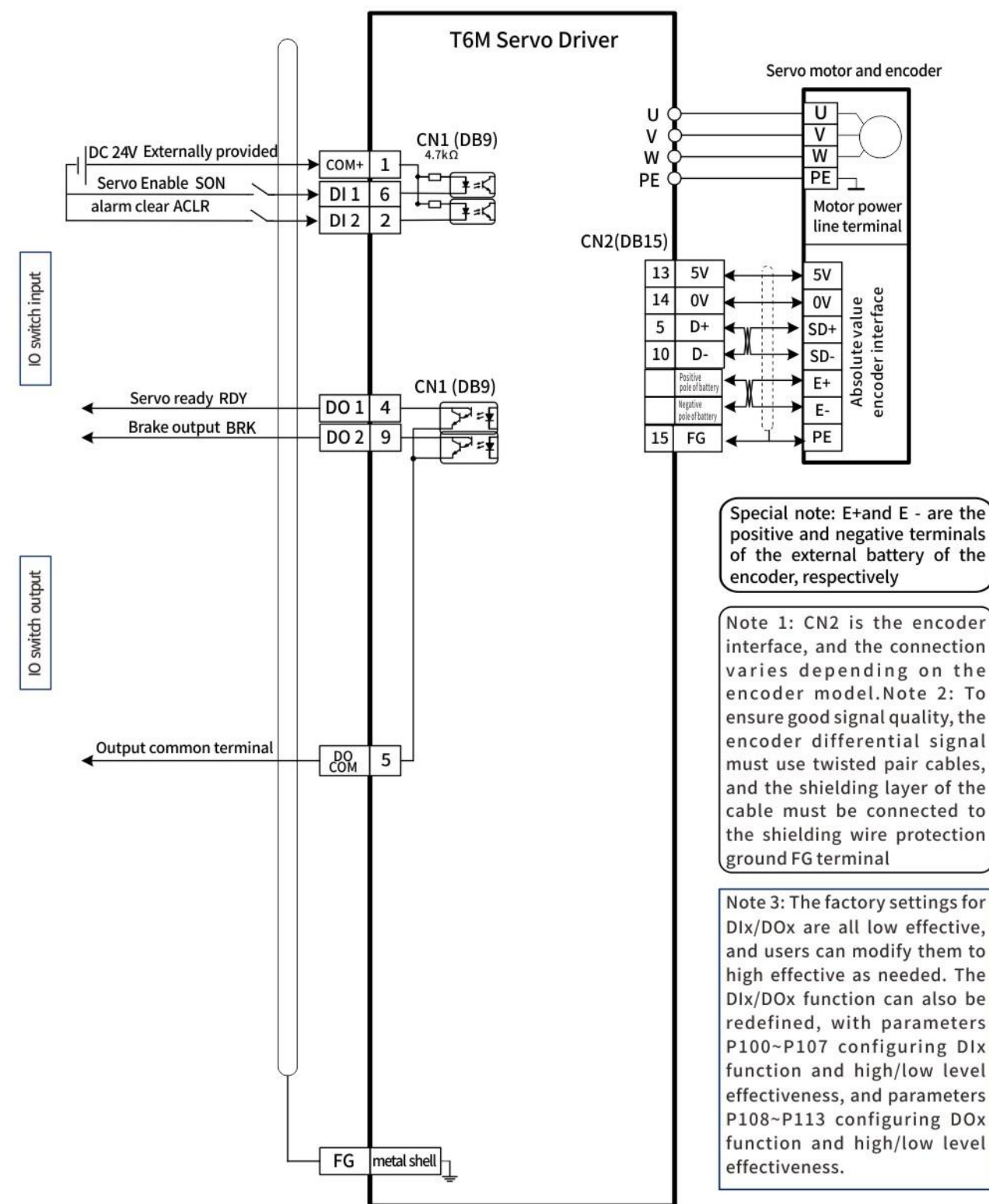
T6M-L20/L30 Installation Dimensional Drawing (Unit: mm)

SERVO DRIVE WIRING DIAGRAM

T6M-L20/L30Wiring diagram



CONTROL WIRING DIAGRAM



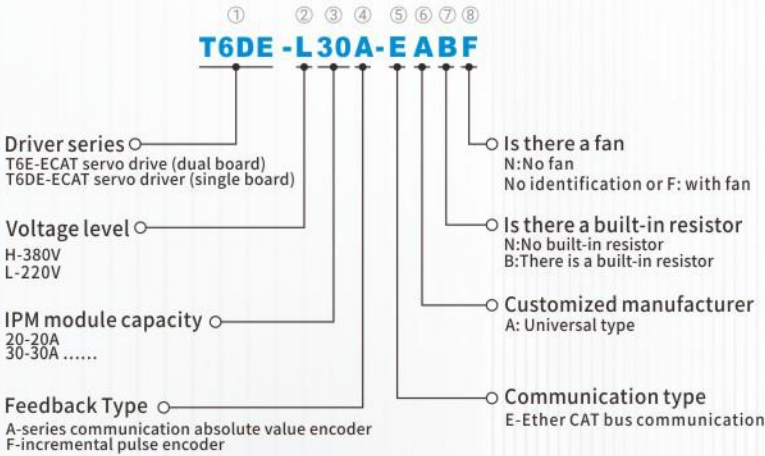
T6E/T6DE(EtherCAT) Bus universal servo drive

EtherCAT (T6E/T6DE) bus absolute value AC servo drive adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability, widely used in metal processing CNC machine tools and other general automation machinery.

Product Features

- ◆ Drive control instructions are transmitted through the EtherCAT bus, making it easy to achieve high-speed and high-precision control
- ◆ Suitable for 17 bit/23 bit multi turn absolute encoders, with high positioning accuracy and good servo shaft rigidity
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Choose the latest industrial grade IPM module with strong overload driving capability
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Specification Description
of Servo Drive Model



ABSOLUTE VALUE SERVO DRIVE STANDARD SPECIFICATIONS AND PERFORMANCE PARAMETERS

project	describe
EtherCAT signal communication	physical layer100BASE-TX
	communication connectorRj45×2 (terminal CN3=IN, CN4 =OUT)
	Network topologyBus type
	Baud rate2×100 Mbps (full duplex)
	Frame data length1484 bytes (Maximum value)
	Synchronization ManagerSm0: Email reception (master to slave) SM1: Email sending (from station to main station) SM2: Process data output (master station to slave station) SM3: Process data input (from station to main station)
	Synchronous modeDC Synchronization (SYNC0)
	Communication ObjectSDO: Service Data Object PDO: Process data object
	LED indicationL/A IN(Link/Activity IN)×1 L/A OUT(Link/Activity OUT)×1 RUN×2
	Communication Protocol StandardsCoE: CANOpen over EtherCAT Equipment protocol standardsIEC61800-7 CiA402 Drive Profile
CiA402 MODE	Loop Synchronous Position Mode (CSP), Loop Synchronous Speed Mode (CSV), Loop Synchronous Torque Mode (CST), Contour Position Mode (PP), Contour Speed Mode (PV), Contour Torque Mode (PT), Zeroing Mode (HM)

T6E SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

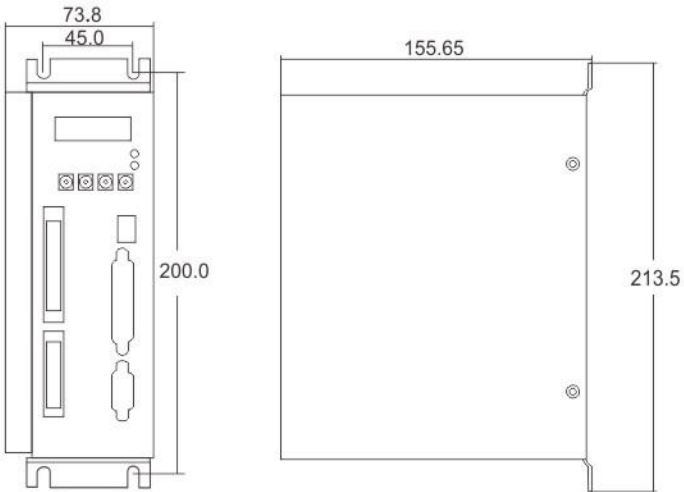


Figure 1 Installation dimension diagram of T6E-L20/L30 (unit: mm)

T6DE SERIES SERVO DRIVE EXTERNAL DIMENSIONS (mm)

驱动器尺寸一 (型号 T6DE-L15/L20)

驱动器尺寸二 (型号 T6DE-L30)



Figure 1 Installation dimension diagram of T6DE-L15/L20 (unit: mm)

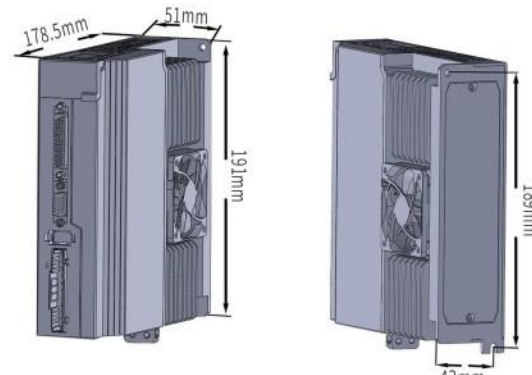
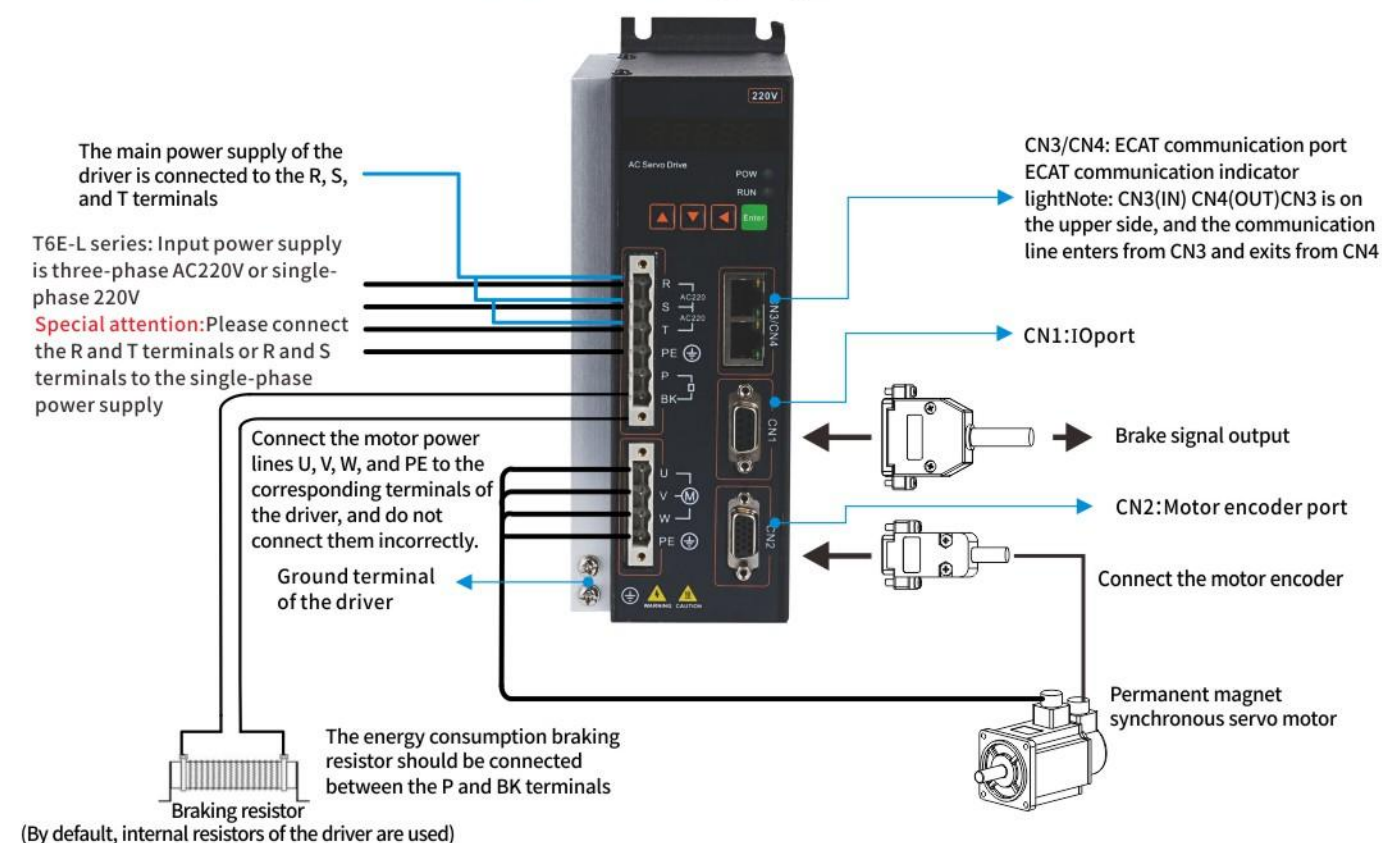


Figure 2 Installation dimension diagram of T6DE-L30 (unit: mm)

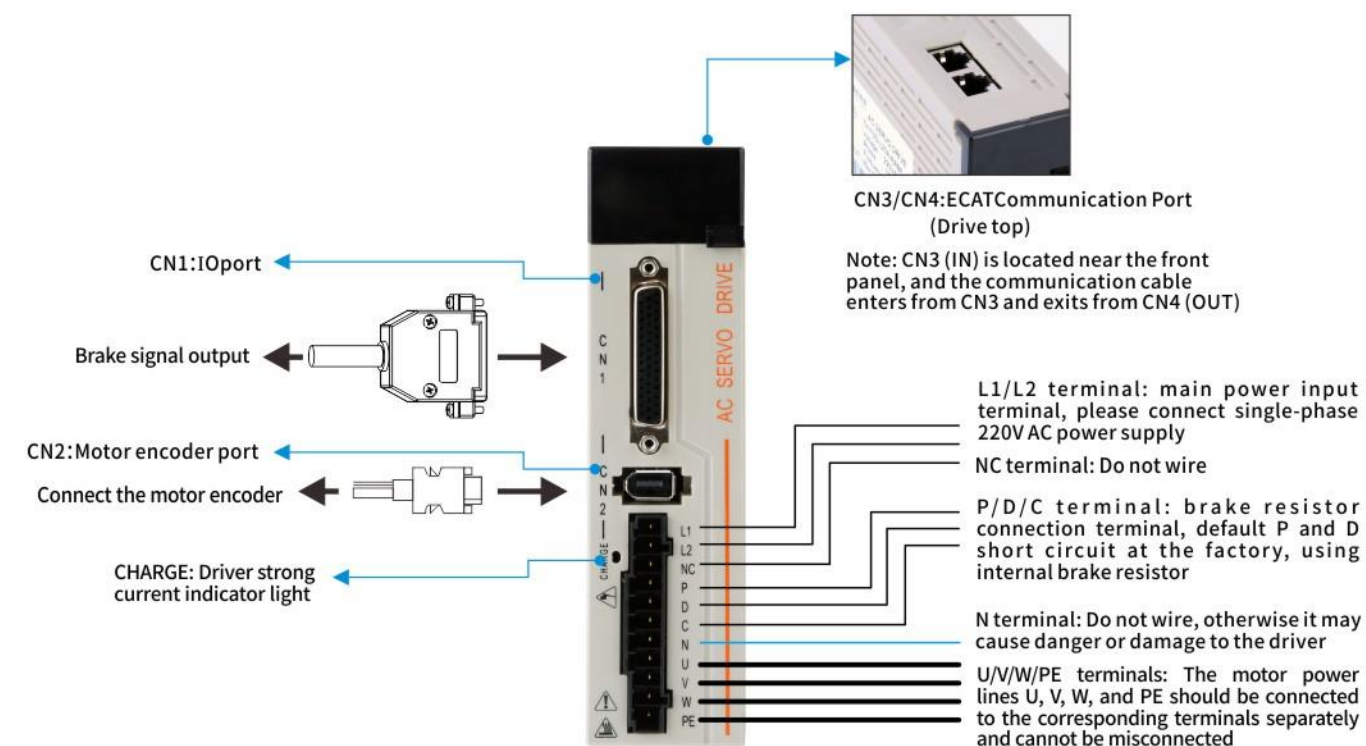
SERVO DRIVE WIRING DIAGRAM

T6E-L20/L30Wiring diagram

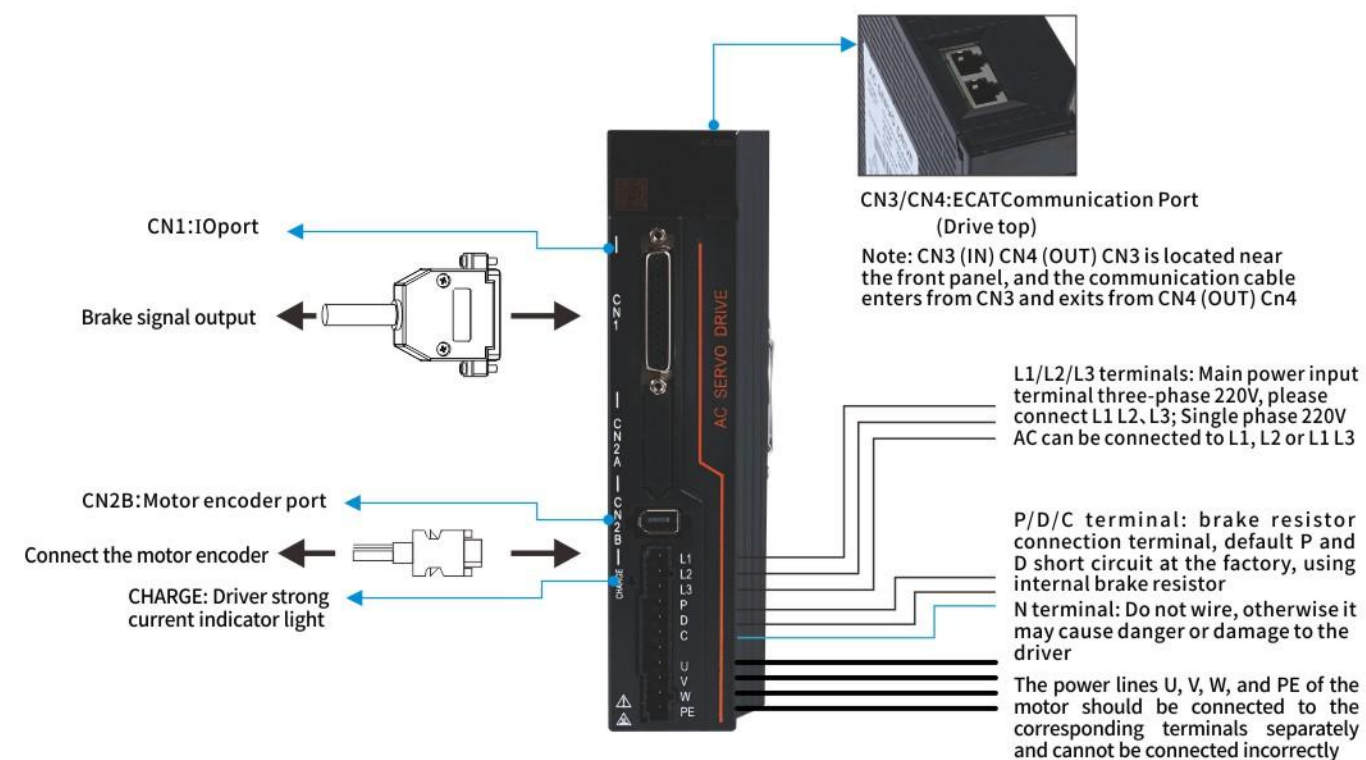


SERVO DRIVE WIRING DIAGRAM

T6DE-L15/L20Wiring diagram



T6DE-L30AWiring diagram



T3DC(CANOpen)Bus universal servo drive

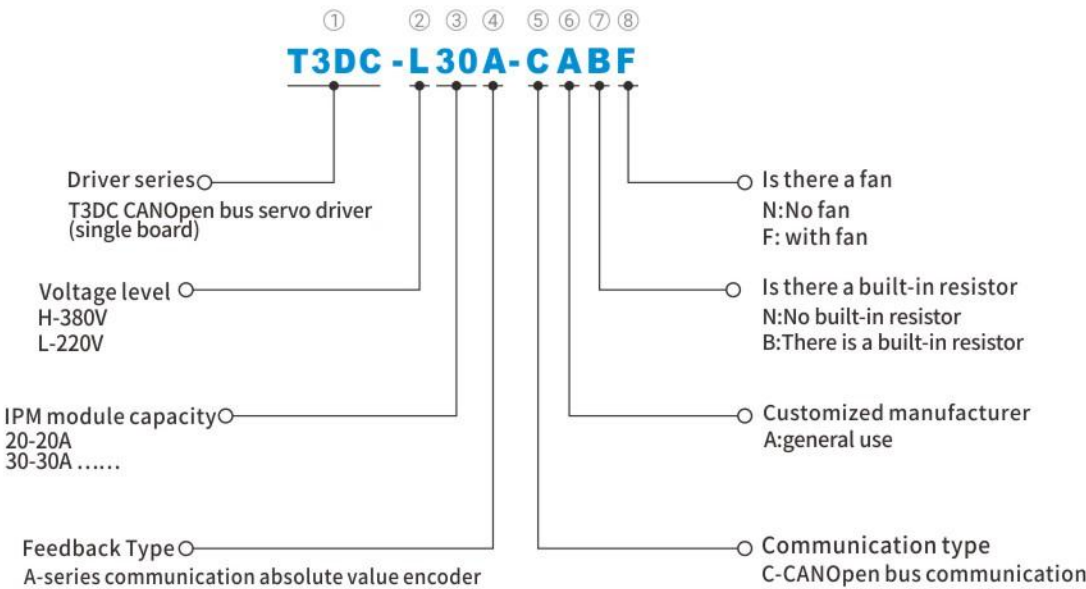
The CANOpen (T3DC) bus absolute value AC servo drive adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability, widely used in metal processing CNC machine tools and other general automation machinery.

Product Features

- ◆ Drive control instructions are transmitted through the CANOpen bus, making it easy to achieve high-speed and high-precision control
- ◆ Suitable for 17 bit/23 bit multi turn absolute encoders, with high positioning accuracy and good servo shaft rigidity
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Choose the latest industrial grade IPM module with strong overload driving capability
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance



SPECIFICATION DESCRIPTION OF SERVO DRIVE MODEL



CANOpen BUS COMMUNICATION SPECIFICATION

project		describe
CANOpen communication	Link Layer Protocol	CAN bus
	application layer protocol	CANOpen agreement
	CAN-ID type	11bit-CAN2.0A
	Baud rate	20Kbit/s, 50Kbit/s, 100Kbit/s, 125K, 250K, 500K, 800K, 1M
	Support sub protocols	DS301, DS402
	supporting service	SDO, PDO, NMT, EMCY, HeartBeat
	PDO transmission type	Event triggered, synchronous triggered, time triggered
	PDO quantity	4个TPDO、4个RPDO, Using synchronous mode is the fastest1ms
Support operation mode	SDO transmission method	Accelerated transmission
	Contour Position Mode (PP), Contour Velocity Mode (PV), Contour Torque Mode (PT), Zeroing Mode (HM), Interpolation Mode (IP)	

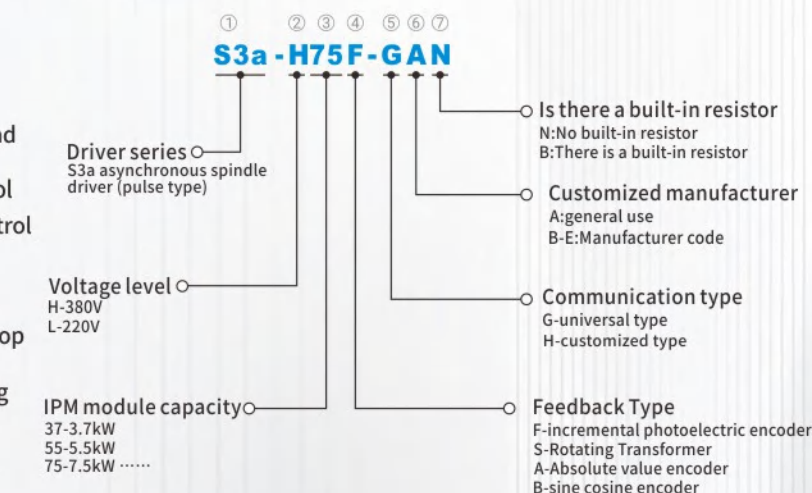
S3a Series spindle servo drive

The S3a series digital AC servo spindle drive is independently developed and produced by Zhejiang Hanghang Electric Drive Technology Co., Ltd. It is a domestically leading AC servo spindle drive product. This series of products adopts advanced fully digital control and AC motor servo control theory, with excellent system performance and high reliability, widely used for spindle drive of various CNC machine tools, textile machinery and other related industrial machinery.

Product Features

- ◆ High cost-effectiveness: servo performance and variable frequency price
- ◆ Adopting high-performance ARM main control chip, with excellent control performance
- ◆ Select a new industrial grade IPM module with strong overload driving capability
- ◆ Integrating position control, speed control, and torque control
- ◆ Can drive various types of spindle motors and be used to control built-in electric spindles
- ◆ Support multiple position feedback components such as incremental photoelectric encoders and rotary transformers
- ◆ Supporting dual encoder control, it can achieve full closed-loop high-precision control of machine tool spindle
- ◆ Capable of spindle accurate stop, rigid tapping, thread cutting and other functions
- ◆ Featuring excellent low-speed characteristics and industry-leading dynamic acceleration and deceleration performance

Specification Description of Servo Drive Model



UNIVERSAL SERVO DRIVE STANDARD SPECIFICATIONS AND PERFORMANCE PARAMETERS

model		S3a-H37	S3a-H55	S3a-H75	S3a-H150	S3a-H220	S3a-H370
Input power supply		three-phase AC380V (85%~110%) 50 Hz /60Hz±1Hz					
Maximum effective value of output current (A)		20	30	42	60	84	113
Standard motor rated power (kW)		3.7	5.5	7.5	15	22	37
Specification	Main circuit mode	Three phase full wave rectification, IPM SVPWM inverter bridge					
	Feedback method	Incremental encoder 1024p/rev, 2500 p/rev, rotary transformer, sine and cosine encoder					
	way of working	S3a: Manual, jog, self run, speed, position, speed/position					
	Communication method	RS485					
speed control way	performance	speed range 1~30000r/min Maximum output frequency 1kHz (speed closed-loop control)					
	speed ratio	1: 30000 constant power speed range>4:1 (corresponding to standard motor speed range of 1500r/min~6000r/min)					
	steady speed accuracy	When the load change rate is 0-100%, the speed change rate is less than (0.1% of the rated speed)					
	simulation speed command voltage range	① DC 0-10V+forward/reverse signal+enable signal; ② DC-10V~+10V+enable signal;					
	speed command input	Simulate instruction multiplication factor:- 32767~32767; Analog instruction frequency division coefficient: 1~32767;					
Position control method	Position command input pulse type	① Pulse+direction; ② CCW pulse/CW pulse; ③ A/B two-phase orthogonal pulse;					
	Position command electronic gear function	Position instruction multiplication factor: 1~32767; Position instruction frequency division coefficient: 1~32767;					
input output signal	Position feedback signal output	Motor encoder or spindle encoder signal 1:1 output, A/B/Z differential signal, line driver output					
	Status signal output	Spindle ready, zero speed output, position/speed reached, alarm output, quasi stop completed					
	Control signal input	Main spindle enable, CCW forward rotation signal, CW reverse rotation signal, alarm clearing, main spindle quasi stop, speed/position switching					
basic function	Regenerative energy processing function	The driver is equipped with a built-in regenerative energy processing circuit and an external regenerative braking resistor					
	operative function	4 buttons to select status display, parameter settings, parameter management, function operation, etc					
	monitoring function	5-digit LED digital display, capable of displaying speed, torque, speed command, feedback position, position deviation, load current, output voltage, IO status, control mode, command pulse frequency, alarm code, etc					
	Protection function	Overvoltage, undervoltage, overcurrent, overload, overspeed, position deviation, quasi stop alarm, position deviation, brake abnormality, encoder abnormality, etc					
	Internal speed command function	Multiple speed commands pre-set internally can be selected					
	Spindle stop function	Support high-speed operation with direct quasi stop, flexible setting of quasi stop direction and position					
	Rigid tapping function	Can participate in interpolation control of coordinate axes, complete functions such as rigid tapping and thread cutting					
	Feed servo function	Optimized control algorithm enables the spindle to have all the functions of a servo axis					

S3a SERIES SPINDLE DRIVE EXTERNAL DIMENSIONS (mm)

Driver size one (model S3a-H37:3.7kW)

Driver size two (model S3a-H55: 5.5kW / S3a-H75: 7.5kW)

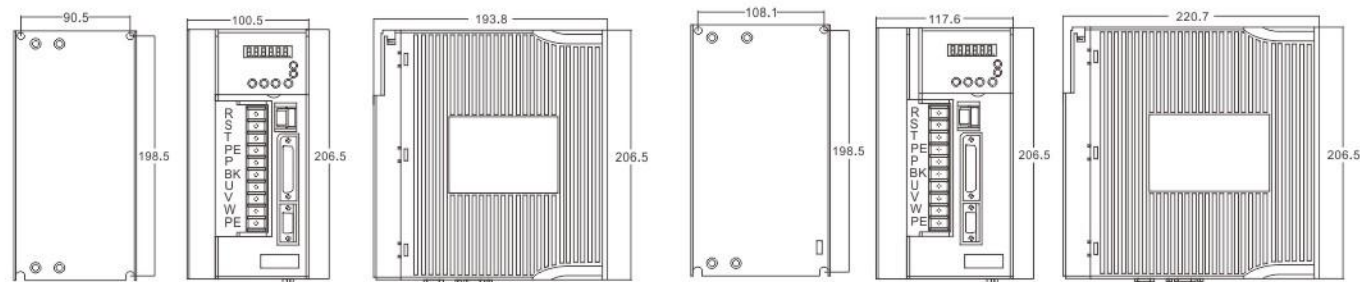
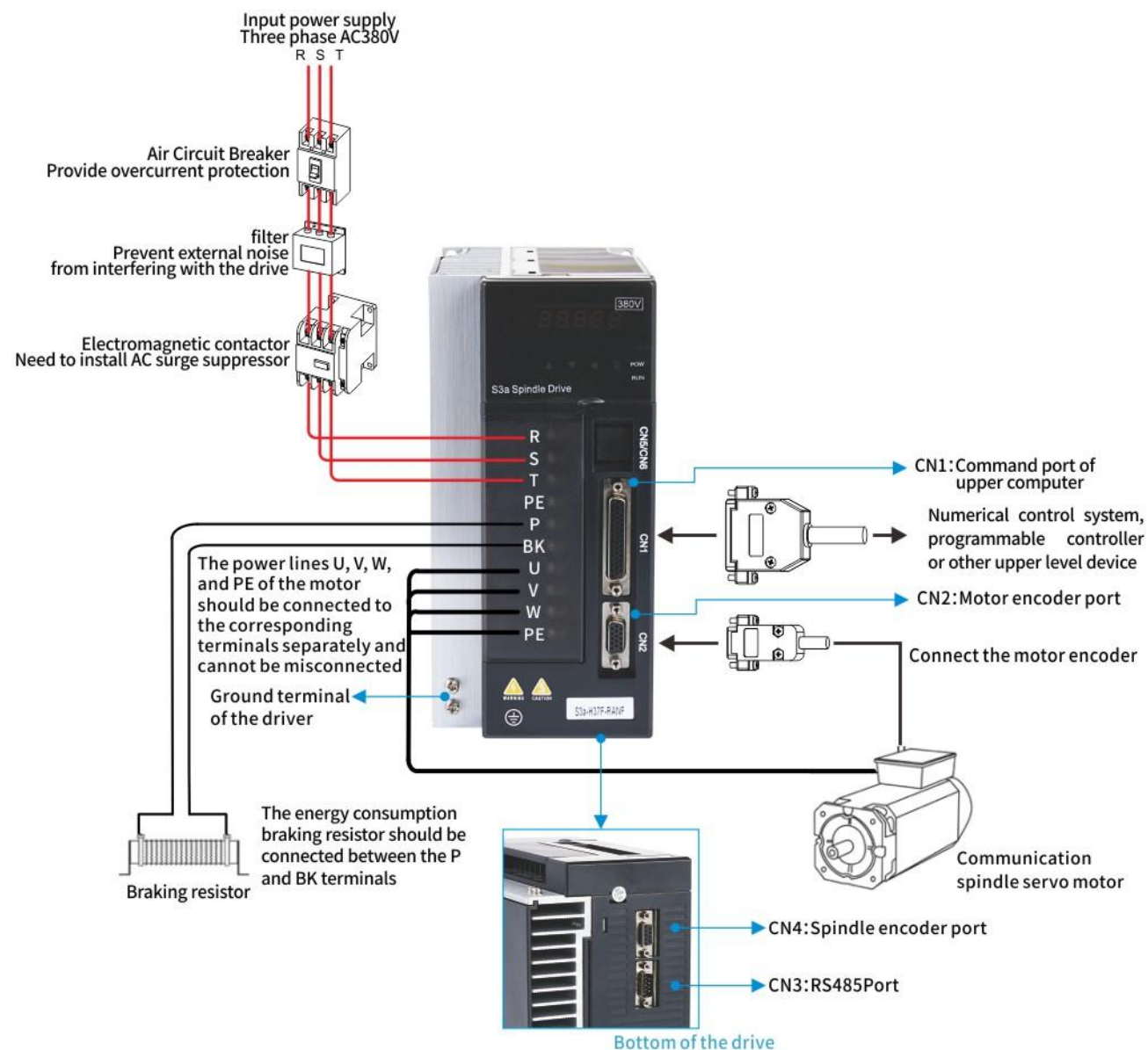


Figure 1 Dimension diagram of S3a-H37 spindle drive (unit: mm)

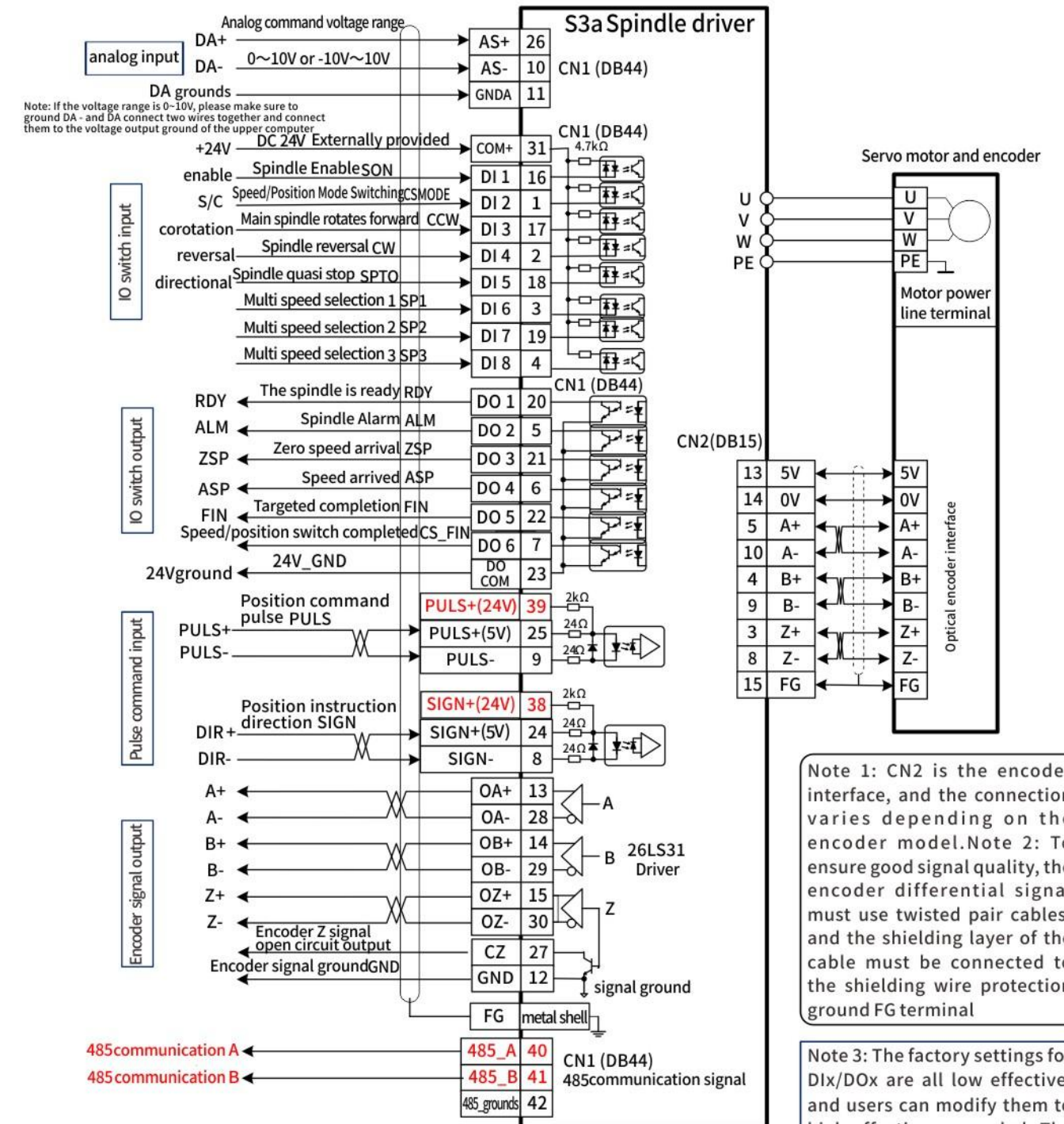
Figure 2 Dimension diagram of S3a-H55/S3a-H75 spindle drive (unit: mm)

S3a SERIES SPINDLE DRIVE WIRING DIAGRAM



CONTROL WIRING DIAGRAM

This side signal is connected to the CNC system



Note 1: CN2 is the encoder interface, and the connection varies depending on the encoder model. Note 2: To ensure good signal quality, the encoder differential signal must use twisted pair cables, and the shielding layer of the cable must be connected to the shielding wire protection ground FG terminal

Note 3: The factory settings for DIx/DOx are all low effective, and users can modify them to high effective as needed. The DIx/DOx function can also be redefined, with parameters P100~P107 configuring DIx function and high/low level effectiveness, and parameters P108~P113 configuring DOx function and high/low level effectiveness.

INTRODUCTION TO SERVO DRIVE PORTS

name	Terminal symbol	detailed description
Main circuit power supply	R, S, T	Connect external AC power supply: three-phase AC380V (85%~110%) 50/60Hz
Braking resistor terminal	P, BK	When using an external braking resistor in parallel with an internal braking resistor, please pay attention to selecting the resistance and power to avoid burning out the driver.
Motor connection terminal	U	Output to motor U-phase power supply
	V	Output to motor V-phase power supply
	W	Output to motor W-phase power supply
ground terminal		Grounding terminal of motor casing
	或 PE	Driver grounding terminal

CONTROL CIRCUIT TERMINAL

type	name	function	detailed description
External 24V control power supply	COM+	DC24V control power output terminal	The positive pole of the power supply at the input terminal is used to drive the optocoupler at the input terminal, DC12~24V, Current ≥ 100mA.
	DO COM	Control power supply 0V terminal	Power ground of input terminal
Control signal input (Programmable)	DI1(SON)	enable input	DI1~DI8 digital input functions are all programmable. The examples shown in this table are factory default configurations, which can be flexibly configured by driver software parameters and can be adapted to various types of upper computers, such as CNC systems PLC. Motion controllers, etc.
	DI2(CSMODE)	Switching between speed/position control modes	
	DI3(CCW)	Positive rotation allowed/enabled	
	DI4(CW)	Reverse allowed/enabled	
	DI5(SPTO)	Spindle quasi stop	
	DI6(SP1)	Multi speed selection 1	
	DI7(SP2)	Multi speed selection 2	
	DI8(SP3)	Multiple speed options 3	
Control signal output (Programmable)	DO1(RDY)	The spindle is ready	All DO1~DO5 digital output functions are programmable, and the examples shown in this table are factory default configurations. They can be flexibly configured by driver software parameters and can be adapted to various types of upper computers, such as CNC systems PLC. Motion controllers, etc.
	DO2(ALM)	ALARM IN	
	DO3(ZSP)	Zero speed arrival	
	DO4(ASP)	Speed arrived	
	DO5(FIN)	The spindle has been stopped accurately	
	DO6(CS_FIN)	Speed/position switch completed	
Analog command input	AS+	Simulate command input positive	Positive connection with the analog command output of the upper computer
	AS-	Simulate instruction input negative	Negative connection to the analog command output of the upper computer
	GNDA	Analog signal ground	Connect to the analog signal ground of the upper computer
Pulse quantity command input	PULS+	Command pulse PLUS input	External command pulse input terminal. Note: The pulse input mode is set by parameter P035 and supports three types of pulse input modes ◆P035=0: Command pulse+direction mode (default state); ◆P035=1: CCW/CW command pulse mode; ◆P035=2: Orthogonal pulse mode.
	PULS-		
	SIGN+	Command pulse SIGN input	
	SIGN-		
Encoder output signal (Sent to the upper computer)	OA+ / OA-	Encoder A-phase signal output	◆Encoder ABZ signal differential drive output (26LS31 output, equivalent to Rs422); ◆Non isolated output (non insulated).
	OB+ / OB-	Encoder B-phase signal output	
	OZ+ / OZ-	Encoder Z-phase signal output	
	CZ	Encoder Z-phase collector electrode open circuit output	◆The Z-phase signal of the encoder is output by an open collector. When the Z-phase signal of the encoder appears, it outputs ON (output conduction), otherwise it outputs OFF (output cutoff); ◆Non isolated output (non insulated);
Encoder input signal (From spindle motor)	A+/A-	Encoder A-phase signal input	The signal input terminal of the spindle motor encoder has different definitions for different types of encoder signals. The table shows the photoelectric encoder, and subsequent products support photoelectric encoders, rotary encoders, sine cosine encoders, and absolute value encoders
	B+/B-	Encoder B-phase signal input	
	Z+/Z-	Encoder Z-phase signal input	
	5V/GNDD	Encoder 5V power/ground	
ground terminal	PE / FG	Shielded wire protection ground	Connect to the shielded wire of the signal cable

DRIVER ACCESSORIES (BRAKING RESISTOR)

Recommended Selection Table for Braking Resistors

drive model	Maximumbrakingcurrent	Braking resistor resistance value	Braking resistor power	Is the driver equipped with a built-in braking resistor
S3a-H37 (3.7kW 25A module)	20A	≥50Ω	≥1500W	no
S3a-H55 (5.5kW 50A module)	20A	≥50Ω	≥1500W	no
S3a-H75 (7.5kW 75A module)	35A	≥30Ω	≥2000W	no
S3a-H150 (15kW 100A module)	35A	≥27Ω	≥2500W	no
S3a-H220 (22kW 150A module)	50A	≥15Ω	≥3000W	no

TECHNICAL PARAMETERS OF SUPPORTING SPINDLE MOTOR

Motor code (P-002)	Adapted Driver model	spindle model	rated power (kW)	Rated current (A)	rated torque (Nm)	Rated speed/ maximum speed (r/min)	Overload multiple
0	S3a-H37 (20A 3.7kW) (380V)	MY165-070C	1.1	2.6	7.0	1500/8000	2
1		MY165-096C	1.5	3.4	9.6	1500/8000	2
2		MY165-140C	2.2	4.9	14.0	1500/8000	2
3		MY165-235C	3.7	7.8	23.5	1500/8000	2
6		MY204-235C	3.7	9.1	23.5	1500/8000	2
7		MY204-350C	5.5	13.0	35.0	1500/8000	1.6
10	S3a-H55 (28A 5.5kW) (380V)	MY204-235C	3.7	9.1	23.5	1500/8000	2
11		MY204-350C	5.5	13.0	35.0	1500/8000	2
12		MY204-478C	7.5	17.8	47.8	1500/8000	1.7
13	S3a-H75 (42A 7.5kW) (380V)	MY204-235C	3.7	9.1	23.5	1500/8000	4
14		MY204-350C	5.5	13.0	35.0	1500/8000	3
15		MY204-478C	7.5	17.8	47.8	1500/8000	2
16		MY204-605C	9.5	23.8	60.5	1500/8000	1.7
17		MY265-700C	11	21.5	70	1500/8000	2
19		MY204-605C	9.5	23.8	60.5	1500/8000	2
20	S3a-H150 (60A 15kW) (380V)	MY265-700C	11	21.5	70	1500/8000	2
21		MY265-960C	15	29.0	96	1500/8000	2
22		MY265-A18C	18.5	35.2	118	1500/8000	1.6
23	S3a-H220 (84A 22kW) (380V)	MY265-A18C	18.5	35.2	118	1500/8000	2
24		MY265-A40C	22	42.3	140	1500/8000	2

T3M/T3G Machine tool specific servo drive

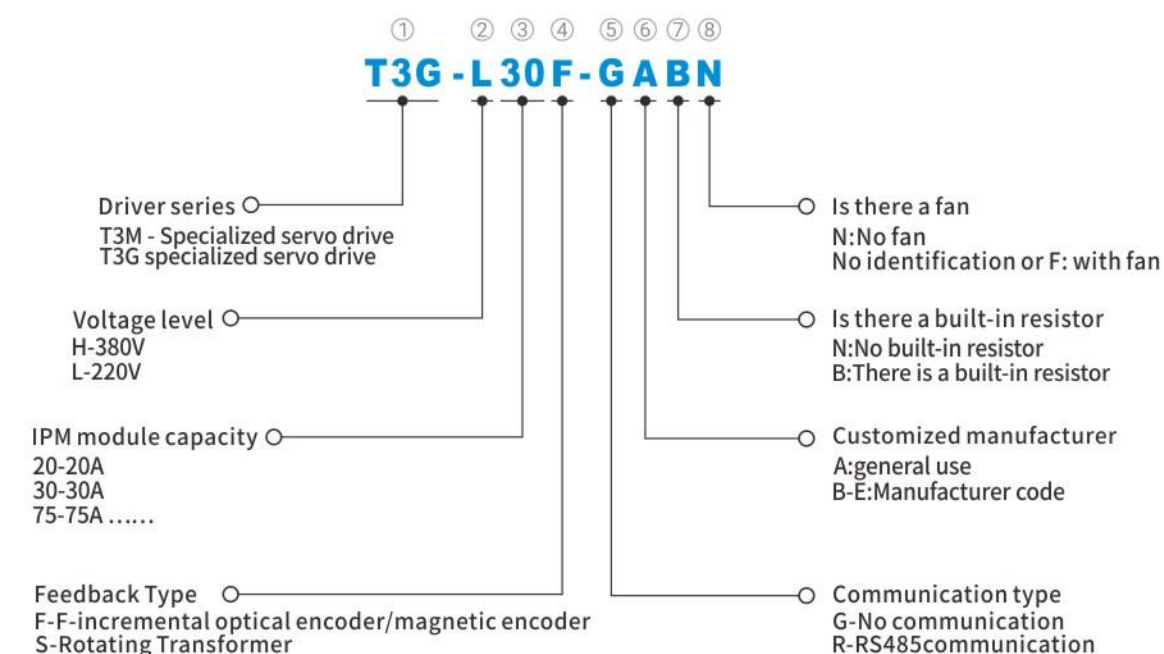
The T3M/T3G series digital AC servo drive is independently developed and produced by Zhejiang Hanghang Electric Drive Technology Co., Ltd. It is a leading AC servo drive product in China. This series of products adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability. It is widely used for servo shaft drive of various CNC lathes, milling machines, grinders, machining centers and other related automation industry machinery.

Product Features

- ◆ The driver instruction type is pulse type, and the motor position feedback is pulse incremental encoder
 - ◆ Adopting high-performance ARM main control chip, with excellent control performance
 - ◆ The control interface adopts the mainstream definition method in the machine tool industry, which can achieve seamless replacement
 - ◆ Select a new industrial grade IPM module with strong overload driving capability
 - ◆ Integrating position control, speed control, and torque control
 - ◆ Can drive various types of permanent magnet synchronous servo motors
 - ◆ Equipped with comprehensive fault protection and status monitoring functions
- Featuring excellent low-speed torque characteristics and industry-leading dynamic acceleration and deceleration performance



SPECIFICATION DESCRIPTION OF SERVO DRIVE MODEL



T3x (T3x: represents T3M and T3G) servo drive external dimensions (mm)

Driver size (model T3x-L20/L30)

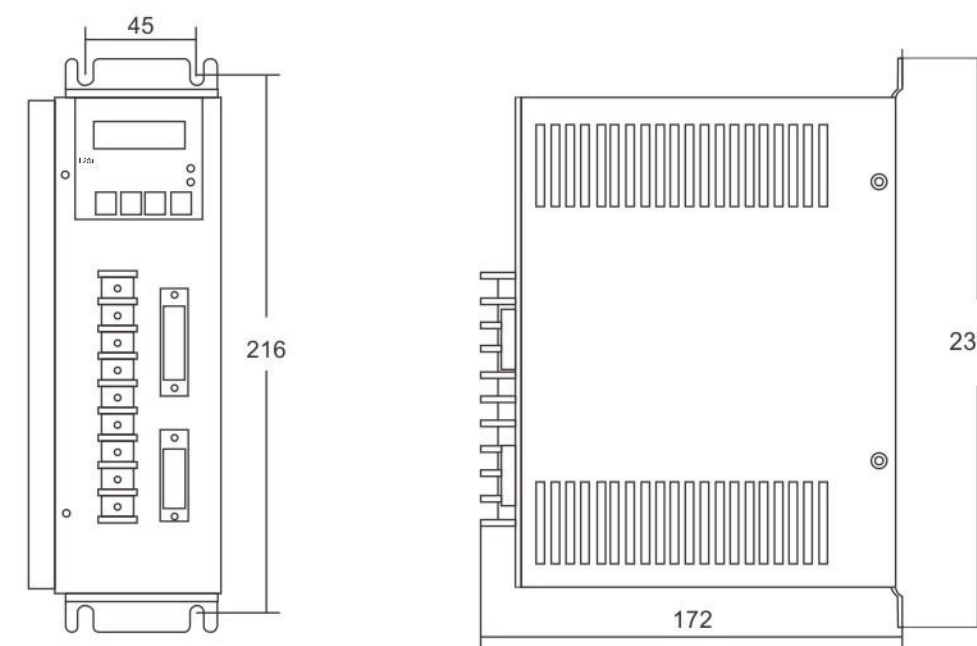
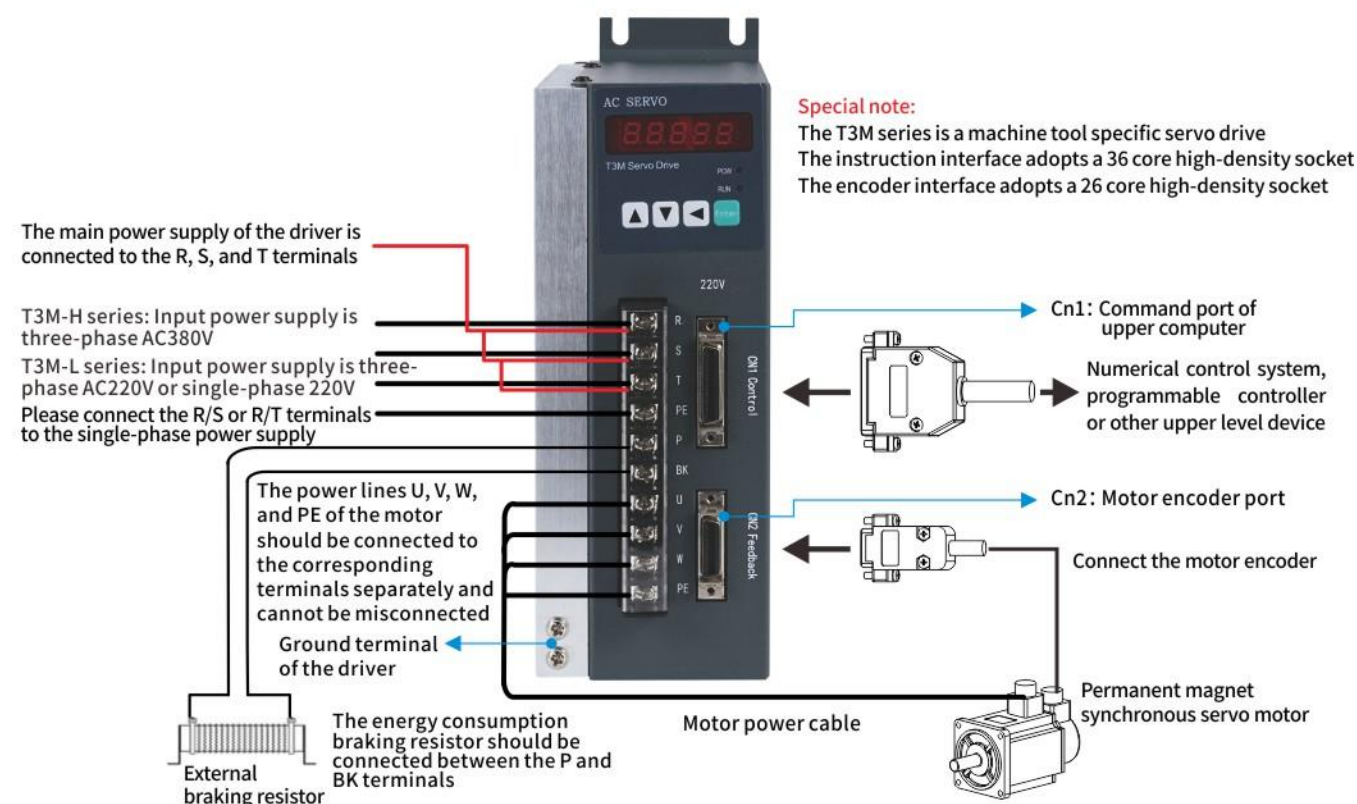


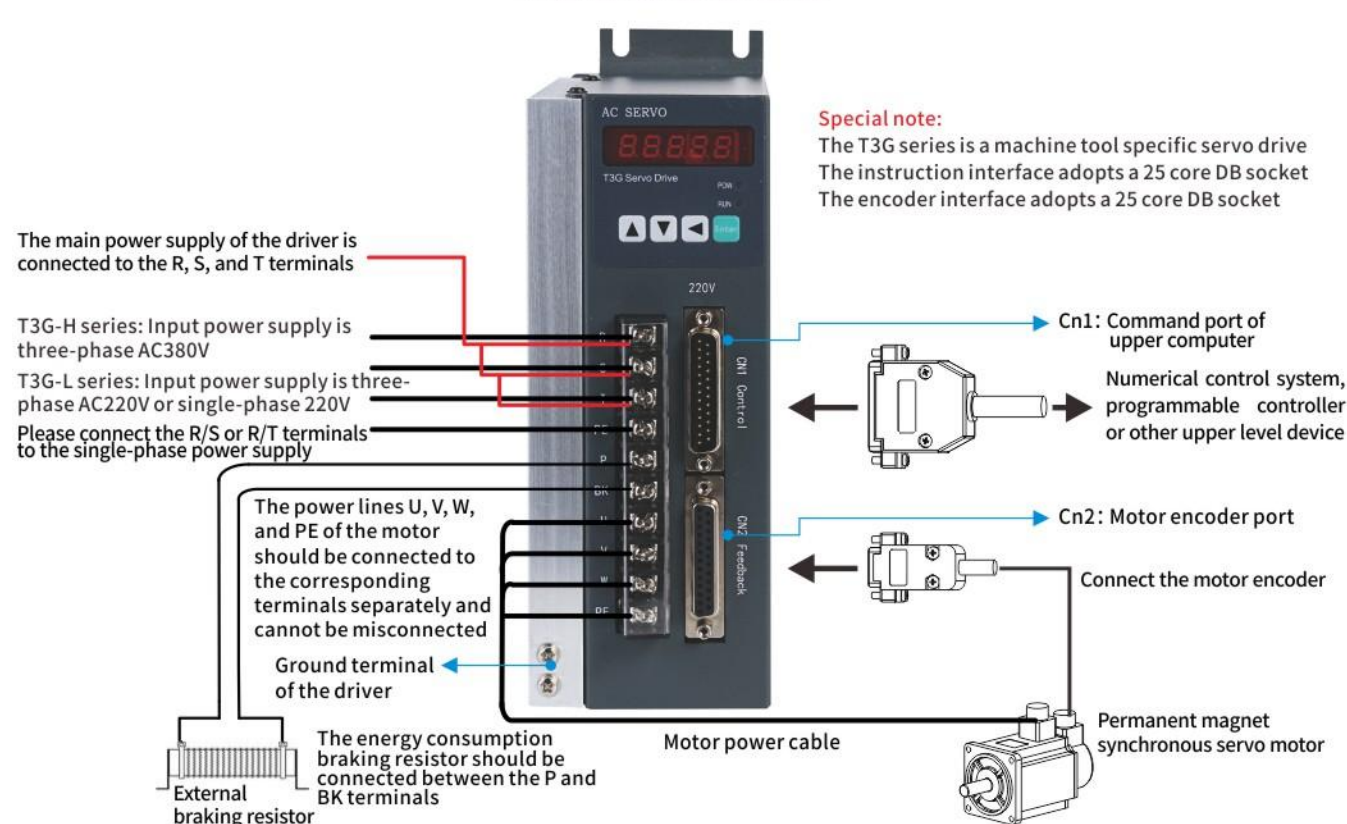
Figure 1 Dimension diagram of T3x-L20/L30 servo drive (unit: mm)

SERVO DRIVE WIRING DIAGRAM

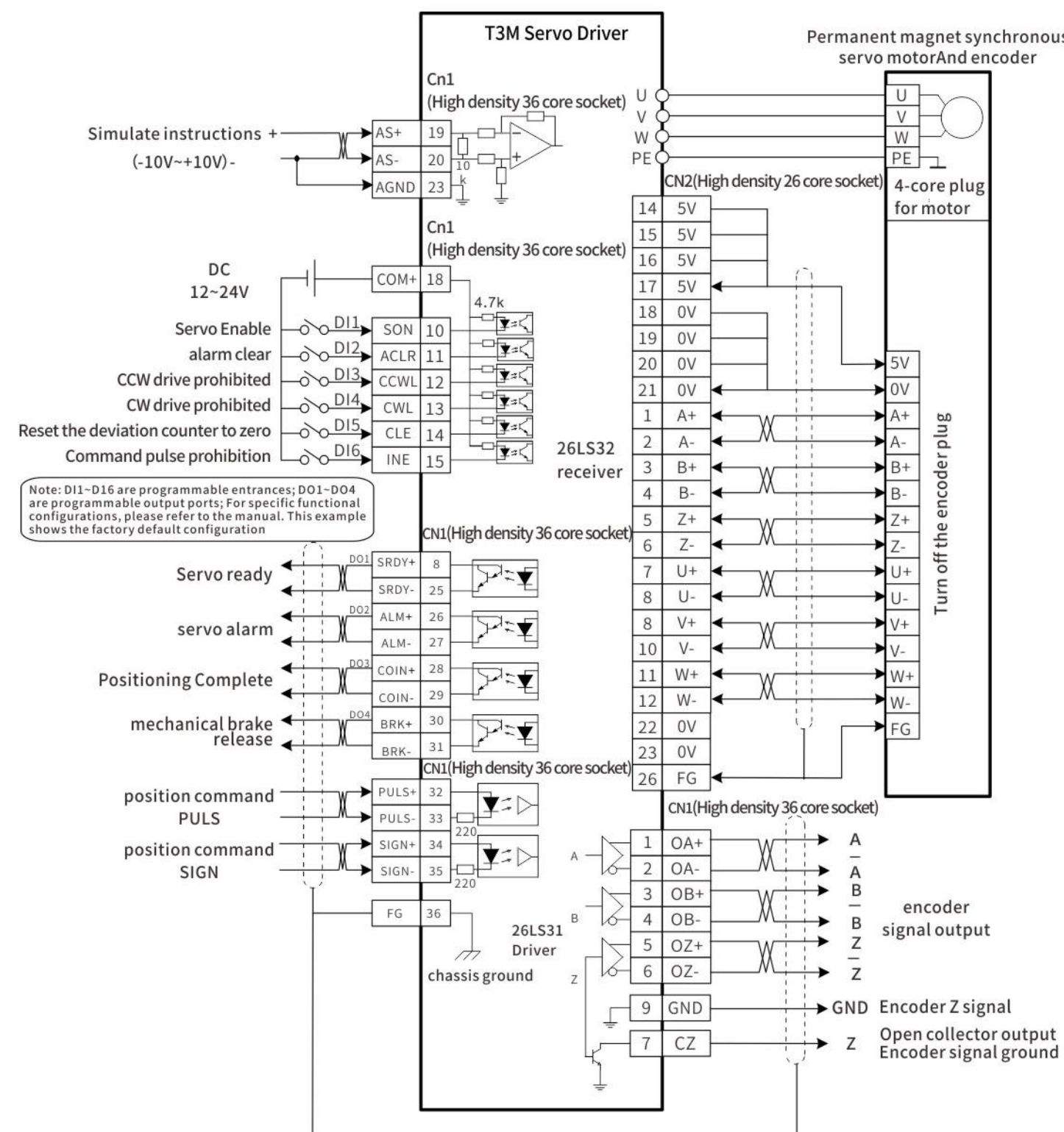
T3M Series servo drive



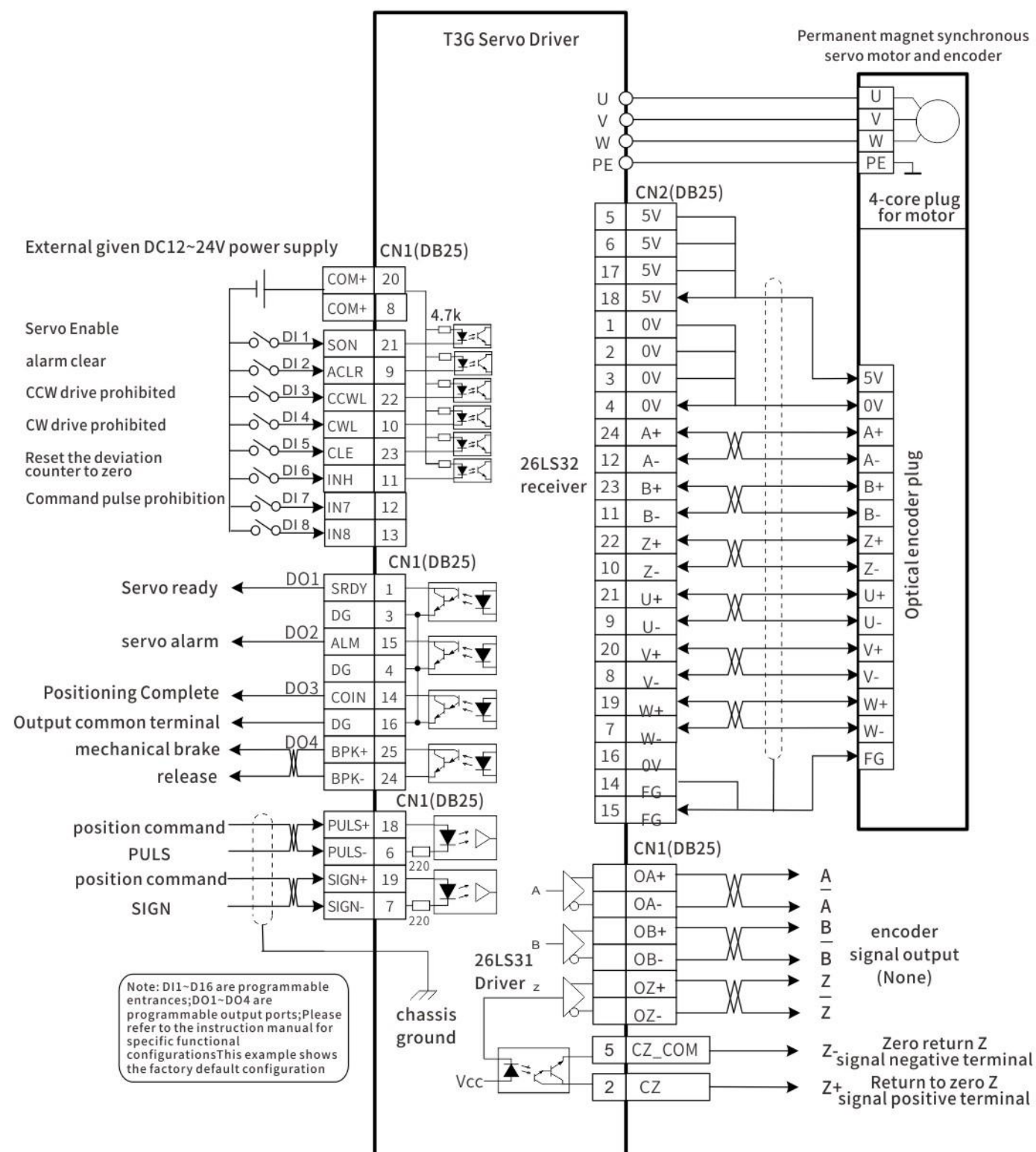
T3G Series servo drive



CONTROL WIRING DIAGRAM



CONTROL WIRING DIAGRAM



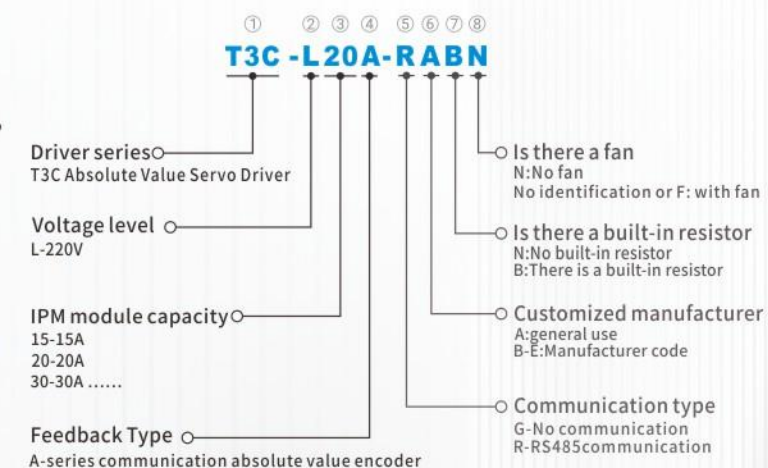
T3C Series Absolute Value Servo Drivers

The T3C series serial communication absolute value servo drive adopts advanced fully digital control and AC motor vector control theory, with excellent system performance and high reliability, widely applicable to various general automation machinery.

Product Features

- ◆ The driver instruction type is pulse type, and the motor position feedback is a serial communication absolute value encoder
- ◆ Adopting high-performance ARM main control chip, with excellent control performance
- ◆ Suitable for 17 bit/23 bit serial communication absolute encoder, with high positioning accuracy and good servo axis rigidity
- ◆ Fully sealed design, good protection, strong anti-interference ability
- ◆ Choose the latest industrial grade IPM module with strong overload driving capability
- ◆ Integrating position control, speed control, and torque control
- ◆ Can drive various types of permanent magnet synchronous servo motors
- ◆ Equipped with comprehensive fault protection and status monitoring functions
- ◆ Has excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Specification Description of Servo Drive Model



DRIVER INSTALLATION DIMENSION DIAGRAM (unit: mm)

Driver size one (model T3C-L15/L20)

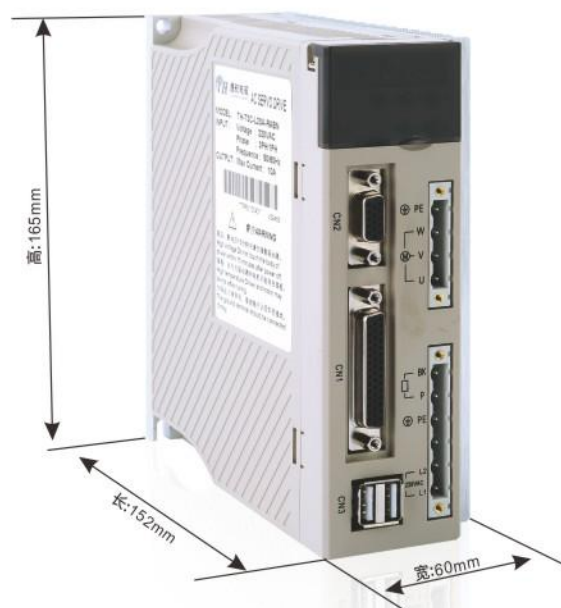


Figure 1 Size diagram of T3C-L15/L20 servo drive (unit: mm)

Driver size two (model T3C-L30)

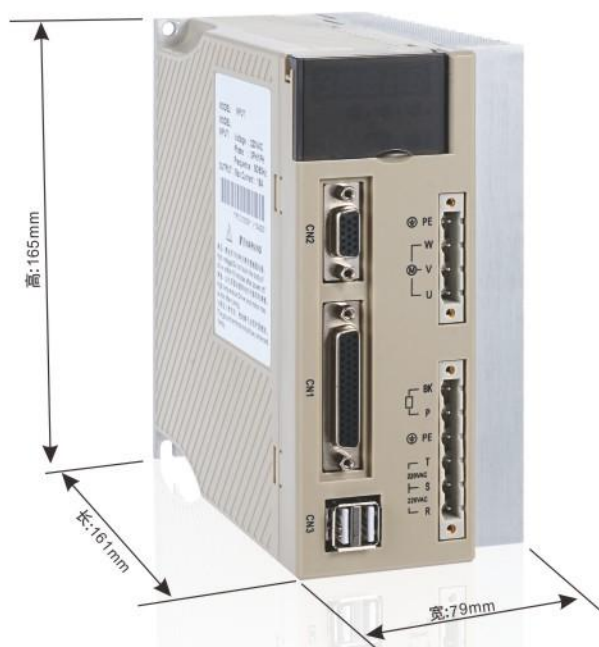
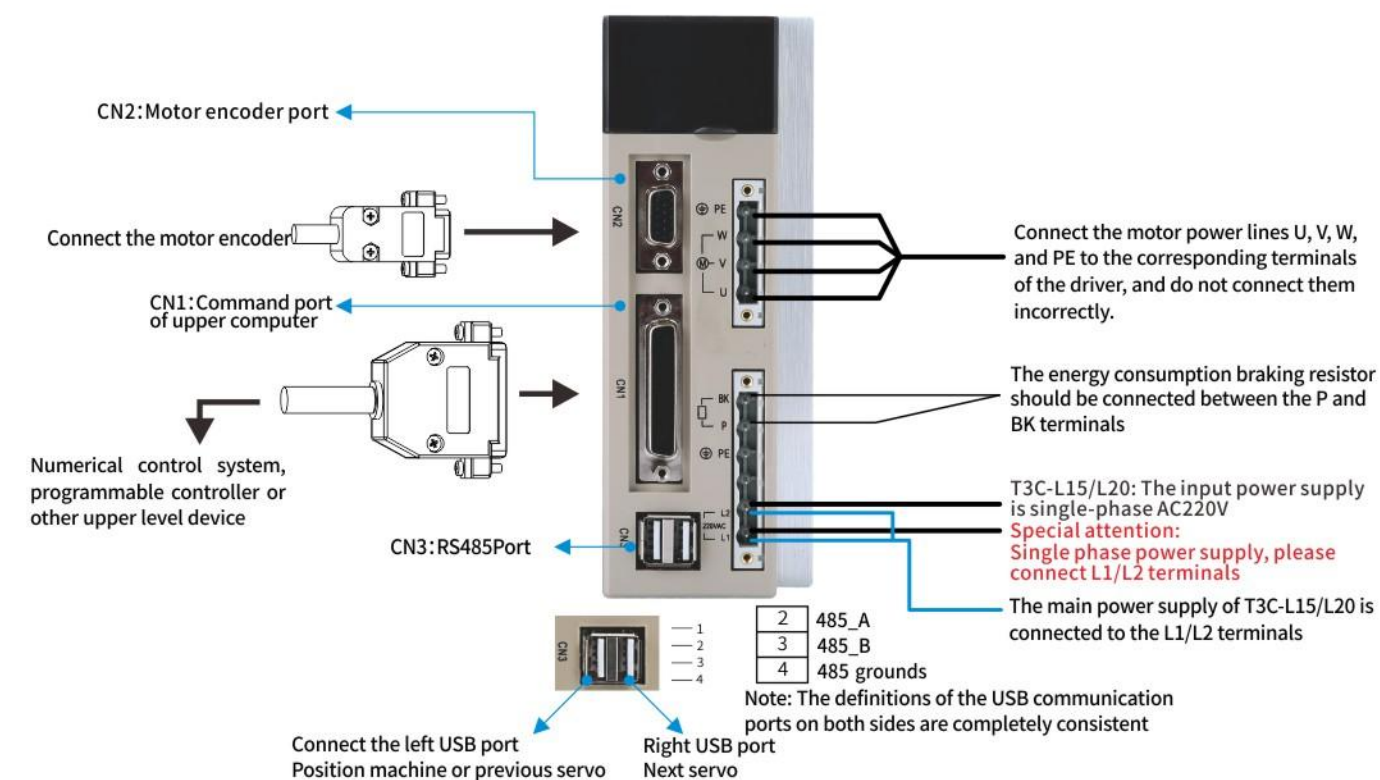


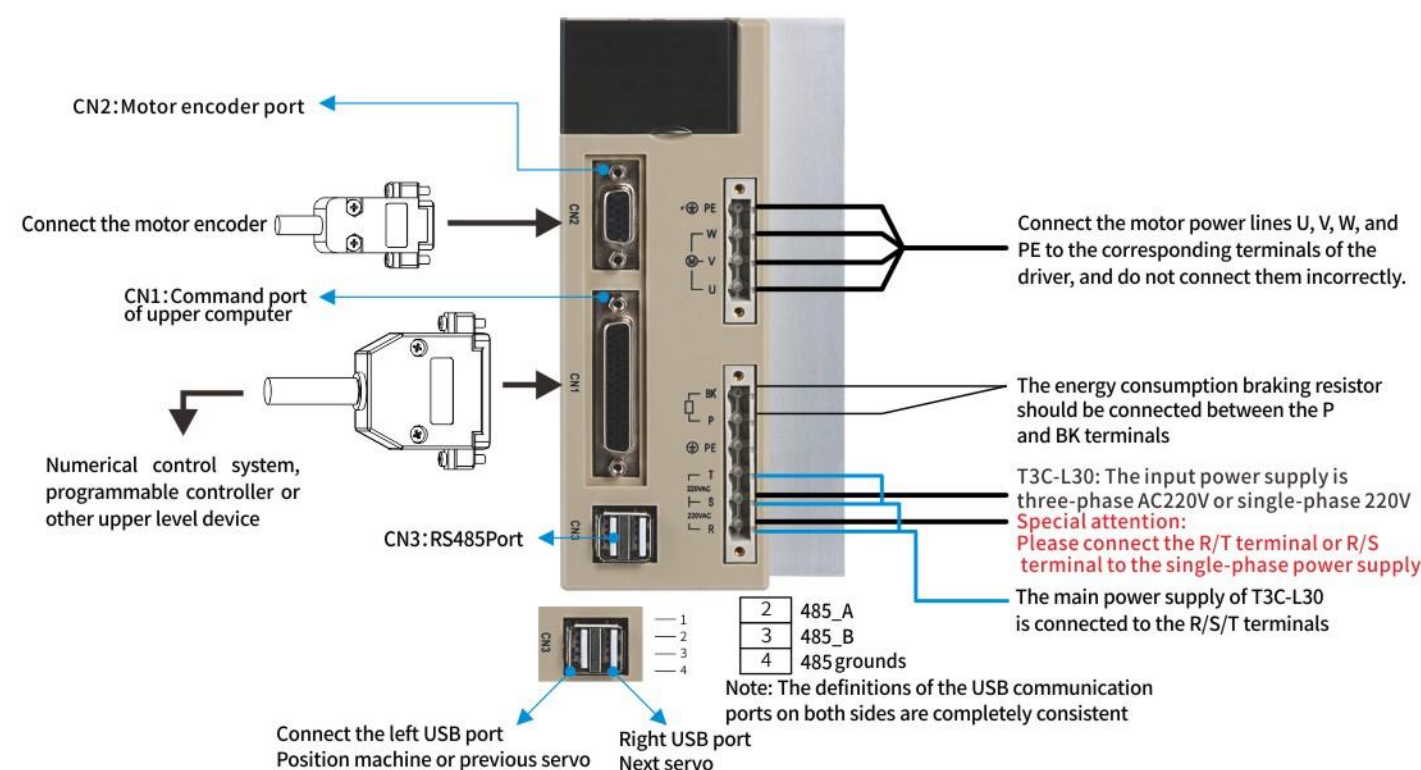
Figure 2 Dimension diagram of T3C-L30 servo drive (unit: mm)

SERVO DRIVE WIRING DIAGRAM

T3C-L15/L20Wiring diagram

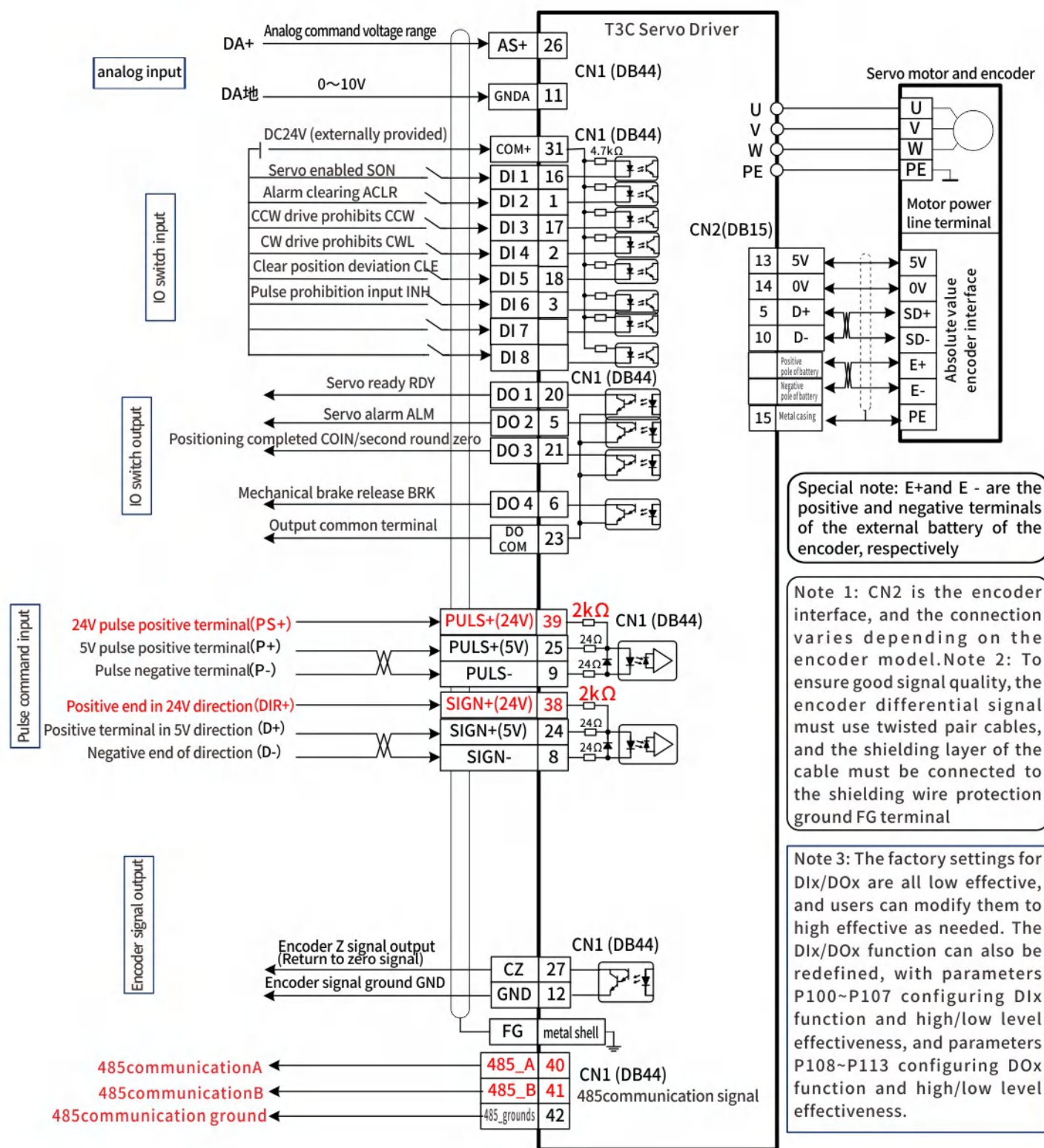


T3C-L30Wiring diagram



CONTROL WIRING DIAGRAM

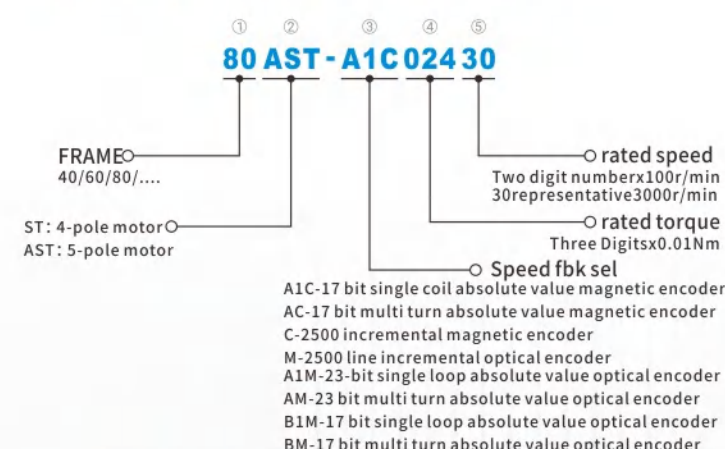
This side signal is connected to the upper computer controller



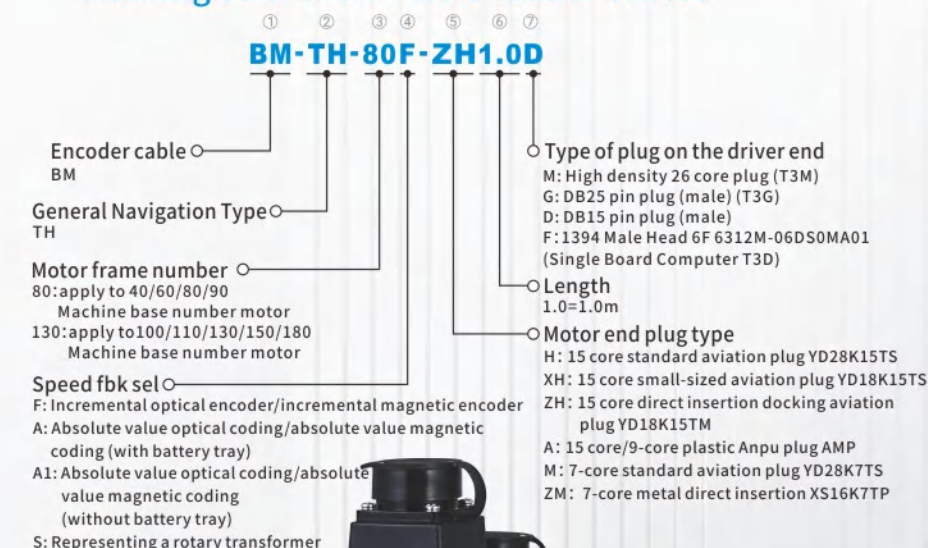
Special attention:
If the pulse command signal is a 24V power supply, PULS+ must be connected to pin 39 of CN1, and SIGN+ must be connected to pin 38 of CN1, otherwise it may damage the pulse command port;
If the pulse command signal is a 5V power supply, PULS+ must be connected to pin 25 of CN1, and SIGN+ must be connected to pin 24 of CN1, otherwise there may be abnormal pulse reception;

Naming rules for servo motors and supporting cables

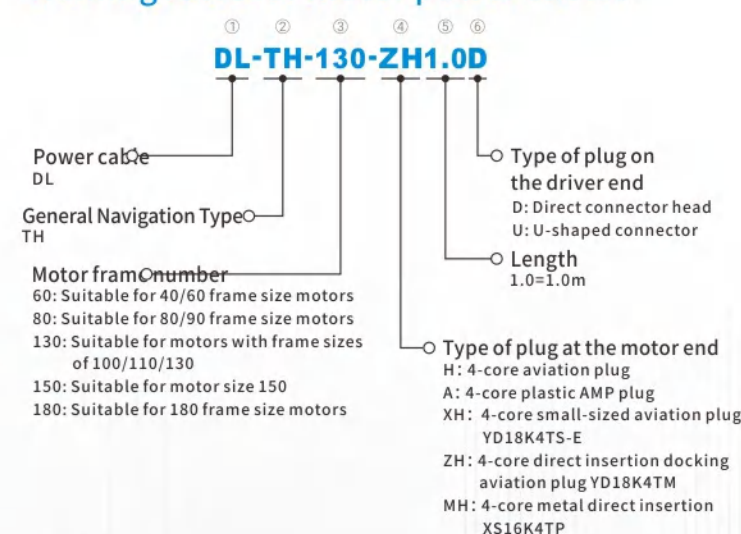
Naming rules for motor models



Naming convention for encoder cables



Naming rules for motor power cables



Annotation on plug types at the motor end

Naming rules for power cables (types of plugs at the motor end)	
H	4-core aviation plug (110/130/150 aviation plug YD28K4TS, 180 aviation plug YD32K4TS)
A	4-core plastic Anpu plug AMP (suitable for motors with frame sizes of 40~90)
XH	4-core small-sized aviation plug YD18K4TS-E (applicable for incremental motor sizes 40~90)
ZH	4-core direct insertion docking aviation plug YD18K4TM (applicable for incremental motors of 40~90 base sizes)
MH	4-core metal direct insertion XS16K4TP (dedicated for absolute value of 40~90 frame size motors)

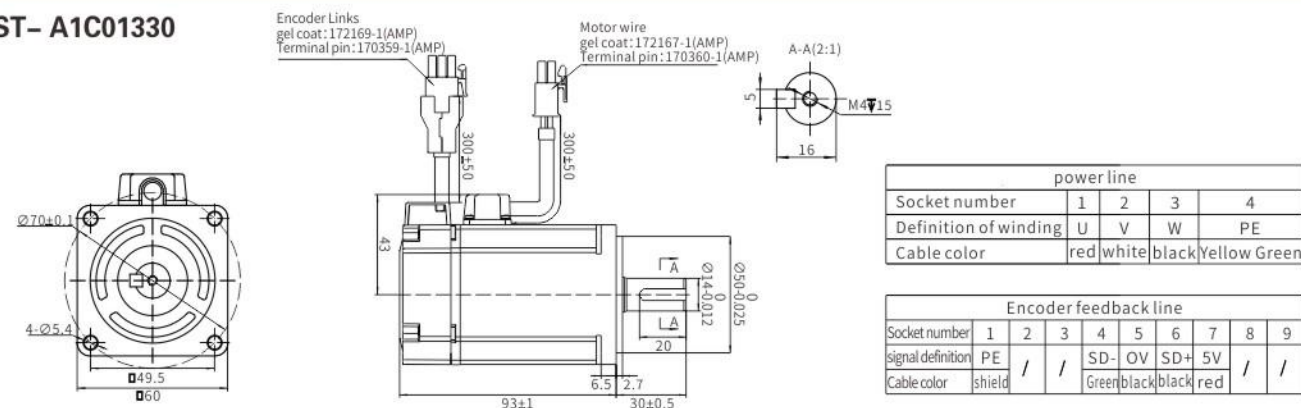
Naming convention for encoder cables (plug type at motor end)

H	15 core standard aviation plug YD28K15TS (applicable to incremental encoders for 110~180 base size motors)
XH	15 core small-sized aviation plug YD18K15TS (suitable for incremental encoders of 40~90 machine base size motors)
ZH	15 core direct insertion docking aviation plug YD18K15TM (applicable to incremental encoders for 40~90 base size motors)
A	15 core/9-core plastic Anpu plug AMP (suitable for incremental/absolute encoders for motors with frame sizes of 40~90)
M	7-core standard aviation plug YD28K7TS (dedicated for absolute value encoder of 110~180 seat motor)
ZM	7-core metal direct insertion XS16K7TP (dedicated for absolute value encoders of 40~90 machine base size motors)

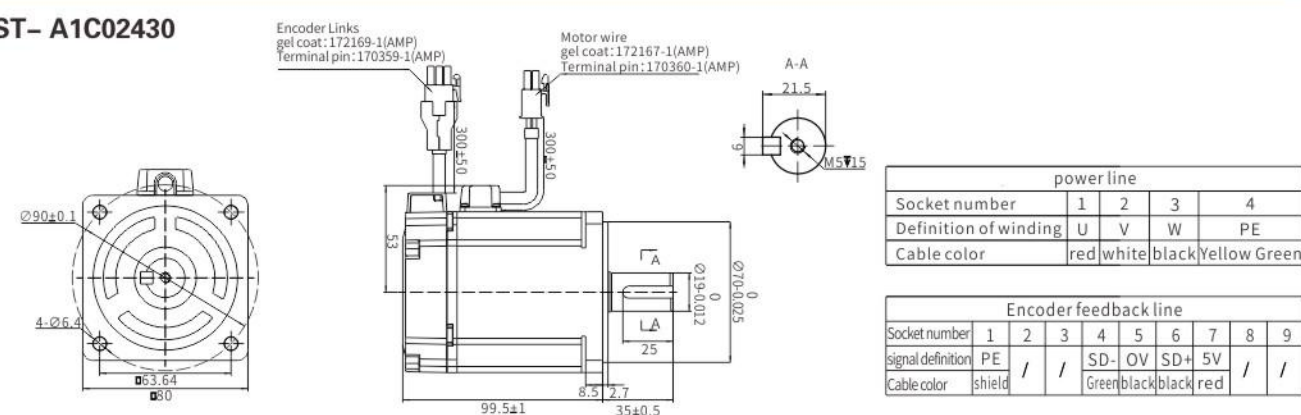


MECHANICAL DIAGRAM OF SERVO DRIVE WITH 5-POLE MOTOR

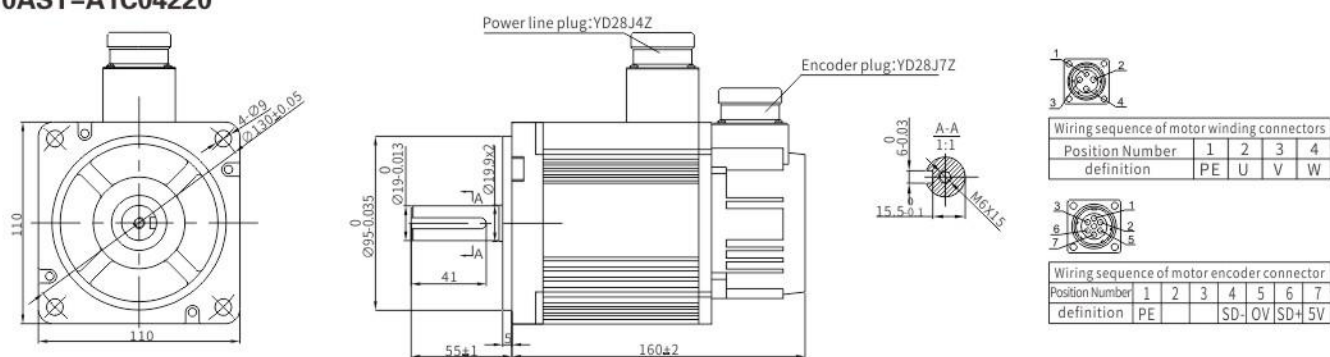
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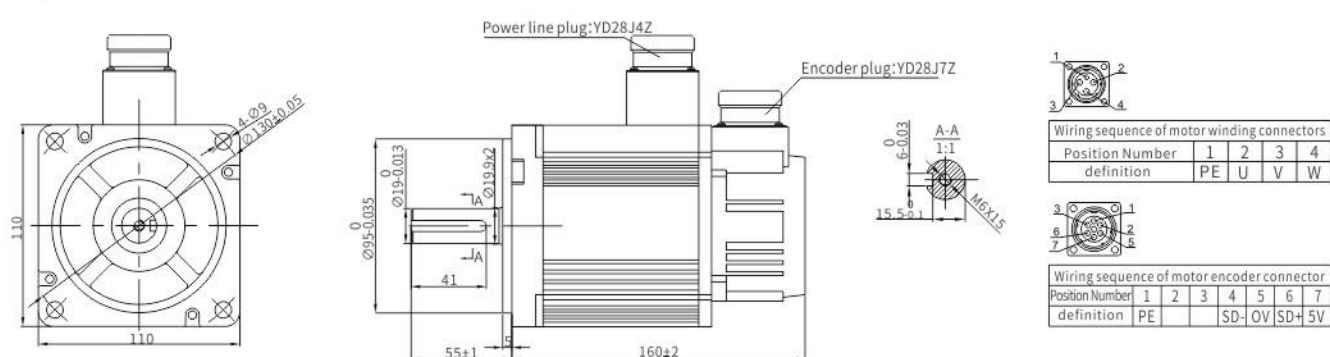
80AST-A1C02430



110AST-A1C04220

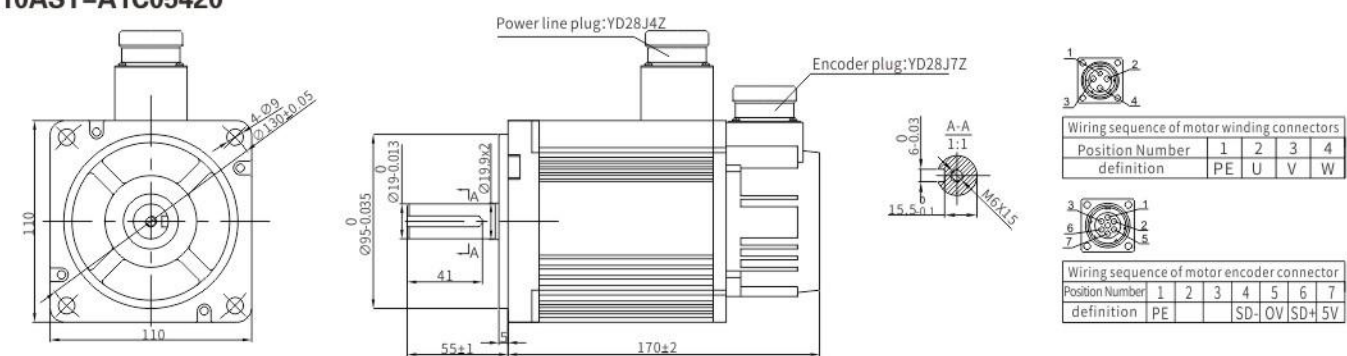


110AST-A1C04230

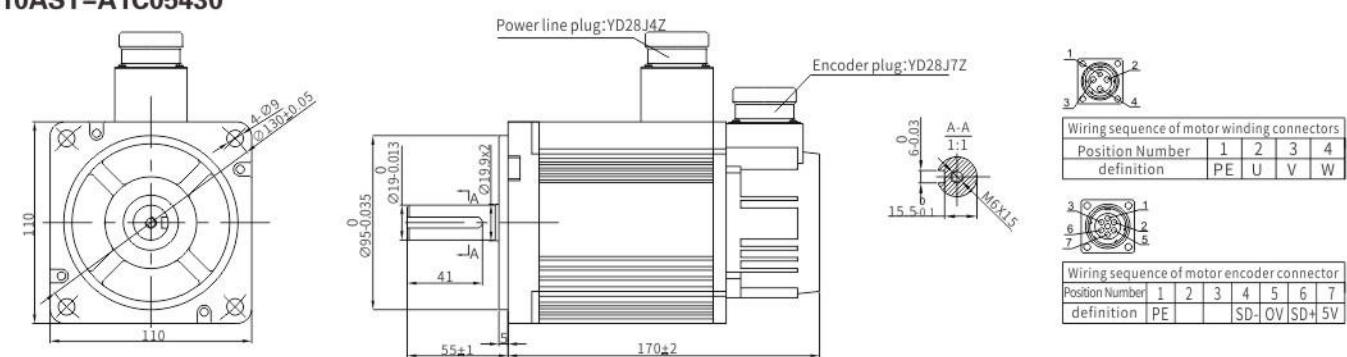


MECHANICAL DIAGRAM OF SERVO DRIVE WITH 5-POLE MOTOR

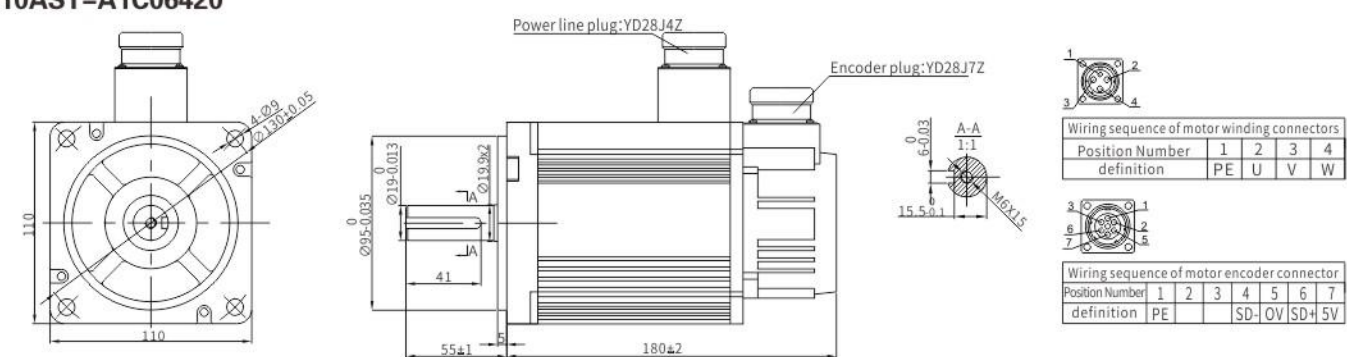
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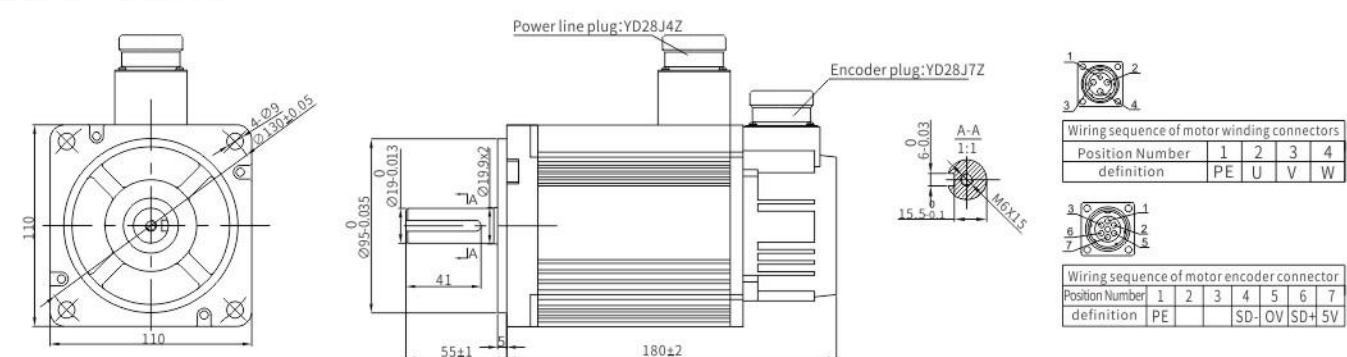
110AST-A1C05430



110AST-A1C06420

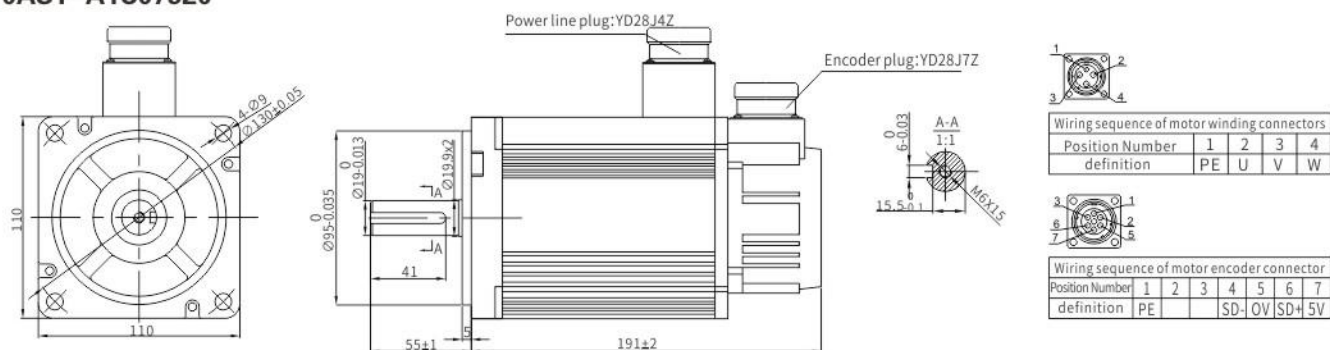


110AST-A1C06430

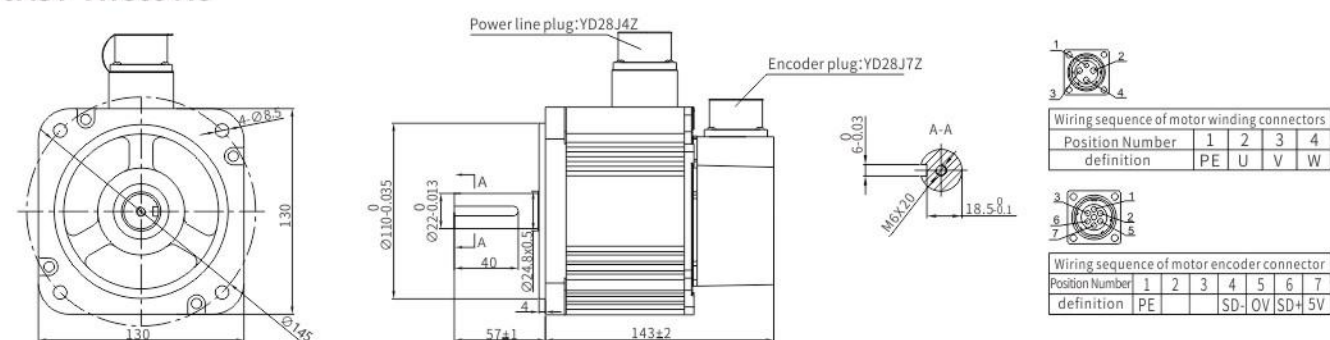


MECHANICAL DIAGRAM OF SERVO DRIVE WITH 5-POLE MOTOR

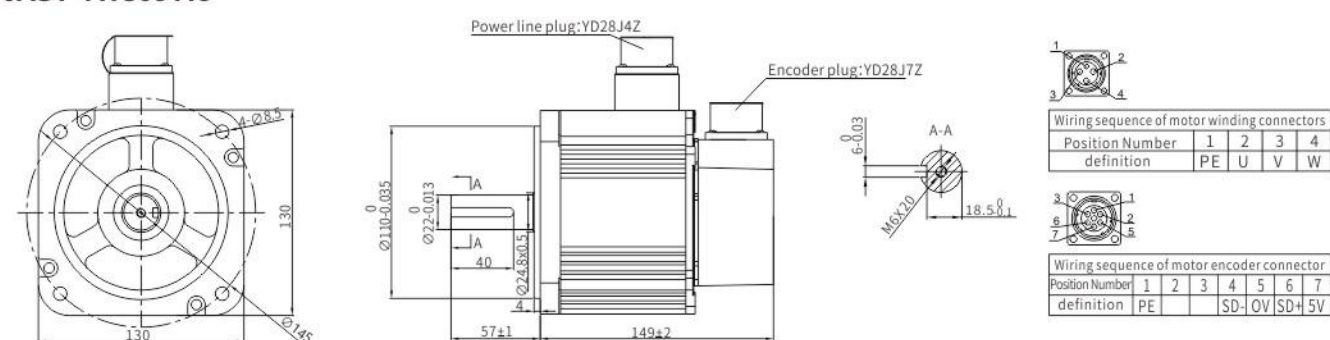
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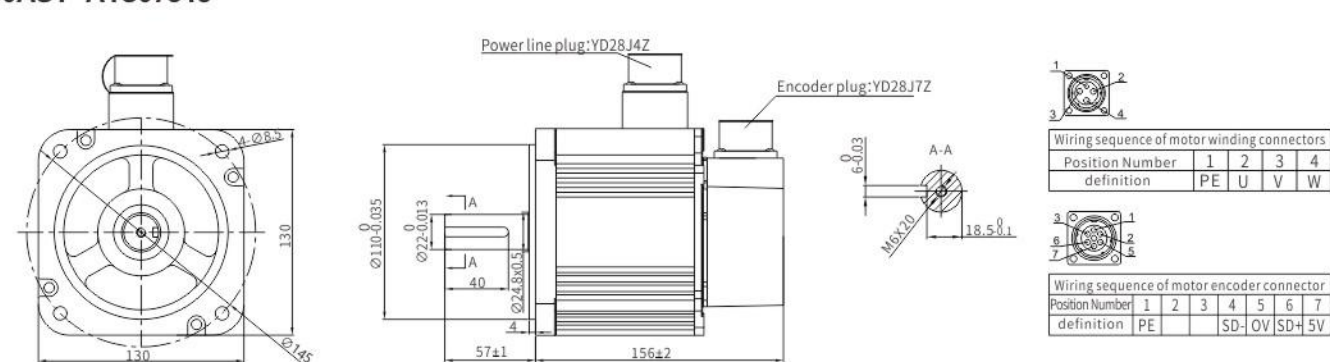
130AST-A1C05415



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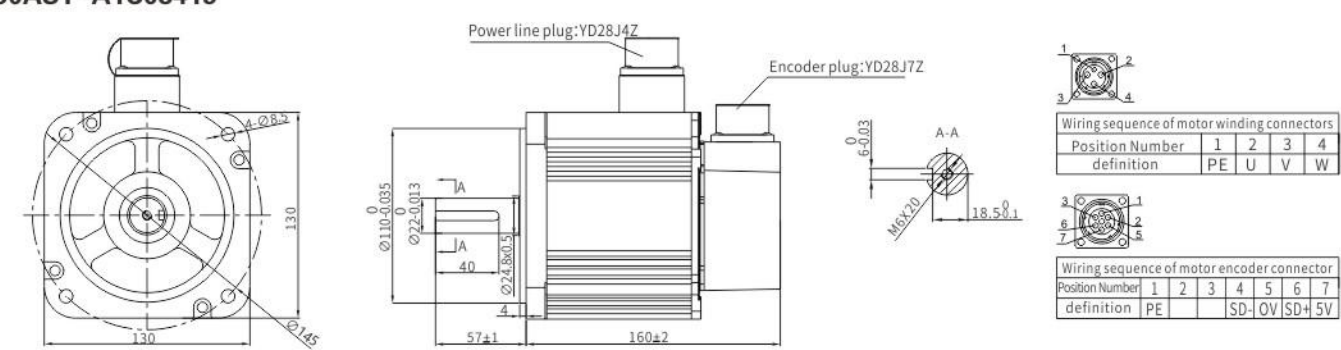


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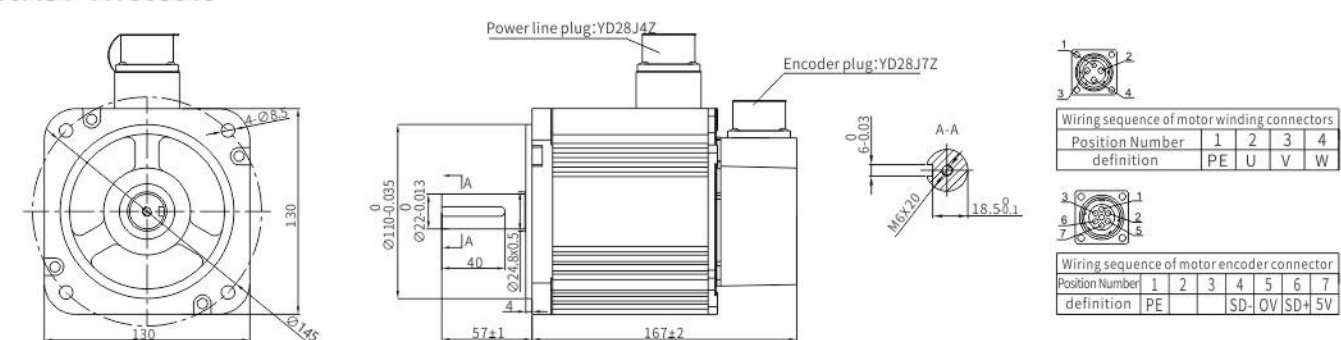


MECHANICAL DIAGRAM OF SERVO DRIVE WITH 5-POLE MOTOR

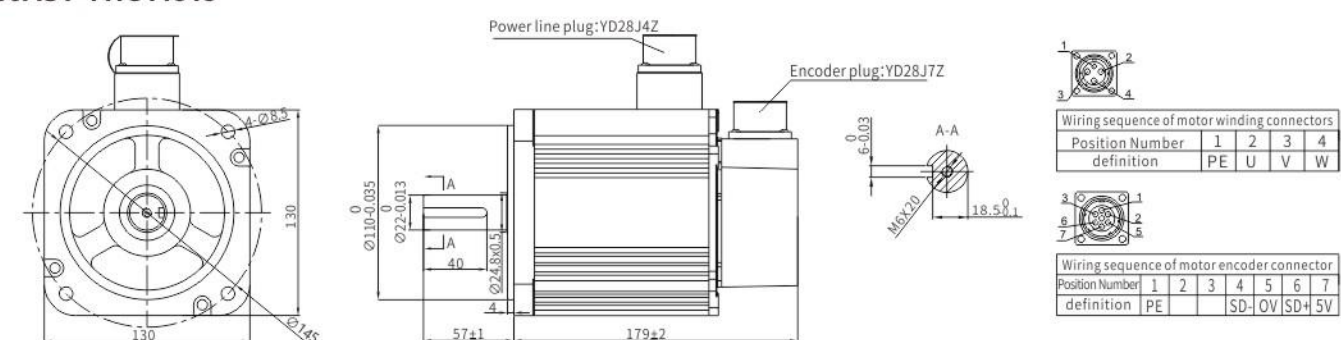
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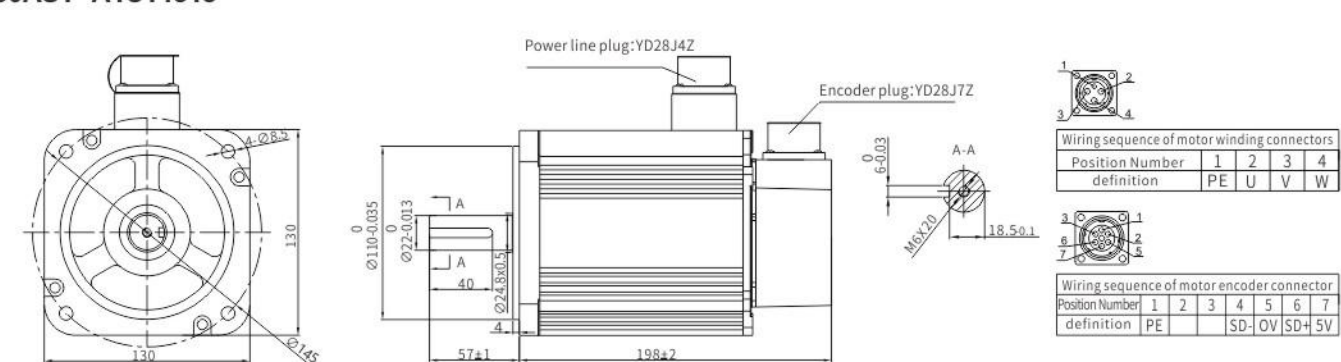
130AST-A1C09615



130AST-A1C11515



130AST-A1C14615



COMPARISON TABLE OF CABLE MODEL CONFIGURATION

Driver series	drive specification	Standard model of power line	Encoder line standard model
T3a/T3L/T3DF	L15/L20	DL-TH-80-A1.0D	BM-TH-80F-A1.0D
	L15/L20	DL-TH-80-ZH1.0D	BM-TH-80F-ZH1.0D
	L15/L20	DL-TH-80-XH1.0D	BM-TH-80F-XH1.0D
	L30(含6NM以下)	DL-TH-130-H1.0D	BM-TH-130F-H1.0D
	L30/L40		
	L50/H25/H50	DL-TH-130-H1.0U	BM-TH-130F-H1.0D
	L40	DL-TH-130-H1.0D	BM-TH-130F-H1.0D
	L50/L75/H25/H50	DL-TH-180-H1.0U	BM-TH-130F-H1.0D
	L40	DL-TH-130-H1.0D	BM-TH-130F-H1.0D
	L50/L75/H25/H50/H75	DL-TH-180-H1.0U	BM-TH-130F-H1.0D
C30G	L20	DL-TH-80-A1.0D	BM-TH-80F-A1.0G
	L20	DL-TH-80-ZH1.0D	BM-TH-80F-ZH1.0G
	L20	DL-TH-80-XH1.0D	BM-TH-80F-XH1.0G
	L30	DL-TH-130-H1.0D	BM-TH-130F-H1.0G
T6DE/T3D/T3DC/C30/ T3DL/SE30 (single board computer)	L15/L20	DL-TH-60-A1.0D DL-TH-80-A1.0D	BM-TH-80A1-A1.0F(No battery box/Anpu) BM-TH-80A-A1.0F(With battery box/Anpu)
	L30	DL-TH-130-H1.0D	BM-TH-130A1-M1.0F (Equipped with battery box/7-core direct insertion) BM-TH-130A-M1.0F (With battery box/7-core aviation plug)
T6E/T6M/T5M/T5a/ T5ML/T5aL/T3C	L20	DL-TH-80-MH1.0D DL-TH-80-A1.0D	BM-TH-80A1-ZM1.0D (Equipped with battery box/7-core direct insertion) BM-TH-80A1-A1.0D (No battery box/9-core Ampu)
	L15/L20		BM-TH-80A-ZM1.0D (Equipped with battery box/7-core direct insertion) BM-TH-80A-A1.0D (Equipped with battery box/9-core Ampu)
	L30(含6NM以下)	DL-TH-130-H1.0D	BM-TH-130A1-M1.0D (Equipped with battery box/7-core aviation plug)
	L30/L40		BM-TH-130A-M1.0D (Equipped with battery box/7-core aviation plug)
	L50/H25/H50	DL-TH-130-H1.0U	BM-TH-130A-M1.0D (Equipped with battery box/7-core aviation plug)
	L40	DL-TH-130-H1.0D	BM-TH-130A-M1.0D (Equipped with battery box/7-core aviation plug)
	L50/L75/H25/H50	DL-TH-130-H1.0U DL-TH-180-H1.0U	BM-TH-130A-M1.0D (Equipped with battery box/7-core aviation plug)
	L40	DL-TH-130-H1.0D	BM-TH-130A-M1.0D (Equipped with battery box/7-core aviation plug)
	L50/L75/H25/H50/H75	DL-TH-180-H1.0U	BM-TH-130A-M1.0D (Equipped with battery box/7-core aviation plug)
T3M	L30	DL-TH-80-A1.0U	BM-TH-80F-A1.0M
	L30	DL-TH-80-ZH1.0U	BM-TH-80F-ZH1.0M
	L30	DL-TH-80-XH1.0U	BM-TH-80F-XH1.0M
	L30(含6NM以下)	DL-TH-130-H1.0U	BM-TH-130F-H1.0M
	L30/L40		
T3G	L20	DL-TH-80-A1.0U	BM-TH-80F-A1.0G
	L20	DL-TH-80-ZH1.0U	BM-TH-80F-ZH1.0G
	L20	DL-TH-80-XH1.0U	BM-TH-80F-XH1.0G
	L30(含6NM以下)	DL-TH-130-H1.0U	BM-TH-130F-H1.0G
	L30		

ASSEMBLY PARTS LIST

Driver series	drive specification	Encoder line standard model
control line	standard configuration1.0m/1.5m/2.0m	
communication line	Length is available0.15m/0.3m/0.5m	
power line	Anpu plug	standard configuration3.0m 
	Direct insertion docking type aviation plug	standard configuration3.0m 
	Small size aviation plug	standard configuration3.0m 
	Aviation plug	standard configuration3.0m 
	Anpu plug	standard configuration3.0m 
encoder links	Live streaming docking aviation plug	standard configuration3.0m 
	Small size aviation plug	standard configuration3.0m 
	Air broadcasting	standard configuration3.0m 
	Comes with battery box (in the shape of a battery box)	standard configuration3.0m 

Note: This table only displays a partial diagram of cable types